

UNIVERSITY of LIMERICK

OLLSCOIL LUIMNIGH

Internet Performance Profiling of Countries and Destinations

Nikolay Todorov B.Eng. (Technical University of Varna)

A Thesis Submitted to the University of Limerick for the Award of the Degree of Master of Engineering

The research work reported in this thesis was undertaken under the supervision and direction of

Dr. Ivan Ganchev
DipEng (Honours), PhD, SMIEEE

Dr. Máirtín Ó Droma,
BE, PhD, CEng, FIEE, SMIEEE

Telecommunications Research Centre

Department of Electronic and Computer Engineering
Faculty of Science and Engineering
University of Limerick

Ireland

November 2014

ABSTRACT

The thesis presents research into data capturing techniques and data analysis in order to discern the Internet performance profiles of countries, both spatially and temporally. The methodology may be described as implementing flexible yet tractable models, which simultaneously account for long-term traffic dependence, latency, dynamic random routing, and spatial dependence. Active Internet probing and traffic monitoring strategies are used in collaboration with the European RIPE Atlas project to capture raw Quality of Service (QoS) measurements. The research goal is to contribute towards developing large-scale network testing methods for superior performance detection, e.g. bottleneck detection as an application of reconstructed delay or loss estimation and build Internet QoS profiles for the countries of interest, as well as obtain indicators of the current performance state of Ireland's access to the global Internet.

This thesis is organised into six chapters. Following an introduction and literature review, two major global Internet performance measurement systems CAIDA and RIPE are reviewed and compared. This is followed by two chapters detailing research methodology used in this project and research results and analysis. These chapters include details how and when the measurements were conducted, an example of the results obtained, together with an explanation of how these results can be used in order to acquire the needed information. The final chapter presents conclusions with a summary of the findings and ideas for future work.

From the results collected, Internet QoS performance profiles for 37 countries are constructed. The worst case scenarios are explored and a reason for the low performance is given. The final results may be analysed and serve as a benchmark for mapping evolving and future performance profiling over longer periods of time, different paths or more destinations within a specific country, depending on the purpose of the experiment conducted. So for some of these outcomes and results have been published in 2013 at an Irish inter-University Engineering, Informatics and Science Research Day [95].

DECLARATION

This thesis is written to fulfil the requirements for the degree of Masters. I hereby declare that the thesis is a result of my work and effort and that it has not been submitted anywhere else for any award. Where other sources of information or diagrams have been used, they have been acknowledged.

Signature: ..... **Date: .....**

Name: Nikolay Todorov B.Eng.

UL ID: 11062916

ACKNOWLEDGEMENTS

I would like to thank everyone that has assisted me to date. My family deserve a special mention for giving me the opportunity to follow this pursuit and giving me the inspiration to do so. Special thank has to be given to everyone that has guided me towards the end goal of completing this research to date, those people are Drs Ivan Ganchev, Máirtín Ó Droma, Eoin O'Connell and Damien Meere. The Telecommunications Research Centre postgraduates have also always been great to work with too. Over the years the willingness within the group to help anyone that needs assistance has always been something that I've admired.

Finally, I'd like to acknowledge all those people both here in UL and beyond who have been there to encourage and motivate me while I've been doing this research. They have been a great influence on me through this process and I'm sure that only for them I would not have completed as of yet; they are Janina Yotova, Conor Hogg, David O'Connell, Yavor Nanev, Tomás Walsh, Davis Gould-Duff and Caoimhe Roche.....to you all a sincere "Thank You!!!".

TABLE OF CONTENTS

ABSTRACT ..... **I**

DECLARATION ..... **III**

ACKNOWLEDGEMENTS ..... **V**

TABLE OF CONTENTS ..... **VII**

TABLE OF FIGURES ..... **X**

TABLE OF TABLES ..... **XI**

LIST OF ABBREVIATIONS ..... **XII**

CHAPTER 1 ..... **1**

INTRODUCTION **1**

1.1 OVERVIEW 1

1.2 STRUCTURE OF THESIS 3

CHAPTER 2 ..... **5**

LITERATURE REVIEW **5**

2.1 INTRODUCTION 5

2.2 PASSIVE TOMOGRAPHY 9

2.2.1 Passive Measurements *9*

2.2.2 Typical Passive Measures *10*

2.2.3 Related Work *12*

2.3 ACTIVE PROBING TOMOGRAPHY 23

2.3.1 Active Measurements *23*

2.3.2 Typical Active Measures *24*

2.3.3 Introduction to Active Probing *27*

2.3.4 Related Work to Date *29*

CHAPTER 3 39

CAIDA VERSUS RIPE 39

3.1 CAIDA 39

3.1.1 Overview 39

3.1.2 Projects and Infrastructure 40

3.2 RIPE NETWORK COORDINATION CENTRE 44

3.2.1 Overview 44

3.2.2 RIPE NCC Activities 45

3.2.3 Coordination Activities 47

3.2.4 Services 48

3.3 CONCLUSION 52

CHAPTER 4 53

RESEARCH METHODOLOGY AND DESCRIPTION OF THE EXPERIMENTS 53

4.1 RECEIVING AND CONFIGURING THE PROBE 53

4.2 THE PROBE SELECTION PROCESS 54

4.3 THE EXPERIMENT 57

CHAPTER 5 60

RESULTS AND ANALYSIS 60

5.1 INTRODUCTION 60

5.2 CLI 61

5.3 DAY-TO-DAY AND TIME INTERVAL ROUND-TRIP-TIME 68

5.4 RESULTS ANALYSIS 83

CHAPTER 6 93

CONCLUSION AND FUTURE WORK 93

REFERENCES 96

APPENDIX LISTING 101

APPENDIX 1: CLI RESULTS 101

APPENDIX 2: JAVA CODE 101

APPENDIX 3: PING RESULTS FOR THE COUNTRIES WITHIN THE EU..... 101
APPENDIX 4: PING RESULTS FOR THE DESTINATIONS OUTSIDE THE EU 102
APPENDIX 5: TRACEROUTE RESULTS FOR THE COUNTRIES WITHIN THE EU 102
APPENDIX 6: TRACEROUTE RESULTS FOR THE DESTINATIONS OUTSIDE THE EU 102

TABLE OF FIGURES

Figure 2.1: Layout of a small computer network Figure 2.2: The corresponding logical topology 8
Figure 2.3: The basic components of an end-to-end active measurement infrastructure. 23
Figure 3.1: Ark Raspberry Pi-based Network monitor..... 43
Figure 3.2: RIPE Atlas probe 50
Figure 5.1: Ping Day-to-Day MIN, AVG and MAX RTTs from North to South for Denmark 69
Figure 5.2: Day-to-Day MIN, AVG and MAX RTTs from South to North for Denmark 69
Figure 5.3: Ping Day-to-Day MIN, AVG and MAX RTTs from East to West for Denmark 70
Figure 5.4: Ping Day-to-Day MIN, AVG and MAX RTTs from West to East for Denmark 70
Figure 5.5: Ping MIN, AVG and MAX RTTs for time intervals from North to South for Denmark..... 71
Figure 5.6: Ping MIN, AVG and MAX RTTs for time intervals from South to North for Denmark..... 71
Figure 5.7: Ping MIN, AVG and MAX RTTs for time intervals from East to West for Denmark 72
Figure 5.8: Ping MIN, AVG and MAX RTTs for time intervals from West to East for Denmark 72
Figure 5.9: Traceroute Day-to-Day MIN, AVG and MAX RTTs from North to South for Denmark 73
Figure 5.10 Traceroute Day-to-Day MIN, AVG and MAX RTTs from South to North for Denmark 73
Figure 5.11 Traceroute Day-to-Day MIN, AVG and MAX RTTs from East to West for Denmark..... 74
Figure 5.12 Traceroute Day-to-Day MIN, AVG and MAX RTTs from West to East for Denmark..... 74
Figure 5.13 Traceroute MIN, AVG and MAX RTTs for time intervals from North to South for Denmark 75
Figure 5.14 Traceroute MIN, AVG and MAX RTTs for time intervals from South to North for Denmark 75
Figure 5.15 Traceroute MIN, AVG and MAX RTTs for time intervals from East to West for Denmark . 76
Figure 5.16 Traceroute MIN, AVG and MAX RTTs for time intervals from West to East for Denmark . 76
Figure 5.17: Denmark's North to South time interval MIN, AVG and MAX RTTs 85
Figure 5.18: Denmark's South to North time interval MIN, AVG and MAX RTTs 86
Figure 5.19: Denmark's East to West 30 day MIN, AVG and MAX RTTs 86
Figure 5.20: Denmark's West to East time interval MIN, AVG and MAX RTTs..... 87
Figure 5.21: Hungary's North to South time interval MIN, AVG and MAX RTTs 87
Figure 5.22: Hungary's South to North time interval MIN, AVG and MAX RTTs 88
Figure 5.23: Finland's East to West time interval MIN, AVG and MAX RTTs 88
Figure 5.24: Finland's West to East time interval MIN, AVG and MAX RTTs 89
Figure 5.25: Luxembourg's North to South time interval MIN, AVG and MAX RTTs 89
Figure 5.26: Luxembourg's South to North time interval MIN, AVG and MAX RTTs 90
Figure 5.27: Ireland's East to West time interval MIN, AVG and MAX RTTs 90
Figure 5.28: Ireland's West to East time interval MIN, AVG and MAX RTTs 91

TABLE OF TABLES

Table 4.1: Locations and probe IDs of the probes within the EU 55
Table 4.2: Target destination cities for the countries outside the EU 56
Table 5.1: Congestion Level Index results for countries within the EU 63
Table 5.2: Congestion Level Index for measurements from Ireland to countries outside the EU 66
Table 5.3: Example of CLI calculation for United Kingdom 67
Table 5.4: Day-to-Day MIN, AVG and MAX RTTs from North to South for Denmark (ping) 77
Table 5.5: Day-to-Day MIN, AVG and MAX RTTs from South to North for Denmark (ping) 77
Table 5.6: Day-to-Day MIN, AVG and MAX RTTs from East to West for Denmark (ping) 78
Table 5.7: Day-to-Day MIN, AVG and MAX RTTs from West to East for Denmark (ping) 78
Table 5.8: MIN, AVG and MAX RTTs for time intervals from North to South for Denmark (ping) 79
Table 5.10: MIN, AVG and MAX RTTs for time intervals from East to West for Denmark (ping) 79
Table 5.9: MIN, AVG and MAX RTTs for time intervals from South to North for Denmark (ping) 79
Table 5.11: MIN, AVG and MAX RTTs for time intervals from West to East for Denmark (ping) 79
Table 5.12: Day-to-Day MIN, AVG and MAX RTTs from North to South for Denmark (traceroute) 80
Table 5.13: Day-to-Day MIN, AVG and MAX RTTs from South to North for Denmark (traceroute) 80
Table 5.14: Day-to-Day MIN, AVG and MAX RTTs from East to West for Denmark (traceroute) 81
Table 5.15: Day-to-Day MIN, AVG and MAX RTTs from West to East for Denmark (traceroute) 81
Table 5.16: MIN, AVG and MAX RTTs for time intervals from North to South Denmark (traceroute) . 82
Table 5.18: MIN, AVG and MAX RTTs for time intervals from East to West for Denmark (traceroute) 82
Table 5.17: MIN, AVG and MAX RTTs for time intervals from South to North for Denmark (traceroute) 82
Table 5.19: MIN, AVG and MAX RTTs for time intervals from West to East for Denmark (traceroute) 82

LIST OF ABBREVIATIONS

ACK	Acknowledgement
API	Application Programming Interface
Ark	Archipelago
AS	Autonomous System
ASN	Autonomous System Number
BGP	Border Gateway Protocol
bps	bits per second
Bps	bytes per second
CLI	Congestion Level Index
DARPA	Defence Advanced Research Project Agency
DHCP	Dynamic Host Configuration Protocol
DHS	Department of Homeland Security
DNS	Domain Name System
EM	expectation-maximization
EU	European Union
HTTP	Hypertext Transfer Protocol
IBR	Internet Background Radiation
ICMP	Internet Control Message Protocol
IP	Internet Protocol
IPMP	IP Measurement Protocol
IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6
IRR	Internet Routing Registry
ISOC	Internet Society
ISP	Internet Service Provider
IT	Information Technology
LAN	Local Area Network
MINT	metric-induced network topologies

MLE	maximum likelihood estimation
NDN	Named Data Networking
NSF	National Science Foundation
OD	Origin Destination
QoS	Quality of Service
RIPE	Réseaux IP Européens
RIR	Regional Internet Registry
ROA	Route Origin Authorisation
RPKI	Resource Certification
RPSL	Routing Policy Specification Language
RTP	Real-time Transport Protocol
RTT	round-trip time
SDSC	San Diego Supercomputing Center
SNMP	Simple Network Management Protocol
SYN	Synchronize
TCP	Transmission Control Protocol
TRC	Telecommunications Research Centre
TTL	Time to Live
UDP	User Datagram Protocol
UL	University of Limerick
USB	Universal Serial Bus
VoIP	Voice over IP
WAN	Wide Area Networks

CHAPTER 1

INTRODUCTION

In this chapter the role of Internet tomography and Quality of Service performance measurements is introduced. What specific research is reported in these along with brief summaries of the contents of each of the thesis chapters is presented.

1.1 OVERVIEW

The Internet has evolved rapidly, in a largely unregulated and open environment, and became a massive, distributed network over the last decade, which continues to expand exponentially. This has left network engineers and service providers with a need to continuously assess and monitor the Quality of Service (QoS) of the networks and backbone environments which comprise the Internet. This includes monitoring packet loss rates, bandwidth utilization, delay distributions, and other measures of throughput and performance [1, 2]. Quantitative assessment of network performance is extremely challenging because of the heterogeneous and largely unregulated structure of the Internet. Tasks such as dynamic routing, optimized service provision, service level verification, and detection of anomalous/malicious behaviour are difficult as one cannot rely on the cooperation of individual servers and routers to aide in the collection of network traffic [1].

The aim of this research is to obtain the Internet performance profiles of countries, both spatially and temporally, by implementing flexible yet tractable models which simultaneously account for long-term traffic dependence, latency, dynamic random routing, and spatial dependence. Active Internet probing and traffic monitoring strategies are used in collaboration with the European RIPE Atlas project to capture raw Quality of Service (QoS) measurements. These are then used to construct and analyse the countries' Internet QoS profiles.

The research goals of capturing Internet performance profiles are made possible through collaborations with major coordinate global activities in this area. Of particular importance is the access to measurements collected in the very large integrated Internet probing and QoS measuring "RIPE Atlas" project coordinated by the independent, not-for-profit Réseaux IP Européens (RIPE). This project encompasses thousands of active probes concentrated in the RIPE NCC service region (Europe, Middle East and parts of Central Asia), capturing, for instance, dynamic profiles of the Internet connectivity and reachability in coarse or fine grained, spatial and temporal resolution Internet paths or logical connections. RIPE's main goals are to produce Internet "traffic maps", provide users with active measurement tools for baseline and on-demand measurements, and also to act as a trusted source, and exchange point, of data regarding real-life, active measurements. Through this, an unprecedented understanding of the state of the Internet in real time is possible. The project itself is discussed further in chapter 3.

The minimally invasive RIPE measurement methodologies, based on software tools for generating probe traffic and monitoring traffic flows, statistical modelling of the measurement process, and sampling strategies for online data collection, are used in this research. Model validation is focused on the application to real network data. By using available databases and probing tools, monthly one-way delay trends on selected routes were obtained as per the methodology described in [3]. *Ping* and *traceroute* measurements were conducted for the duration of one month each.

After the raw data from the measurements was extracted, it was then used for building a performance profile of 26 preselected countries within the European Union as

well as defining the current state of Ireland's Internet connectivity to 11 international destination outside the Union. This was made possible with the generation of bar graphs and the calculation of Congestion Level Index. Both of the above helped to visualise behaviour and traffic dependencies. The full set of results including bar graphs generated, the table with all Congestion Level Index results and all raw data files can be found in the appendices on the attached DVD.

1.2 STRUCTURE OF THESIS

Chapter two provides a detailed literature review on the topic of this project. The chapter is divided into three main parts: Introduction, Passive Tomography and Active Probing Tomography. In the passive and active tomography parts there is a detailed description of both strategies accompanied with the typical measures related to those methods as well as the related work to date in the areas.

Chapter three provides an overview of the CAIDA and RIPE, the two largest organisations involved in this area of research. A description of their current projects is given and further details surrounding the most relevant projects to this research are specified. The chapter finishes with a conclusion as which project would be more suitable for the purpose of the experiment and why.

In chapter four, an explanation of the experiment and the research methodology is provided. The chapter starts with a description of the process on becoming a RIPE Atlas member and the procedure for installation of the probe. In order for the experiment to be conducted, a number of probes in the countries of interest had to be selected and this is described next, with a full list of all countries and cities where those probes are located. A description of the experiment itself is then provided with details on how the measurements were conducted, the limitations behind them, the time frame and location of each measurement. Finally, there is an explanation of how the raw data was manipulated in order to acquire the needed information.

Chapter five provides an overview of the results and analysis of the same. It starts with a definition of the Congestion Level Index (CLI) used for the purpose and the way it was calculated, followed by a separation of the countries in different groups depending on the CLI results. Summarised tables with all the CLI results are provided for the countries within the European Union and the destinations outside, as well as a single example in the form of a table consisting of all the values used and calculations made in order to get the summarised one. The chapter continues with an explanation of how the results are visualised with two different types of bar graphs: one visualising the day-to-day minimum, average and maximum round-trip-times, and one summarising those but for specific time intervals. A full example is provided with all the bar graphs generated for Denmark as well as the tables with the extracted data used for the creation of the graphs. The last part of the chapter is the analysis of the results. Here the main obstacles for building an Internet profile of the countries are explored and an example and discussion regarding the worst case scenarios extracted, based on the CLI index, is made.

The sixth chapter is the conclusion which gives a brief overview of the whole project. This overview goes through the whole process, the limitations met and an explanation of how and why the results are useful. A recommendation for future work and development on the same topic is given.

CHAPTER 2

LITERATURE REVIEW

In this chapter the field of Internet Tomography is introduced in detail. Specifically Passive and Active Probing Tomography are explored as methods for capturing the overall network performance. There is a detailed description for both of these strategies along with their typical measures and a detailed overview of the related work to date in the areas.

2.1 INTRODUCTION

As the Internet has evolved very rapidly in a largely unregulated and open environment and became a massive, distributed network, over the last decade, which continues to explode in size, the network engineers and service providers are keenly interested in continuously assessing and monitoring the quality of service of the networks. This includes loss rates, bandwidth utilization, delays, and other measures of throughput and performance [1, 2]. Quantitative assessment of network performance is extremely challenging because of the heterogeneous and largely unregulated structure of the Internet. Tasks such as dynamic routing, optimized service provision, service level verification, and detection of anomalous/malicious behavior are very hard as one cannot

rely on the cooperation of individual servers and routers to aid in the collection of network traffic [1].

The need of network engineers and internet service providers (ISP) for tools and techniques for monitoring network utilization and performance stimulates the work in network tomography. This need has increased due to the number and range of services supported continuing to grow exponentially in recent years. The new services such as VoIP, multimedia streaming and on-line games require high-level QoS guarantees [4, 5]. For network administrators, it is very important to keep abreast of what is happening in the network so they can identify for example bottlenecks or network misconfiguration.

The term network tomography was introduced in Vardi (1996) to characterize two classes of inverse problems in computer and communication networks. The goal is to recover higher-dimensional network information from lower-dimensional data [6]. The first class deals with passive tomography where aggregate data is collected at the router level and the goal is to recover path-level information. The most common application is estimation of the origin-destination traffic matrix of a network. The second is active tomography which deals with reconstructing link-level information from end-to-end path level measurements by sending packets from a source to several receiver nodes, all located on the periphery of the network. The goal is an estimation of quality-of-service parameters such as loss rate and delay distributions [4]. In any of these cases, statistical models should be developed for the measurement process.

In passive tomography monitoring agents are placed at internal routers and switches in the network. This is a node-oriented method that collects packet and network flow information passively through monitoring. Aggregate level data is collected on total traffic flowing into and out of the monitored nodes. The measurements consist of counts of packets that pass through the nodes in the network. The information collected is used from network operators for capacity planning and management decisions. The main problem is that they need access to all the network elements in order to deploy monitoring agents to collect information. The volume of

data can be substantial so information about the origin-destination cannot be collected at the individual packet level. The inverse problem is to recover the origin destination information of the traffic patterns in the network [4, 6, 7].

Active network tomography refers to a different class of inverse problems. As the Internet is a collection of subnets that are controlled by many different entities with almost no collaboration between each other, this results in restricted access to the internal nodes of the network. The solution is for probe packets to be sent from a source node to several receiver nodes located on the periphery of the network of interest and measure packets transmitted and/or received between nodes or time delays between transmissions and receptions. The probe packets are sent using the multicast protocol where a single packet is sent from the source and is duplicated at each branching point on the way to the receivers. The inverse problem is to recover the delay distributions and loss rates associated with the interior network nodes from the end-to-end measurements collected [2, 7]. The time delays are due to both propagation and router processing delays. The measured path delay is the sum of the delays on the links comprising the path. The delay on a single link is the propagation delay on that link and the queuing delay at the routers lying along that link [7]. Characterizing these parameters is very important for detecting faults, congestions, and other anomalous behavior as well as ensuring compliance of service level agreements with users [5] and management of overlay network [6, 8].

Development of new applications with higher QoS requirements such as VoIP and online games demand greater need for fast and efficient algorithms for assessing and responding to changes in network performance [6]. Due to the complexity and size of modern networks the traditional approach for assessing network performance based on detailed queuing models at the individual router level is not enough for capturing overall network performance. Estimating link-level QoS parameters requires access to the internal links and routers which makes the collection of detailed information at this level very difficult. As a result the reconstruction of link-level data from path-level end-to-end measurements involves sending probe packets from a source to one or more receiver nodes [6].

Consider the network depicted in Figure 2.1. Each node represents a computer terminal or a router. A connection between two nodes is called a path. Each path consists of one or more links representing the connection between the intermediate nodes. Messages are transmitted by sending packets of bits from a source to a destination node along a path which generally passes through several other nodes[7].

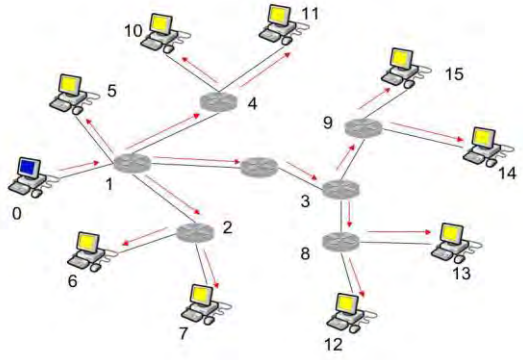


Figure 2.1: Layout of a small computer network [7]

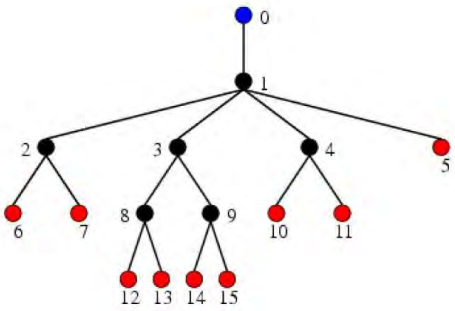


Figure 2.2: The corresponding logical topology [7]

If one wants to send a file from one location in the network to another, the file's content is broken into pieces called packets. Additional information on origin-destination, reassembly instructions (such a sequence numbers), and error-correcting features are added. The group of packets associated with a particular file is referred as a 'flow'. The origin destination information is used by the network elements (routers and switches) to deliver the packets to the intended recipient. For a better understanding we can imagine that the routers/switches located at the core of the network play a role of

intersections in a road network. They queue up the incoming packets and forward them to the next router according to some protocol which is usually along the most effective route. Due to the fact that the queues consist of a physical block of computer memory with a finite buffer size, if the queue is full when the packet arrives it is dropped. Depending on the transmission protocol, the sender may or may not get noticed in order to arrange a re-transmission. In case the packet is accepted it waits until it reaches the front of the queue and is forwarded to the next router on the way to its destination. This queuing mechanism is responsible for observed packet losses and, to a large extent, for packet delays. The physical topology of the network in Figure 1a can be transformed to a logical topology. Usually the logical topologies have the structure of a tree. When making the transformation from a physical to logical topology a combination of two or more links without a split is represented by just one chain and the information of the separate links cannot be estimated separately (Figure 2.2) [2, 6]. If for example we send a packet from source node 0 to destination nodes 6, 7, 10 and 11 we are going to obtain path-level losses information for paths 0-6, 0-7, 0-10 and 0-11. The big problem of active network tomography is to estimate the link level parameters of the individual links comprising these paths: 0-1, 1-2, 1-4, 2-6, etc. [6].

2.2 PASSIVE TOMOGRAPHY

2.2.1 Passive Measurements

When talking about passive tomography measurements we mean the standard process of monitoring the traffic passing by the predefined measurement points (monitors) by observing the performance and behaviour of packet streams. For this goal statistically useful quantities of data should be obtained from the destination addresses of interest so a proper conclusion about the desired characteristics could be made. The traffic flowing through the observed node can be filtered in different ways to isolate different streams, according to criteria such as destination address, protocol, or application. At one extreme this involves discounting address information entirely to

obtain at-a-point aggregate link statistics such as packet size distributions or traffic composition by protocol. This class of network measurements is usually referred to as *passive at-a-point* measurement [9].

End-to-end performance metrics like loss and delay may be obtained from the so called passive end-to-end measurements which are actually combined passive measurements at an origin and destination. Here packets such as acknowledgements are captured as a function of known Origin Destination (OD) pairs at each end [9]. For the whole process to be possible problems like timestamp synchronisation and packet identification must first be solved, as well as the logistics of retrieving, comparing and merging of the various data sets. In this case even bigger quantities of traffic must be obtained between the OD pairs examined compared to the simple passive at-a-point case, although this is less critical in the round-trip case than in the one-way case. There are cases where bidirectional passive measurement at a single physical point on a link inside the network of interest could provide such data, instead of physically separate monitors at different nodes [9].

2.2.2 Typical Passive Measures

There are metrics typically collected using passive measurement techniques a lot of which refer to a certain network element. However, some end-to-end measures like round-trip performance, are also quite commonly obtained using passive measurement [9, 10].

Throughput is the average rate of data carried over a link or a path through the network, or passed through a certain node. It is usually measured in bits per second (bps) or bytes per second (Bps) and sometimes in data packets per second or data packets per time slot. It is measured by counting the amount of data transferred during a specified time interval. Usually throughput refers to the total amount of data carried over a link. In certain cases when needed it could be refined by filtering the traffic based on application type, protocol, destination, origin, etc. [9].

Choosing the time interval is not an easy task for these types of measurement. If they are too short or too long, the data collected might not be useful as there might be decreased confidence or hidden burstiness and possibly the non-stationarities of the traffic on smaller time scale [9]. That's why in practice one to five minute intervals are used and from them are generate daily and weekly plots.

The construction of traffic matrices is an example of throughput measurement where passive measurements are collected for both incoming and outgoing traffic at nodes around the edge of the network.

Utilisation - The link utilisation is basically how much of the capacity of the network is actually used. It can be defined as the ratio of measured throughput and the capacity (maximum data rate) of the link [9]. This gives service providers information about how efficiently resources of the network are being used and how close they are of being overloaded.

Availability can be defined as the percentage of time when the system/service is in functioning condition. *Barlow and Proschan [1975]* define the probability of e repairable system as "the probability that a system is operating at a specific time t". An example of an availability metric is *connectivity*, defined in IETF IPPM standard, RFC 2678 [9].

Latency - it represents the delay across a network link. It could be measured either one-way or round-trip. Usually active methods are used for measuring latency but passive approaches exist as well [9, 11].

Although for a high level view of the network performance, the above metrics are used, for more detailed analysis statistics like **Packet size distribution, Length of packet trains,** and **IP address prefix length distribution** are collected to help in router design and configuration [9, 12]. In addition *raw traffic traces* are commonly used for diagnosis of network performance problems and for resolving security related issues [9].

2.2.3 Related Work

We can generally classify the tomographic techniques into two categories: continuous and boolean [13, 14].

In the continuous tomographic methods sophisticated statistical algorithms are used to solve linear system of equations that relate path measurements to individual link characteristics. For example in [15] and [16], the expectation-maximization algorithm is used to infer link-level loss rates from end-to-end multicast probe packet measurements, while in [17] the least-squares method is applied to compute delays for smaller path measurements to individual link characteristics [13]. Markov Chain Monte Carlo algorithm is used in [18] for identifying lossy links based on passive observations made on the traffic at a server. In [19] is proposed a method where minimal set of linearly independent paths are selectively monitored so for example loss rates and latency of all paths can be inferred [13].

In Boolean tomographic methods [20-22] when a path has loss rates or delays exceeding a certain threshold, it is assigned a bad label. The rest of the paths are labeled as good paths. In [20] is proposed a simple algorithm that attributes the bad paths to the smallest set of consistent bad links. The authors of [21] assume that some paths may be wrongly labeled which complicates the link diagnosis problem. They use passive measurements of end-to-end application traffic in a sensor network setting to identify links with high loss rates [13]. [22] proposes a polynomial-time algorithm for determining the smallest independent subset of paths that can differentiate identical anomalous-link sets as distinguished by the bigger input set. They assume that randomly large subset of links can experience an anomaly in the same time and this leads to many paths being monitored even though in practice it is very unlikely this to happen [13].

2.2.3.1 Diagnosing Link-Level Anomalies

Finding the best location for monitoring devices in passive probing has been studied in [23], [24] and [25]. The goal there is to cover as many IP flows as possible and maximize the flow sampling rate. In contrast the work in [13] is focused on diagnosing anomalies and answering questions like where to locate measurement points and which paths to monitor to make the identification of problematic links in a passive monitoring setting easier.

Mesh and tree topologies are considered in [13] as both can be found in practical settings. Mesh topologies are generally used in Wide Area Networks (WANs) while tree topologies are preferred in Enterprise environments. The passive monitoring approach relies on continuously observing IP packets traversing network paths to detect anomalies. A set of passive monitoring devices are placed at various points in the network.

As there is a variety of technologies available for non-intrusively observing network traffic, for this experiment are used the so called network taps. They are hardware devices that hook directly onto the network cable and send a copy of the traffic passing through them to one or more other network devices. These network taps are placed on the network links and connected to off-the-shelf PC or server either directly through the NIC or via specially designed for this purpose high performance cards like DAG from Endace [26]. Another technology for passive monitoring is the Port mirroring. This is a method that forwards a copy of each packet, either incoming or outgoing, from one port of a network switch to another which is connected to a monitoring device [13].

Packet losses and delays can be detected using passive taps for observing the traffic. If we have two taps just before the start and after the end of a path, then by counting the packets seen at the probes at previously set time intervals we can make a conclusion if there were losses or not. If the difference between packet counts at the taps is large and exceeds a pre-specified threshold then we can say that packets are being lost

along some links on the path. If timestamps are associated with the packets then it is possible to detect delays for the same path of interest by keeping track of the difference between packet timestamps averaged over an interval or for sample packets at the taps [13].

As a path may contain a couple of links, for diagnosing the anomalies, they must be distinguished. The condition specified in [13] as Theorem 1.1 is necessary. It states that "A set of monitored paths Q is sufficient to diagnose the anomalous link if for every pair link (e1,e1) in E, there is at least one monitored path in Q that contains exactly one of the two links (that is, a path that distinguishes between the links e1 and e2)".

The passive monitoring infrastructure is built in two phases. In the first phase a minimal set of links in the core network is chosen where the additional probes are going to be placed so that all single-link anomalies can be correctly identified. The goal is to minimize the installation and operation cost as much as possible. A minimum set of paths to monitor is determined in the second phase so that all single-link anomalies may be detected and diagnosed.

The problems associated with the two phases are as shown in [13]:

Problem statement 1 (Probe Placement): Given a directed graph $G=(V,E)$, set of paths P, and set of directed link pairs L that cannot be distinguished by paths in P, select the smallest subset of (bidirectional) edges F on which to place probes so that every link pair in L is distinguished by some edge F. As V includes the set of routers and switches and E is a set of bidirectional edges representing the communication links that connect the network nodes.

Problem statement 2 (Path Monitoring for Detection): Given a directed graph $G=(V,E)$, and a set of paths P' that can be monitored, select the minimum subset of paths $Q_{det} \subseteq P'$ such that every link in E belongs to some path P' .

Problem statement 3 (Path Monitoring for Diagnosis): Given a directed graph $G=(V,E)$, and a set of paths P' that can be monitored, select the minimum subset of paths $Q_{diag} \subseteq P'$ that satisfies the anomaly diagnosis condition (Theorem 1.1).

The probe placement problem for both mesh and tree topologies turns out to be NP-hard. For mesh topologies a logarithmic factor approximation algorithm is obtained and the main idea is to reduce a Probe Placement problem instance to an instance of Set Cover [27] by mapping each undistinguished directed link pair in L to an element, and each edge (potential probe location) to a subset of link pairs in L that it can distinguish. For solving the problem in tree topologies a 3-factor approximation algorithm is presented. As mentioned before a link pair can be distinguished by placing a probe anywhere on the path segment between the pair. As an example, in a tree with paths between the leaves, we can move bottom-up without distinguishing a pair until the closest node to the root is reached. We see that the pair cannot be distinguished further up in the tree and we say that the link pair becomes ripe at this point. The Lazy Placement algorithm presented in [13] delays selecting a probe location to distinguish a pair until it becomes ripe.

Selecting the minimum number of paths for anomaly detection for mesh and tree topologies is also NP-hard. A logarithmic factor approximation algorithm may again be applied for selecting a subset of paths to cover all the elements in the network graph for mesh topologies. Thus, for the path cover problem for tree topologies we have a 2-factor approximation to the optimal number of paths needed to cover links. The algorithm starts by choosing randomly node in the tree as the root node. After this for each leaf node it adds to Q_{det} two paths from P': the outgoing path that reaches closest to the root, and the incoming path that reaches closest to the root.

After the anomaly detection, identification of the link is needed. Approximation algorithm for computing a minimum number of paths to diagnose single-link anomalies is proposed in [13]. For mesh topologies a logarithmic factor approximation algorithm is applied and for computing the minimum number of paths in tree topologies a constant factor approximation algorithm is used.

2.2.3.2 Obtaining the Origin-Destination Matrix using Passive Tomography

The goal in estimating the origin-destination matrix is to obtain the information about the distribution of traffic flowing from one node to another for all pairs of nodes in a network. The obtained information of course must be studied in different variations such as different time-of-day and day-of-the-week, as there might be a change [4].

Here monitoring stations are placed at the individual nodes and the data collected is the total number of packet counts passing through the node. These packets contain origin-destination (OD) information but the amount of data is too big to look at the information of every packet individually. This is the reason why only the data from the total packet count is obtained using the Simple Network Management Protocol (SNMP) [4].

Let's examine the example in [4]. In this case $|\nu|$ indicates the number of nodes and D represents the OD pairs in the network. A fixed time period is examined where the traffic patterns are reasonably homogeneous. Y_t is a $|\nu|$ column vector containing the number of packets going through all the nodes in period t for $t = 1, ..., T$ during the time of study, and $\bar{y}$ is the average number of packets for the entire period. R will be a $|\nu|$ x D routing matrix, and X will be D column vector that represents the unknown flows crossing the network. The routing matrix changes over time, usually every few hours [4, 28]. It is also usually estimated by computing shortest path using the Interior Gate Protocol link weights that indicate congestion levels, and the obtained information about the network topology. If we write:

$$\bar{y} = RX. \qquad (1)$$

The statistical inverse problem is to reconstruct the distribution of X from the aggregate level data $\bar{y}$.

Vardi (1996) [29] modeled the traffic flows as Poisson, i.e., the X_j's are independent Poisson random variables with means λ_j's [4]. In the Poisson assumption a higher order information can be used for estimation as the variance is equal to the mean. It also provides additional estimation equations because of this. Vardi (1996) studied maximum likelihood estimation (MLE) using EM algorithm but it becomes computationally inflexible for large networks. He studied several heuristic methods as alternatives and the method-of-moments estimation turns out to be the most promising. He proved identifiability for all practical networks [4]. Under the same framework a Bayesian approach is considered in [30]. Even though the goal here is dealing with prediction of the actual OD traffic counts, "The Poisson assumption implies that the variance of X is proportional to the mean" [4]. This assumption was relaxed by [31].

The first generation of models though didn't work well when trying to estimate the distribution of X in high speed networks as they are very sensitive to the assumption, which was not very practical for real network traffic data [11] and because of that a new generation of models were developed, which employ extra information and different assumptions. There are two significant approaches. They are the tomogravity model [32, 33] and the method of routing changes [34]. The tomogravity model assumes that the OD flow X (i,j) between the nodes i and j is proportionally related to the total amount of traffic going out from node j, and the total amount going in node i [32]. The same models assumes independence between the sources and destinations, but as stated in [11] this is likely to be violated in backbone networks due to the so-called hot-potato routing policies adopted by their operators (operators offloading peering traffic at the nearest exit point). In [32] a variation of the simple tomogravity model was also proposed.

In the route change method [34] an attempt to overcome the under-constrained nature of the problem is made using manipulating the link weights and in that way inducing additional routing matrices R.

In the third generation of traffic matrix estimation methods, data is obtained during $t = 1, ..., T$ periods. By integrating temporal considerations, the purpose is to

assess the temporal evolution of the OD flows over time. Two approaches are developed in [31] for estimating the parameters of X as they change over time. The first is based on a moving window with a locally time-homogeneous approach within each window while the second one is using more formal temporal model [4].

[34] suggested the following a state-space model:

$$Y_t = RX_t + u_t,\ t = 1\ ,..\ ,\ T \qquad (2)$$

$$X_t = CX_{t-1} + e_t,\ t = 1,\ ...\ ,\ T \qquad (3)$$

Equation (2) represents the traditional traffic matrix model, where u_t is the measurement error. Equation (3) sets a temporal model for underlying traffic state and a spatial dependence between OD flows through the non-diagonal elements of the C matrix, where e_t captures the traffic system noise process [4]. A modification of this model is suggested in [35]. Therefore, assisting in calibrating the parameters of the model, direct measurements are allowed occasionally for selected OD flows.

The problem of traffic matrix estimation has proved useful to network operators, particularly for capacity planning purposes. As the problem has ill-posed nature it is a necessity to impose different modeling assumptions that in many cases have had a negative impact on the accuracy of the results [36]. The focus over the years has shifted from temporal to spatial variability understanding of the OD flows. The quest for models that can capture more precisely the characteristics of the OD flows in high speed networks, fast and scalable estimation techniques, and the simulation of realistic traffic matrices [36] are only some of the current research challenges.

2.2.3.3 Network Discovery from Passive Measurements

Doing experimental measurements for understanding the Internet structure is quite important for developing new protocols, traffic engineering, troubleshooting and many others. Although most of the studies in the network topology discovery are based on active (Traceroute-like) measurements, passive measurements of packet traffic offer an

expanded perspective of Internet structure using much lower impact and network resources [37].

Discovering and characterizing Internet structure and topology through experimental measurements has been an active research area for some time [37-42]. Most prior work for measuring Internet's structure has been based on active measurements but as with every technique it has certain limitations. Some of them are:

- as the Internet is a huge network this means that a set of measurement hosts M and target host N where N>>M must be established as the various features of the infrastructure must be captured in the measurements (especially on the edges of the network [6]) [37]
- the probes sent from measurement hosts to all destinations results in a significant amount of traffic and complex management issues [37]
- as the ISP often try to prevent structure discovery for example by filtering ICMP packets, this makes tools like Traceroute ineffective [37]

The authors of [37] explore the problem of Internet-wide structure and topology discovery using passive measurement. There are limitations that make this method really challenging. First, the individual measurements themselves include almost no information about the network structure and second, the source IP addresses are often subject to privacy restrictions. Despite the limitation, two capabilities are demonstrated in [37]:

1. Internet sources can be automatically and precisely clustered into meaningful groups corresponding to shared network topology;

2. Network topology can be accurately recovered from large amounts of passive information collected when coupled with very small number of additional active measurements.

The methodology in [37] begins with using a standard technique to determine the hop-count distance between the sources and the passive monitors [37, 43]. This results in a hop-count distance vector for the particular source. This provides an indication of the topological location of the source relative to the monitors. The sources can then be clustered using the gathered information from the hop-count vectors. When sources have similar hop-count vectors then they most probably are close to each other and in one client group [44] or autonomous system. Observed is also the fact that sources from common topological location might have different hop-count vectors. This difference is a constant offset caused from the fact that the routes to the passive monitors are shared but the paths to network egress points differ. Next, a lightweight method for discovering the network topology is developed by connecting sources and monitors by adding to the passively collected data a very small number of active measurements. "They provide ground-truth assessments of the number of shared hops between pairs of sources and passive monitors or pairs of monitors and a source" [37]. Combining the knowledge of shared hops and the clustering inferred from the passive measurements, is enough to reconstruct the logical network topology.

2.2.3.4 Hop-count Distance Measurements

For evaluating the hop-count distance measurements in [13], three different data sets are used. The first one is Orbis [45] which is a network topology generator. The graphs it creates have properties consistent with many of those observed in the Internet. Orbis enables the analysis of methods' capabilities with full ground truth and over a variety of sizes.

The second data set is Skitter [46]. In Skitter the measurements are based on Traceroute-like active probes and it generates a router-level connectivity map of the Internet. The maps are openly available and represent measurements sent from a set of 24 monitors to nearly 1 million target hosts over the Internet. The idea is to have one responding node in each /24 prefix.

The third data set used in [13] is a collection of data for a 24 hour period from 15 honeypot sensors with different topology. Over 37 million packets were collected and evaluated in the study. The segments of IP address space observed from the honeypots varied from /25 to /21 and one /16. For inferring the number of hops between the honeypot monitor and the host, the technique described in [19] is used.

In the overall analysis, it is assumed that the only data that will be used to infer network structure is the source IP address and TTL extracted from the heather of each packet.

2.2.3.5 Clustering

The goal in [13] is developing a method for clustering of IP sources that are topologically close to each other from a layer 3 hop-count point of view. The key to the algorithm is that the location of a source is defined by its relative distance to multiple monitors so sources with similar relative distances will be topologically close. For this purpose, packets from the sources must be observed in a satisfactory number of monitors.

Clusters of IP sources can be generated using unique hop count contrast vectors and the K-Means algorithm. A large number of small clusters with minimal difference between the contrast vectors might be considered a good choice with this approach but the problem comes from the fact that small clusters miss the case where sources might have different hop-count vectors even if they are located in the same area of the network. This happens when the subnet has multiple egress points or some of the monitors are located in the same subnet as the cluster. According to Theorem 2 in [13] if we examine a subnet with two egress points, all sources contained in the subnet will have collinear hop-count vectors. From the proof of this theorem we see that sources in subnets with multiple egress points may have small differences in the hop-count vectors which depends on the number of egress points and the nature of the paths to the egress points.

The distribution of hop-count vectors in general can be represented with a mixture of Gaussian models, in which each Gaussian component represents the distribution within one subnet, as the hop-count data contains sources from many different subnets. These models can be applied to data using Expectation-Maximization algorithm, and particularly the version proposed in [47] as it automatically finds out the right number of clusters using an information-theoretic criterion [13]. Each hop-count contrast vector will be related with a given Gaussian component, after the Gaussian mixture model is determined, which provides a clustering of the sources and the number of clusters is equal to the number of Gaussian components inferred by the EM algorithm [13, 47]. From the comparison of prediction from the Gaussian mixture model and the one based on randomly clustered set of sources in [13], we can see that the clusters determined by the Gaussian Mixture model result in an obviously better prediction and are actually grouping sources that share similar paths to the measurement nodes.

2.2.3.6 Topology Discovery

Putting together the passive hop-count data and a small number of active measurements is enough for identifying the topological relations between the clusters. The active measurements are traceroutes from the measurement nodes to a small subset of target hosts which essentially act as representatives of the clusters [13].

Although it is a straightforward approach, there are two potential problems. The first problem comes with the fact that some of the clusters may not be completely correct, as there might be multiple egress points or missing hop counts. Secondly, the number of clusters may be too large for complete traceroute probing (Internet-wide perspective). A new method for imputing the missing data is developed in [13]. For more information see section 6 in [13].

2.3 ACTIVE PROBING TOMOGRAPHY

2.3.1 Active Measurements

The term active measurement actually means the injection of artificial probe traffic into the network, and the measurement of its characteristics at different points. The main idea is that the initial structure of the probe traffic is known, and by measuring how it is affected by the section of the network it goes through, network conditions can be inferred. In the active end-to-end measurements the probes are sent between controlled OD pairs, and the measurements are made at those origins and destinations only [9, 48-51].

In Figure 2.3 are shown the components of an end-to-end measurement infrastructure where the origin of the artificial probe stream is the Sender and the destination is the Receiver.

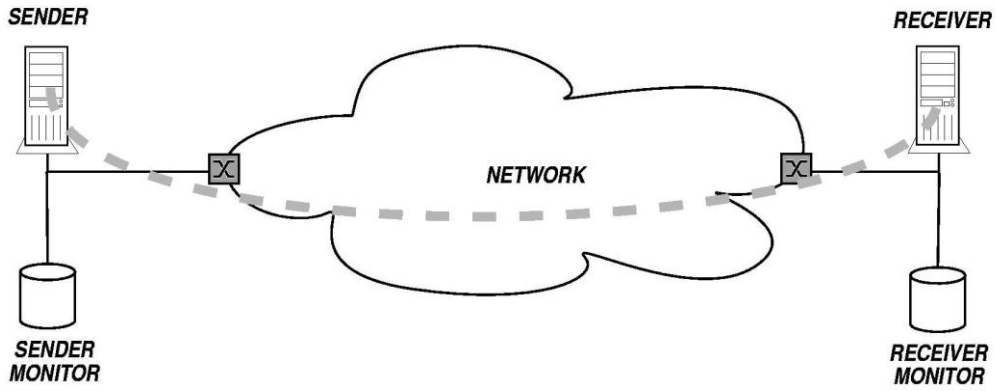


Figure 2.3: The basic components of an end-to-end active measurement infrastructure. [9]

An active measurement like this usually consists of two parts:

"(i) the sending process at the sender, which is necessarily invasive and involves distinct software and hardware challenges

(ii) the monitoring process both at the sender and the receiver, which can be viewed as simply a passive measurement, with a filter to select the expected probe stream." [9]

The breakdown above is equally correct for the round-trip and one-way forms of end-to-end streams. The biggest disadvantage of this approach is the invasive nature of sending probe traffic. On the other hand, it has a great flexibility as the OD pairs can be chosen and many different kinds of probe traffic can be selected, including packet streams which imitate different traffic types [10].

Another advantage of the active measurements is the flexibility to plan the probe streams in a way that they match different measurement requirements. The measurement focus can be shifted from determining delay and loss on a route, to estimating bottleneck [52], available bandwidth [53] and the rate of the background or cross traffic [54]. A bottleneck link is usually the link with the lowest physical transmission rate, while the links of a route that have the lowest available free transmission capacity are characterized by the available bandwidth. Other advantages are the possibility of performing, repeating and varying experiments as often as desired, the amount of measurement data which is greatly reduced compared to the passive monitoring of high bandwidth links and the avoidance of data privacy issues. Depending on the deterministic and/or statistical criteria the design parameters can be chosen with different sequence of packet types, sizes and inter-departure times [9].

The biggest problem with conducting active measurements, and at the same time the main disadvantage, is their invasive character. That's why usually in measurement project, probe streams of average bandwidth in the range of 10kbps are used (corresponding to the traffic produced by one user via a dial-up modem connection [55]). As the low rate streams generated are periodic or pseudo Poisson the sampling is at low time resolution [9].

2.3.2 Typical Active Measures

Network Latency - the time that a packet needs to travel from the source generating it to its destination. When the case is a round trip time (RTT) then the source and destination are identical but there is a distant node in the network that echoes the

packet back. "The RTT measured between a client and a server is one of the key factors determining the performance of interactive applications and the efficiency of feedback based control mechanisms" [9].

Latency has several components:

1. *Physical network delay or propagation delay* is basically the time a packet travels through the Internet. It is the sum of all the fixed delays in routers and switches along the path the packet traverses as well as the time it takes photons or electrons to traverse the physical links and is estimated usually by the minimum of the measured RTT-s.
2. *Queueing delay* - delay cause by the queueing in the network elements. This component is determined by the properties of the real traffic flowing through the network and changes over time, while the propagation delay on a given path can be accepted as a constant.
3. *Server processing delay or service* time - the time taken by the server to process the incoming data and send it via one of its outgoing links.

Various active measurement methods for network latency do exist and the most common and simple is the ping application available on a variety of operating systems. It sends ICMP echo messages and measures the response time. The ICMP echo based ping uses automatic process for generating responses. As the ICMP messages require special handling the probe packets are very often generated using different protocols such as UDP, TCP and UDP/RTP. When using other protocols there is a need of special applications on the measured servers and the installation and running can give rise to some complex security related problems [9].

One-way delay measurements are very useful for some applications that are intrinsically one-way, as well as for detecting bottlenecks. In contrast to the RTT measurements the one-way delay ones require highly synchronised network clocks [9].

Metrics for the measurement of RTT and one-way delay are standardised by IETF IPPM in RFCs 2679 and 2681.

The portion of packets lost traversing the path between a client and a server is called **one-way packet loss**. It is usually expressed as a percentage and the one-way loss is a more important metric than the round trip loss, unlike the case of latency. This metric is very important in term of applications as if the end-to-end loss exceeds a certain limit then the performance drops significantly. For different applications the limit may vary but it is very important for real-time applications as VoIP [9].

The packet loss measurements are performed mainly using active probing. This is as a result of the lack of guarantee of traffic flowing between the measurement points as well as the complexity of matching packets from the traces obtained at two different locations.

It is desirable to be able to measure the **TCP performance** as it is the main protocol found at the statistics collected on the composition of Internet traffic. Packet loss and network latency are some of the factors influencing the TCP performance but even knowing them, it is still hard to predict the performance of TCP connections. Attempts for analysing TCP performance [56-60] and establishing standard metrics for it have been made before [9].

As well as passive measurements, active measurements are also used for **Topology discovery**. A classic method is conducting traceroute-like measurements. This application like the ping one, uses ICMP and this means that it bears the same disadvantages. An example of topology discovery using network delay measurement can be found in [61].

Routing dynamics is another field employing active probing measurements. RTT measurements are usually used, but as a possible alternative passive monitoring of the Border Gateway Protocol (BGP) messages can be used as well [9, 46, 62].

2.3.3 Introduction to Active Probing

The main idea in active probing is sending controlled streams of packets or probes across the network and measuring of their patterns of arrival at the destination. As the characteristics of the send probes are controlled and known in advance, we can reveal fine details of network state and traffic condition form the intrusions caused by the network [9].

The basic variables in the active probing approach are the protocol type of the probes, the packet contents and the sending process. There is big variety of possible probe streams as the sending process can be chosen from many stochastic and deterministic models.

Selecting the protocol type is very important, as they can be treated differently by the network. For example generating ICMP packets is often rate limited and that's why for measuring variable metrics, different kinds of probe streams will be generated with their own properties. For measuring loss and delay many protocols can be used, yet the ICMP packets are much better suited for other more specific purposes [9].

UDP - UDP packets have been the primary choice for measuring one-way delay, delay variations and one-way loss because it is a connectionless protocol and there is no higher protocol controlling its release into the network. It is also not difficult to quickly write a UPD probe 'engine' [9]. The probe size can be kept minimal containing information only for the sequence number we can keep track of the identity of packets. A huge disadvantage is that the sending and receiving software must be written and installed at each end for this kind of active probing measurements as well as the related problems of access, installation delay, cost, security and accurate timing. UDP can be used for round-trip measurements as well but issues like turnaround time of packets may arise as user software must be installed.

DNS servers are the major users of UDP streams, but as a whole they are a small percentage of total Internet traffic. As real-time ('non-elastic') traffic becomes more

common, a remarkable growth in the proportion of UPD traffic may be noticed. In case the separation of elastic and non-elastic traffic by different mechanisms come into effect at the same time, the use of UDP probes may become specialized to the measurements of the real-time component of traffic [9].

"The possibility of using header fields to carry additional state information, or to collect information from the routers traversed, has not been explored. The risk is that any such scheme would depend upon the specific workings of routers, information about which may not be available from the vendor, may change over time, be unreliable, and different for different vendors" [9].

TCP - measuring various metrics like throughput and loss characteristics using TCP connections, one can make inference for the network condition, specifically for TCP connections [63]. Because of the various TCP implementations and the difference of their performance, there has been a standardisation effort (RFC 3148) toward a special TCP implementation designed for measurement purposes.

The same programs used for generating UDP streams can be used for generating TCP/IP controlled probe streams of traffic. SYN packets will be used for one-way measurements, and SYN/SYN-ACK pairs for round-trip measurements. The TCP packets will be treated by routers as if they are the beginning of a TCP connection and measures obtained from them will be for the conditions found by new connections starting up. Using SYN/SYN-ACK pairs has the benefits such as the availability of software in most of the end nodes which will allow round-trip measurements without the need for access, installation etc. They also may be realised at a fairly low level, which will reduce delays at the end system which dilute the network level delays one is trying to measure [9].

One of the disadvantages of this method is that servers may interpret a sequence of SYN packets as an attack, and after detecting them, start filtering them out in the future. A discussion for the likelihood and outcomes of such complaints is given in [63]. Another disadvantage is the heterogeneous nature of the systems, as they will generate

SYN-ACKs after a delay which is inconsistent and dependable on many uncontrolled factors, such as server load.

ICMP - ICMP packets are designed for information exchange between end host and core routers. An example for their active measurement uses are the ping and traceroute applications. The ping program gives RTT values and tests the reachability and availability of nodes, while the traceroute reconstructs path information using a sequence of packet with incremented TTL fields. Simple measurement efforts, both small and large scale, have been made using ping [64, 65]. A discussion of the problems associated with this method can be found in [66].

A huge disadvantage of ICMP based measurements in the public Internet is that most sites have limited or even blocked ICMP packets from their networks to prevent attacks, or to hide internal topology. The ICMP packets directed from routers are also frequently given lower priority than TCP and UDP packets. For overcoming these problems the IP Measurement Protocol (IPMP), a very simple protocol for low impact measurement, is under development [67]. Another tool using ICMP is pathchar [68]. It relies on routers generating ICMP messages in response to UDP probes with limited TTL and combines the control and probe based approaches.

2.3.4 Related Work to Date

2.3.4.1 The Statistical Inverse Problem

The Inverse Problem as mentioned above is to reconstruct the distribution of the link-level information from path-level measurements. One can study the performance of the internal links of a network by sending "probe" packets from a source to all nodes in the periphery of the network. Of course special equipment is used on the source and receiver nodes for sending packets and collecting information about losses and delays. After this, the inverse problem has to be solved to reconstruct link-level loss and delay data from the end-to-end measurements. Unicast or multicast transmission protocols

may be used for sending the packets. In the unicast scheme, packets are sent to one receiver at a time while in multicast protocols, packets are sent simultaneously to a specified subset of receivers. The information in multicast measurements is from higher-order and allows the reconstruction of internal link-level information. In case a network has turned off multicast transmission for security reasons, back-to-back unicast scheme is used, where packets are sent within nanoseconds of each other to two or more receivers. This scheme have been proposed in order to mimic multicast transmissions [4, 5, 69].

Although in practice the number of nodes can vary from a small number to several hundred, we can reduce the size of the network by combining links and nodes if we are interested in just a coarse investigation of the network. Looking at all the nodes is compulsory if a detailed study is needed [4].

The usual approach to probing experiments is based on multicast transmissions, where the probes are sent to all the receivers in the network of interest but the scheme seems to be inflexible as one would rarely want to send the same number of probes to all the nodes and at the same time. The goal in many cases is to investigate different regions of the network with different intensities and if possible different times [4]. A flexicast experiments scheme was proposed in [6] and [70], for active tomography. This consists of $C = \{C_h, N_h\}$, where C_h is a collection of sub-experiments with probe size N_h. As the individual sub-experiment covers only part of the network it cannot estimate all the parameters of the sub-network. However, by thoughtfully designing the sub-experiments, we can estimate all the parameters in the network of interest [4]. This is very useful in network monitoring, where different regions of the network are studied depending on where anomalies arise.

In [15] the problem was formulated for loss rates and multicast transmission scheme was introduced as well as an algorithm that computes estimators that are asymptotically equivalent to the MLES was derived. In [71] was discussed the problem for delays and a heuristic algorithm was proposed for calculating a nonparametric estimate of the link delay distributions. A pseudo-likelihood approach for the delay

problem was developed in [72] by considering all possible pairwise probes from the full-multicast experiments. An estimator for the back-to-back probing scheme that represents link delay using a point mass at zero and a finite mixture of Gaussian distributions was presented in [73].

2.3.4.2 Design of probing experiments

As the unicast scheme is not very useful for estimating link-level parameters, multicast scheme is used, which sends packets to all the receivers in the network simultaneously.

However, this scheme has a number of disadvantages. The first one is that it creates a huge amount of traffic which is more than necessary for estimating the link-level parameters and the data generated is very high-dimensional. The other very important drawback is that it doesn't allow investigation of sub-networks with different intensities and at different times, as in cases of probing sensitive parts of the network we may want to probe them as lightly as possible so disturbance will be avoided [69].

The inflexibility of the unicast and multicast schemes raises the need for new probing experiments. In [6] and [74] is introduced a class of flexible probing experiments called flexicast experiments. It consists of a combination of schemes with each scheme designed to study a sub-network. The data from the different schemes have to be combined in order to estimate the link-level parameters of sub-networks.

Conditions necessary and sufficient for identifiability of link-level loss rates are shown in [6]. They are as follows: (a) all receiver nodes are covered; (b) every internal node in the tree is a splitting node for some k-cast scheme in the flexicast experiment. The same conditions turns out to be enough for estimating discrete delay distributions as studied in [74].

Many flexicast experiments satisfy the identifiability requirements and the selection between these has to be based on other criteria. If the experiments are based

only on bicast and unicast schemes, they will have minimal data complexity (just 1- and 2-dimensional outcomes) but these provide information on just first and second-order dependencies which makes it less efficient (in statistical sense) to k-cast schemes with higher values of k. The full multicast scheme will be the most efficient in this case. The conclusion is that the overall choice of flexicast experiments has to be a compromise between statistical efficiency and flexibility and also including the ability to adapt over time to accommodate changes in networks conditions [69].

2.3.4.3 Inference for loss rates

In the literature, inference for loss rates has been studied under the following stochastic framework. Let $Z_r(m)$ = 1or 0 depending on whether the packet sent from the source node reached the receiver node r $\in$ R, or not, where R is the set of all receivers. Defining hypothetical random variables: X_i (m) = 1 or 0 depending on whether the packet m traverses successfully link $i \in \varepsilon$,or not, where ε is the set of all links. The collected data is analyzed under the models showed in [7], [15], [71], and others.

[15] was the first who studied inference for loss rates for the multicast scheme. In [6] was *developed* MLEs based on the EM algorithm for flexicast experiments.

As mentioned before the flexicast experiment C is made up of a collection of independent sub-experiments C_h. Each sub-experiment is a k-cast experiment and can be viewed as a k-dimensional multinomial experiment [4, 69]. Specifically, each outcome is of the form $\{Z_{r1},...,Z_{rk}\}$ where Z_{rj} = 1 or 0 depending on whether the probe reached receiver node r_j or not. Let $N_{(r1,...,rk)}$ represent the number *of* outcomes corresponding to this event, and let $\gamma_{(r1,...,rk)}$ be the probability of this event. The log-likelihood for the experiment C_h is then proportional to $\gamma_{(r1,...,rk)}\log(N_{(r1,...,rk)})$. This means that for C the overall log-likelihood will be the sum of the log-likelihoods for the individual experiments. However the $\gamma_{(r1,...,rk)}$ are complicated functions of α_k, the link level loss rate, so one has to use numerical methods to obtain MLEs.

This belongs to the class of missing data problems. The EM algorithm is a natural approach for computing the MLEs[1, 7, 75, 76]. The structure of the EM-algorithm for *general* flexicast experiments was introduced in [6]. The EM-algorithm works well when it is coupled with bicast and unicast experiments but only for small and moderate networks. It is computationally not practical for large networks.

In [77] is developed a class of fast estimation methods based on least-squares. This is done by transforming the loss-estimation problem to a linear problem. The general least-square *algorithm* offers a non-iterative and very fast estimation scheme. A more efficient estimation scheme is based on iteratively-reweighted LS. In [77] are studied the properties of these and other schemes. The conclusion is that IRWLS estimators are very close to the MLEs even in realistically small samples. Based on the LS schemes, one can also compute the asymptotic variance-covariance matrix of the estimators leading to clear construction of standard errors and hypothesis tests. This is another advantage of these LS schemes over the MLEs obtained using EM algorithm [4].

2.3.4.4 LAN Analysis Using End-to-End Delay Tomography

Over the last couple of years, computer networks have experienced exponential growth in terms of number of user, amount of traffic and the number and complexity of the applications. The lack of centralized control and the multi-layered structure are other important features. They make the assessment of performance of the network a very difficult task. It is impractical to monitor a large number of links directly due to the fact that it may affect network performance and the data generated will be in massive amounts. If the network of interest spans several subnets, then assistance from one provider will not be enough. This is the reason why active probing is used. Discrete statistical model for estimating link delay distributions based on active probing of a Local Area Network is presented in [78].

The data collected in [78] is based upon flexicast probing [79]. The flexicast scheme consists of experiments involving groups of receivers of different sizes. Each

experiment is from a series of probes simultaneously sent to each member of the group using multicast or back-to-back unicast probing. Each end-to-end delay is the sum of the delays occurring on each link. Two independent assumptions are made. First is that consecutive probes are independent of each other and the second one is that the delay a probe experiences along one link is independent of the delay it experiences along any other link.

In order to guarantee the estimation of the discrete delay model the following proposition for design of the probing experiment is made: If C is a collection of probing experiment and T is a general tree network topology then for every internal node $s \in T$, let there be at least one $c \in C$ such that c probes at least two receivers and s is a branching node for c. Also for each receiver r there is at least one $c \in C$ such that c covers r. The above conditions are necessary and sufficient to guarantee that C identifies the discrete delay distribution for every link in T.

The key concept is that the grouped probing leads to correlated end-to-end results. The correlations provided information about the shared paths and by using all internal link as branching points, a shared-path information from the source to every internal node is collected.

The computationally challenging aspect in this setting is to partition the observed end-to-end delays into the set of possible link delay combinations. This differs from previous estimators in two ways. First, the estimator is built especially for the flexicast probing of which multicast is a special case. Second, the maximum likelihood approach has advantages with regards to efficiency. The MLE presented in [78] results in more precise estimates for multicast data than previous procedures.

An alternative to the full EM is considered, presented in computing the local MLE on the subtrees and combining the results in a process called peeling. It is called that way because a path-level distribution is used and the distribution of a subpath to solve for, or peel away, the distribution for the other subpath. The combination of trees is called grafting. The EM algorithm to solve for the MLE of the logical links on the

subtree is used and then peeled to get estimates for the single links. This scheme scales very well for collection of bicast and unicast measurements as the EM algorithm is applied to a series of three-link, two-layer trees.

The capability to quickly estimate the probability of large delays is one of the advantages of the discrete delay model and its estimation schemes. This is very important information for capacity planning and performance estimation for applications like Voice-Over-IP telephony. By setting the bin size to be some large delay of interest, such as damaging threshold for VoIP quality, the framework can be easily used for estimating the probability of a deteriorating quality of service on each link. This process is equivalent in some ways to tail estimation [81]. Using this procedure improves speed in two ways. First, it reduces the experiential estimation of the max delay: larger bin sizes lead to fewer bins. Second, it takes advantage of the complexity calculation. The complexity of the EM algorithm in the worst-case scenario is based upon observing at least one value for every possible end-to-end delay and by using large bin sizes the observations are collapsed.

The importance of the identifiability condition is that it provides a set of conditions to construct a collection of probing experiments that provably identifies the individual link delay distributions. Such a procedure is described in [78], based on the set covering problems. The attention is restricted only to collection containing bicast and unicast schemes.

2.3.4.5 Routing Topology Identification Based on End-to-end Measurements

The routing topology of a network is related to the physical topology. Having a period of routing stability, the routing topology is describing the paths traversed by packets sent from one end host to another. The logical topology represents only the nodes that differentiate multiple paths. This means that a node with single ingress and egress link is absorbed into the link connecting its neighbors [80].

Without the cooperation of the internal nodes, the traceroute and SNMP-based methods will fail to identify the routing topology of the network. The logical topology identification methodology based only on end-to-end measurements information is the popular choice even though it may lose some information of the routing topology and is less robust [80]. The logical topology identification falls in the framework of metric-induced network topologies (MINT) [81-88]. An improved method based on the hop count of the path from source to the destination for multicast Internet is proposed in [86] for getting more detail information of the topology. This method can only infer hop count differences of the logic link in the same layer of a tree which makes the routing topology identification based on end-to-end measurements meaningful.

A new methodology for inferring the hop count of the shared path for each 1-by-2 component (one source sends probes to two destinations) based on digging measurement framework is presented in [80]. The initial TTL value of the large packet is used in sandwich to control the shared path of the large and small packets. The delay difference of the small packet will reach the maximum value when the initial TTL value is equal to or larger than the hop count.

The main process of the methodology presented in [80] includes three steps. First, the hop count of the path from the source to each destination is calculated. Second, the hop count of the shared path from the source to each destination is inferred. Third, the routing tree is created by the classification algorithm based on hop count information of the three logical links of every 1-by-2 components in the network. A sandwich probe is basically a large packet sent by the source, destined to one destination, between two small packets with another destination. The second small packet always queues behind the large one, until they separate from each other. For the measurement framework in [80] called digging measurement framework, for two destinations they are made a set of sandwich probing experiments. As the large packet is going to be discarded by the internal node when the TTL number is comes to zero, the shared path of the large packet and the small packets in the sandwich probes increases.

For each measurement is assumed that the results are independent and identically distributed. This is practical if the probes are adequately separated in time. Then the delay difference can be calculated by the difference of minimal delay of the small packets or the mean delay difference

2.3.4.6 Active Probing Strategies for Problem Diagnosis in Distributed Systems

A common practical problem in complex, multi-component systems is the accurate diagnosis using inferences based on the results of various tests and measurements. The goal is to achieve high accuracy with a small number of tests. In [89] the probe selection problem for the purpose of network management is studied and the problem of efficient diagnosis from probe outcomes using approximate inference in Bayesian networks is studied in [90], but the probe selection problem and particularly active probing were not considered.

For pre-planned probing in [91] are implemented two approximation algorithms - greedy search and subtractive search. The first one starts with the empty set and adds at each step the "best" of the remaining probes - the probe which maximizes the information gained about the system state. The second algorithm starts with the complete set of available probes and is considering each one in turn. After this it discards it if it is not needed.

For implementing the active probing a probing system is outlined. The results from the probes are analysed to infer what kind of problems may occur and if additional information is needed, the most useful set of probes is determined and sent. When the additional probes are received and analysed if needed the process is repeated until the fault is localised. With this method fewer probes will be used than if the entire probe set is pre-planned. For detailed algorithm see [91].

The results from the simulations made in [91] show that active probing considerably reduces the size of the probe set when compared with pre-planned, or "passive", probing.

CHAPTER 3

CAIDA VERSUS RIPE

In this chapter a detailed overview of the two largest organizations involved in this area of research is provided. The current projects of CAIDA and RIPE are described with more detailed focus on the ones that are of interest and related to this research. A conclusion is derived in the end of the chapter, as which project would be more suitable for the purpose of the experiment, based on the information provided in the projects overview.

3.1 CAIDA

3.1.1 Overview

The Cooperative Association for Internet Data Analysis (CAIDA) is an organization created in 1997 by Dr. KC Claffy and Tracie Monk which is based at the San Diego Supercomputing Center (SDSC), an extension of the University of California at San Diego (UCSD). It is a collaborative undertaking among organizations from the commercial, government, and research sectors. The association is used as a focal point for promoting greater cooperation in the engineering and maintenance of a robust,

scalable global Internet infrastructure. Most of CAIDA's support is currently provided by government agencies, the most notable of which being the National Science Foundation (NSF) and the Defense Advanced Research Project Agency (DARPA).

The primary focus of CAIDA is the development of measurement, analysis and visualization tools in order to benefit the community and advance the robustness and scalability of the Internet. The tools are intended to be made publicly available as well as the data acquired from public sources to be made available to researchers and others through the CAIDA web pages.

CAIDA helps organizations to jointly address areas of common interest on topics related to the infrastructure level measurements and analyses; scaling up the performance of the Internet through the evolution of the supporting technologies; promoting the understanding and characterization of traffic behaviour, using active and passive measurements and the analysis of routing data, affecting individual and global Internet infrastructures; help Internet Service Providers (ISPs) to manage their networks with improved traffic analysis and visualization tools.

3.1.2 Projects and Infrastructure

3.1.2.1 Macroscopic Topology Measurements

Internet macroscopic topology measurements have been conducted by CAIDA since 1998. Using tools like *skitter* and *scamper*, global IP level connectivity has been tracked by sending probe packets from a set of source monitors to millions of geographically-distributed destinations in the IPv4 address space, and since 2003, the IPv6 address space has also been continuously probed as well. The topology maps of the Internet which have been generated are very important for characterizing and analysing the Internet's infrastructure, properties, dynamic behaviour, and evolution. The data collected characterizes connectivity and performance of the Internet, providing

a valuable input for empirically-based modelling of the Internet behaviour and properties.

3.1.2.2 The PREDICT Project

In order to develop hardware and software solutions that are resistant to hacking attempts and malicious software, researchers and developers need current data on Internet security threats which includes samples of normal and malicious Internet traffic, malicious software samples as well as logs from machines compromised in the targeted attacks. There are various concerns over privacy, security and proprietary information that make the collection of the data required more difficult. That's why the Department of Homeland Security (DHS) has developed the Protected Repository for the Defence of Infrastructure against Cyber Threats (PREDICT) project. The project provides datasets with current operational data, to vetted researchers, in a secure and controlled manner. The primary goals of PREDICT are the development, implementation and maintaining of a Web-based portal that catalogues current network and operational data and handles data requests and at the same time enabling secure access to multiple sources of data collected on the Internet, while making it possible for PREDICT participants to share information for the purpose of developing new models, technologies and products that will increase cyber security capabilities.

3.1.2.3 The UCSD Network Telescope

The UCSD network telescope is also knows as black hole, an Internet sink, darkspace, or a darknet. It represents a globally routed /8 network which is approximately 1/256 of all IPv4 Internet addresses. The network itself carries almost no legitimate traffic as there are only a few provider-allocated IP addresses in this prefix. If the genuine traffic is discarded from the incoming packets, then the data remaining would represents anomalous unsolicited traffic, or Internet Background Radiation (IBR). Lately the traffic destined for darknet has evolved to incorporate longer-duration, low-intensity events intended to establish and maintain botnets. The telescope

instrumentation is maintained and expanded by CAIDA, and the data is collected, analysed, archived and access to it is provided to vetted security researchers.

3.1.2.4 Named Data Networking (NDN)

Named Data Networking (NDN) is a project exploring a new Internet architecture where instead of naming the locations it will focus on naming the data. This will shift the Internet from its focus today on "*where*" -- addresses and hosts -- to *what* -- the content that users and applications care about. As part of the project over 20 researchers study the technical challenges and problems that need to be addressed in order to validate NDN as a future Internet architecture: routing scalability, fast forwarding, trust models (content protection, privacy, security and defence), network security (efficiency of signatures for signing and verification), and fundamental communication theory.

3.1.2.5 IPv6 Evolution

The goal of this project is to evaluate the evolution of IPv6 in three dimensions: topology, traffic, and performance. Even though IPv4 address space is running out and the long term benefits of transition to IPv6 was explained to the wide public, the traffic volumes from IPv6 traffic are generally reported to be less than 0.1%. CAIDA is seeking characteristics of current IPv6 deployment in order to find a way how to advance it.

3.1.2.6 Archipelago Measurement Infrastructure

Archipelago is an active measurement infrastructure developed by CAIDA with its central server at CAIDA and several dozen monitors distributed in 39 countries. It is customized specifically for active network measurements with its main goals being the reduction of the effort needed to develop and deploy sophisticated large-scale measurements, and providing a step towards a community-oriented measurement

infrastructure. For measurements requiring highly precise time synchronization over the Internet, Ark monitors are using RADclock.



Figure 3.1: Ark Raspberry Pi-based Network monitor [92]

The Ark monitor itself is a small, inexpensive measurement node, based on the Raspberry Pi. It is as capable as a desktop system of several generations ago and also provides a flexible Linux-based programmable platform for conducting networking research. CAIDA is working on the development of a high-level API to make the writing of measurement tools easier. Using a process called team probing, monitors are put into groups and the measurement work is divided up between the group members. This is done in order to create a coordinated large-scale traceroute-based topology measurements. These measurements are completed using a powerful and flexible active measurement tool called *scamper,* which supports IPv4, IPv6, traceroute, and ping. The results are then distributed as the IPv4 Routed /24 Topology Dataset. IPv4 Routed /24 AS Links Dataset is also provided, which includes Autonomous System (AS) links obtained from the IP paths of the Topology Dataset.

While very powerful, the Ark project does not yet support personalized measurement. Individuals hosting an Ark monitor are not able to configure or perform measurements on their own. If this aspect is incorporated into the future development of the project, it would be a big step forward as it will catch the attention of wider audience, allowing people to conduct their own measurements with almost no effect on the network performance and to various destinations that might not be in the scope CAIDA is covering at the moment. Current measurement datasets and maps can be found on the CAIDA website.

3.2 RIPE NETWORK COORDINATION CENTRE

3.2.1 Overview

The Réseaux IP Européens Network Coordination Centre (RIPE NCC) was established in April 1992. It is an independent, not-for-profit membership organisation and acts as the Regional Internet Registry (RIR). RIPE NCC supports the infrastructure of the Internet and provides global Internet resources and services in its service region such as allocation of IPv4, IPv6 and Autonomous System Number (ASN) addresses. Members of RIPE are mainly Internet Service Providers (ISPs), telecommunication organisations and large corporations with most of them located in Europe, the Middle East and parts of Central Asia. It has thousands of members in over 75 countries distributed globally.

In a broader sense, RIPE NCC also provides services like the RIPE Database and administrative support for the RIPE community. The organization also manages one of the 13 root name servers (K-root), coordinates the support of ENUM delegations, deploys a routing database and manages a neutral measurement network that offers freely accessible and reliable statistics on the operation of the Internet.

3.2.2 RIPE NCC Activities

3.2.2.1 Regional Internet Registry

In parallel with the assignment and allocation of Internet number resources, one of RIPE's key responsibilities is the RIPE Registry. The organisation is maintaining its accuracy and quality as it contains information about the current holders of the resources. The registry is constantly evolving and dealing with changes resulting from the exhaustion of the unallocated IPv4 address pool. The Internet number recourses are distributed according to policies developed by the RIPE community. The RIPE NCC not only allocates and assigns IPv4 and IPv6 address space, but also assigns AS Numbers. The latter are registered with the primary associated routing policies which ensures the uniqueness of AS Numbers. This process facilitates the best possible aggregation of routing information and the efficient utilisation of the IP address space and AS Numbers. The RIPE NCC supervises the complete lifecycle of the resources allocated including registration, update support and deregistration of returned recourses. Internet number recourses are deregistered when they are to be found allocated or assigned under false pretences, breach of RIPE Policies or have been hijacked. The RIPE NCC also checks for unauthorised changes to the RIPE database objects. This way the accuracy of the RIPE Registry is maintained.

3.2.2.2 RIPE Database

The RIPE Database contains information about the IP addresses and ASNs distributed and their current holder's information, such as contact details and related attributes. The RIPE NCC is in the role of a data collector while the responsibility for updating and maintaining the information in the database is on the resource holders. The information provided in the RIPE Database is vital for network troubleshooting and determining abuse contacts. This service is available for anyone who wants information about networks and Internet number recourses and the users vary from network engineers, system administrators and researchers, to End User enthusiasts. The RIPE Routing Registry, part of the global Internet Routing Registry (IRR), is also included in

the RIPE Database. The IRR itself consists of several databases where network operators can publish their routing policies and routing announcements. The IRR also shares information between network operators in order to provide stability and consistency of global Internet routing. The RIPE Database software is being updated in order to better support operations and improve internal processes. This provides an efficient system with high-quality service, with close to 100% uptime. To quickly respond to outages and data loss, RIPE NCC is planning to improve the resiliency of the RIPE Database, reducing downtime and increasing the trust that users have in the service.

3.2.2.3 Resource Certification (RPKI)

The RIPE NCC members can request a digital certificate for the Internet number recourses they hold with the Resource Certification (RPKI) system provided. The certificate is a cryptographic proof for all the resources registered with the RIPE NCC, which are held by a particular organisation. The system brings benefits for the RIPE Registry by increasing its robustness and also makes Internet routing more secure.

3.2.2.4 LIR Portal

Members of the RIPE NCC can manage everything related to their membership and Internet number resources they hold, including creation and management of their resource certificates and it can all be done via the LIR Portal. It is now changing from requesting resources to managing them, as the last /8 addresses are being allocated. There's also an IP analyser incorporated in the LIR Portal, which is geared towards increasing data quality in the RIPE Registry and database. This aids LIRs to get the most out of the IPv4 address space which they currently hold.

3.2.3 Coordination Activities

3.2.3.1 DNS Operations

DNS coordination and support activities as well as DNS services are provided for the IPv4 and IPv6 address space managed by the RIPE NCC. The RIPE NCC provides secondary DNS to support the reliability of reverse lookups and also guarantees the stability and diversity of the DNS root name server system. The scalability of the DNS infrastructure is developed based on the requirements specific for each of the secondary, reverse and primary DNS services.

Since 1997, RIPE NCC is operating the K-root server through a set of globally distributed anycast clusters. It is a very important part of the Internet DNS infrastructure, and so far, five global nodes of the K-root name server have been deployed by RIPE NCC. Anycast instances of the K-root server with local reachability have also been deployed and currently there are 12 active nodes. By collecting and analysing the data from the performance of the K-root anycast nodes, RIPE NCC is determining if there's a need for additional nodes to be deployed.

3.2.3.2 RIPE Labs

Using RIPE Labs, network operators, developers and industry experts test and discuss new and innovative Internet-related tools, ideas, analysis and statistics. This way the community is informed about new ideas and prototypes the RIPE NCC is working on. It also provides the operator community with data analysis regarding the state of the Internet and with the help of those new tools, aid the community in understandings the various aspects of routing, DNS, reachability and other topics.

3.2.3.3 RIPE Meetings

There are two five-day RIPE Meetings each year at various locations throughout the region which RIPE NCC services. The meeting brings ISPs, network operators and other interested parties from around the world while the RIPE NCC provides the administrative, logistical and technical support. Even outside the scope of the meeting the RIPE NCC is committed to maintaining close relationship with its industry partners, including the four other Regional Internet Registries (RIRs), ICANN, the IETF and the Internet Society (ISOC). These relationships are very important for the coordinated administration of resources that each organisation is responsible for.

3.2.4 Services

3.2.4.1 Training

As part of its services the RIPE NCC provides various types of training within its service region in order to support the correct registration and administration of Internet number resources as well as other areas like IPv6 and routing security. The training courses available to the RIPE NCC members and the RIPE community are on Local Internet Registry, IPv6, Routing Security, RIPE Database and Hands-on Technical Workshops. The Local Internet Registry training course helps members to understand the RIPE NCC processes and how to request Internet number resources, and at the same time, making sure that members are aware of the correct procedures for registering and updating registry data and working with the RIPE Database. Members thinking about deploying IPv6 are advised to attend the one-day long IPv6 training course. They will be taught how to create a deployment plan, the implications of IPv4 exhaustion, and the differences between IPv4 and IPv6 addresses.

Another one-day course is the Routing Security training. It covers topics such as usage of the Internet Routing Registry (IRR), Routing Policy Specification Language (RPSL), and Resource Certification (RPKI). Members are taught how to make Internet

routing more secure and reliable with the help of the Routing Registry and the RPKI system, how to set up the RPKI system in the LIR Portal, and use resource certificates to create Route Origin Authorisations (ROAs). The RIPE Database training course provides hands-on experience in parallel with a practical introduction on the usage of the RIPE Database. It educates the members on the RIPE Database's scope and importance and how it operates. With the benefits brought by the Hands-on Technical workshops, the awareness of members for upcoming technologies, best practices, protocols and RIPE Policies is increased as well as their knowledge of the tools and services provided by the RIPE NCC.

3.2.4.2 RIPEstat

RIPEstat is a web-based interface that provides current and historical information about Internet number resources. Information such as IP address space, Autonomous System Numbers, and related information for hostnames and countries can be found in one place. Data sets are collected from the RIPE NCC's internal data sets as well as from relevant external sources. The data gathered includes registration and routing data, DNS data, geographical information, abuse contacts and RIPE NCC's measurements. All that information is presented in the form of a web-based widget, which can be embedded on any webpage. An API for accessing the raw data in advanced applications is also provided. The goal of RIPEstat is to provide a single, consolidated interface for accessing the RIPE NCC's publicly available data. The RIPE NCC is currently in the process of merging all of their public data sets into RIPEstat. This will make retrieving data much easier for the users and will eventually become the sole interface for accessing any of the RIPE NCC's publicly available data.

3.2.4.3 The RIPE Atlas Project

Introduced by the RIPE NCC, RIPE Atlas is the next generation Internet active measurement network. Built from thousands of probes distributed around the globe, it measures connectivity and reachability, collects unique data and provides valuable live and historical information about the reliability of the network. All measurements are

pooled and provide an unprecedented understanding of the state of the Internet in real time. The development of the project started in late 2010, and is built on the previous RIPE NCC's measurement network - the Test Traffic Measurement Service. Since the start of the Atlas project, the goal of RIPE NCC has been to create the world's largest Internet measurement network.

At the moment there are several thousand active probes in the RIPE Atlas network. Most of them are in the RIPE NCC service region of Europe, the Middle East and parts of Central Asia. The network is constantly growing and the data collected is providing useful maps and graphs that help building a big-picture view of the Internet. The name "RIPE Atlas" itself, comes from the word *atlas* - a book of maps as it is believed that it will produce different kinds of maps about the Internet, such as latency maps, reachability maps, etc., from the measurement data collected.



Figure 3.2: RIPE Atlas probe [93]

The probe itself (Figure 3.2) is a small hardware device, running software developed by the RIPE NCC and powered by a USB. Any USB port capable of supplying 500mA can be used. To connect to the Internet the probe has an RH-45 Ethernet interface. Once powered and connected to the Internet the probe acquires an IP address and DNS resolver information using DHCP. A static network configuration can also be assigned if the host doesn't want to use the dynamic one. A host of a probe is considered anyone who applies for, and receives a RIPE Atlas probe, connects it to their network and leaves it running. In order to allow the probe to connect to the network, the minimum array of services that need to be enabled are DHCP, DNS and outgoing TCP

port 443 (HTTPS). Allowing ICMP, UDP and HTTP(S) would enable the probe to do measurements. After that is done, the probe starts performing its built-in measurements which include round trip time (RTT) measurements to the first and second hop, ping and traceroute measurements, and SSL queries to some predetermined destinations, and DNS queries to root DNS servers. The bandwidth consumed by the probe depends on the number of measurements the probe is running at any given time but if worried, the host can specify the maximum bandwidth the probe can use.

Everyone hosting a probe is able to conduct their own user-defined parameters. The host has access to all the probes on the network and can run measurements from them to destinations of his choice. Once the probe is connected, the host starts earning credits for the time that the probe is connected. It should be noted that a probe is considered "connected" only when it was a working Internet connection and is connected to the RIPE Atlas infrastructure. If the probe has an Internet connection but cannot connect to the central infrastructure then it is shown as "disconnected". A host earns credits for each of their active probes, if more than one. The collected credits can be used for user-defined measurement either from the hosted probe itself, or from any other probe in the RIPE Atlas network. When a user logs into his account on the RIPE Atlas website, they can check their credit balance or start a user-defined measurement. Of course there are limitations as to the number of measurements, probes and credits which can be used. The maximum permissible amount of simultaneous measurements is 100, with no more than 500 probes used per measurement and no more than 270 000 credits used per day. There are enhanced RIPE Atlas probes with greater measurement capacity than the regular probes and they are called Anchors. These not only perform more measurements, but also act as stable, cooperating regional targets for measurements within the RIPE Atlas network. They allow users to observe the traffic at both the source and the destination, providing valuable information about the local and regional connectivity and reachability of the Internet. The data collected is made publicly available. Organisations hosting RIPE Atlas Anchors receive additional benefits as a return of their contribution.

The RIPE Atlas provides users with an active measurement tool for baseline and on-demand personalised active measurements from thousands of vantage points across the Internet and records the responses. The technical community can use the data and the Internet "traffic maps" produced for different purposes. This neutral, transparent data collection infrastructure provides data sets that can help analyse the operation and growth of the Internet, using active data collection methods it offers timely hard data such as country-wide Internet outages or measurement data on the state of the Internet. The data collected is of a great value to the network operators, researchers, the technical community, and anyone interested in the healthy functioning of the Internet. It can also help people understand and learn more about the underlying networking infrastructure and the data flows which keep the Internet running globally. For this reason, the development of the RIPE Atlas project has been guided by the users. RIPE NCC continues to expand the command, control and collection infrastructure to accommodate the growing number of probes and create new and enhanced Internet traffic maps, as well as integrating the RIPE Atlas data into other tools, such as RIPEstat.

3.3 CONCLUSION

For the purpose of this research there were two primary considerations as to which data sets to employ. After comparing the CAIDA Archipelago and the RIPE Atlas projects, it was decided that the most suitable data set for this research would be the RIPE Atlas. Currently, from the Ark project, the only measurements collected and therefore publically available would not be of any use for the purpose of this research as they are only designed to accommodate the purposes of CAIDA's research and the project itself doesn't have any personalised measurements. With a primary benefit of user-defined measurements from the RIPE Atlas project it was chosen as the best option to conduct the measurements and fulfil the purposes of this research.

CHAPTER 4

RESEARCH METHODOLOGY AND DESCRIPTION OF THE EXPERIMENTS

This chapter gives a detailed description of the experiment and the research methodology used. It starts with explanation of the process and procedures for configuration of the probe and then continues with a description of the probe selection process providing the full list of the chosen probes and their locations. A detailed description of the experiment is provided in the end of the chapter along with an outline of the limitations encountered throughout the process.

4.1 RECEIVING AND CONFIGURING THE PROBE

The process of becoming a RIPE Atlas member is made as simple as possible for those willing to join. After filling the online application for the probe and following the approval process, the probe itself was delivered to the Telecommunications Research Centre (TRC) at the University of Limerick (UL). It was of common interest to place and connect a probe in the Limerick area, as this was, and currently is the only RIPE Atlas probe operating in Limerick. The packet in which the probe was delivered consisted of the probe itself and the supporting documentation concerning the connecting and configuration of the probe. The probe itself is a small black hardware

device with a short USB cable attached. The software the probe is running is developed by the RIPE NCC. In order to power the device, it is connected to a Desktop PC that is consistently operational, as disconnection of the probe is not advisable as it might affect the measurements and their completeness. A Cat 5E Ethernet cable was used to connect the probe to the Internet. As the Atlas probe is operating within the University's network domain, the cooperation of the IT Division at the University was required, in order to provide a static network configuration with which to configure the probe. This was done on the RIPE NCC website, and the probe was registered and configured. Afterwards a set of services needed to be enabled from the IT Division in order to allow the probe to send and receive traffic and perform the measurements it was intended for, so it won't be stopped from the firewall.

4.2 THE PROBE SELECTION PROCESS

For the purpose of the research a network of RIPE Atlas probes was constructed to capture traffic measurements. The probes selected were in 26 countries within the European Union and 11 countries outside the EU. During the selection process, a number of factors were taken under consideration, such as the geographical location of the probes selected and their up-time. In each one of the 26 countries within the EU, four probes were selected based on their availability and their geographical position at the furthest east, west, north and south point of the country, respectively. In case there was no probe corresponding to some of the directions, measurements were conducted only in the directions where there was a probe available. For the countries outside the EU, the measurements performed were from the probe located in the UL's TRC to the capital and/or financial centre of those countries.

The location and ID of the suitable probes identified within the EU, are listed below:

Table 4.1: Locations and probe IDs of the probes within the EU

Countries	North	South	East	West
Austria	Linz (578)	Klagenfurt (265)	*Vienna (2050, 2178)	Götzis (942)
Belgium	Turnhout (10227)	Athus (45)	Liege (3988)	Nieuwpoor (4525)
Bulgaria	Pleven (2103)	Plovdiv (3074)	Varna (1230)	Sofia (4528)
Cyprus	Nicosia (746)	Limassol (3216)		
Czech Republic	Liberec (218)	Brno (2533)	Karvina (2954)	Plzen (3027)
Denmark	Aalborg (796)	Odense (841)	Lystrup (690)	Lemvig (4752)
Estonia	Tallinn (751)			
Finland	Rovaniemi (2320)	Helsinki (911)	Lappeenranta (546)	Vaasa (505)
France	Lille (573)	Marseille (736)	Strasbourg (2279)	Brest (2566)
Germany	Kiel (789)	Kempten (266)	Dresden (2078)	Aachen (4407)
Greece	Thessaloniki (3126)	Athens (3115)		
Hungary	Budapest (2944)	Pecs (3747)	Debrecen (4242)	Nagykanizsa (3336)
Ireland			Dublin (6007)	Limerick (2327)
Italy	Monza (2751)	Acireale (611)	San Benedetto (3189)	Grosseto (2339)
Latvia	Riga (434)	Daugavpils (3751)		
Lithuania	Panevezys (2532)	Vilnius (2747)	Zarasai (3509)	Kaunas (633)
Luxemburg	Ettelbruck (3456)	Bettembourg (3047)		Capellen (3345)
Netherlands	Leeuwarden (36)	Roermond (3647)	Enschede (1347)	The Hague (2463)
Poland	Gdansk (887)	Krakow (4112)	Bialystok (3016)	Zagan (4746)
Portugal	Porto (2889)	Lisbon (724)	Portalegre (3169)	
Romania	Suceava (2905)	Bucharest (1493)	Pascani (1211)	Timisoara (698)

Slovakia	Zilina (2638)			Bratislava (6008)
Slovenia	Ljubljana (2718)			Logatec (1323)
Spain	Bilbao (4068)	Gibraltar (575)	Girona (47)	Seville (679)
Sweden	Kalix (665)	Malmo (4934)	Norrtalje (759)	Gothenburg (373)
United Kingdom	Dalgety Bay (1425)	Southampton (3378)	Norwich (1422)	Strathfoyle (2481)

The target destination cities for the countries outside the EU, are listed below:

Table 4.2: Target destination cities for the countries outside the EU

Countries	Capital	Financial Center
Australia	Canberra	Melbourne
Brazil	Brasilia	Sao Paulo
Canada	Ottawa	Toronto
China	Beijing	Shanghai
India	New Delhi	Mumbai
Japan	Tokyo	Kanagawa
Russia	Moscow	Saint Petersburg
Saudi Arabia	Riyadh	Jeddah
South Africa	Cape Town	Johannesburg
UAE	Abu Dhabi	Dubai
USA	Washington, D.C	New York City (east); Los Angeles [30]

In case of disconnection or deactivation of some of the probes, a new one was required to be selected in order to continue the measurement. They are marked with a "*" sign in the lists above. The location of the new probe was aimed to be as close as possible to the location of the previous one. The destination IP addresses were selected from the www.shodanhq.com database. In the best case scenario the measurements could've been performed between the same two points (two probes) without the need of selecting different destination IP addresses, but as of now this is not possible because the IP addresses of the probes are not publicly available. The location of the target IP

addresses was aimed to be the same as the one of the probe conducting the measurement in the opposite direction (e.g. a measurement conducted in Bulgaria from east - Varna to west - Sofia would have the probe with ID 1230 for its source and the destination IP address is selected in Sofia, as there is the location of the probe conducting the measurement in the opposite direction). In case the IP address was not responding or blocking the measurements after a period of time, a new IP address was selected in the same area.

4.3 THE EXPERIMENT

The idea behind the project is to build an Internet Quality of Service (QoS) performance profiles for some predetermined destinations by extracting raw QoS data. This data is then used to build the profiles of the selected countries within the EU as well as establish Ireland's up-to-date Internet QoS connectivity performance state of connections with some economical key destinations around the globe. The final results can serve as a benchmark for future Internet performance improvements. In order to get QoS data, *ping* and *traceroute* measurements were conducted for the duration of one month in each of the selected countries in the EU and destinations outside the EU. It was only possible to perform the measurements for that long because of the credit system part of the RIPE Atlas project. It is for this reason that the experiment could not commence sooner. In order to collect the needed credits to perform the desired measurements the probe located in the TRC had to be connected for an extended period of time, as it could only earn a certain amount of credits per day.

The first group of measurements conducted was *traceroute* measurements for the duration of July 2013. This set of measurements was only for the countries within the EU due to the limitations of how many measurements can be performed simultaneously. The final results are presented in bar graphs, taking into account the round-trip-time (RTT) from the last hop. If the last hop was unreachable then the results from the last responsive hop were taken. The only downside of the traceroute measurements is that it can't be checked if the last hop is reachable or not while the measurement is running. The only thing that can be done is to check regularly if the probe conducting the

measurement is active or not. In case it is not, a new probe was selected as soon as possible in order to continue the measurement.

The second group of measurement was performed for the duration of August 2013. This group consisted of *ping* measurements conducted in the countries within the EU with the same source and destination addresses as the *traceroute* ones from the month before that. Again due to the limitations of how many measurements can be performed at the same time, the last measurements to the destinations outside the EU had to be left for the next month. One of the most useful features of the *ping* measurements were that it was immediately noticeable if a destination was reachable or not, as it is shown on the measurement personal page. They had to be checked regularly if all the probes are active and destinations reachable. In case any of the probes was not active or the IP address was unreachable then actions had to be taken quickly, and new ones had to be selected in order for the measurement to be as complete as possible.

During the month of September 2013, both *ping* and *traceroute* measurements were conducted to the destinations outside the EU. These measurements all shared the same source which was the probe located in UL's TRC. The same rules applied here in case of destinations being unreachable. Whenever this was the case, a new IP address was selected in the same area in order to complete the measurement. As the destination IP addresses for both types of measurements were the same, it was possible to observe the *ping* measurements and see if the destination was reachable or not. In case it was not, both *ping* and *traceroute* measurements were restarted with the new destination.

After all the measurements described above were finished, the raw QoS data was extracted from the RIPE website in the form of JSON and txt files. As there are two different types of files from the *ping* and *traceroute* measurements, containing different information, two java programmes (Appendix 2) were designed specifically for the purpose of extracting the correct data as required. For both types of these measurements, in order to help visualise behaviour and understand the patterns and dependencies, bar graphs were generated for the harnessed data. Those bar graphs are presented in two ways and represent two different types of information extracted. Using

a confidence interval, the highest 5% of the RTTs are not taken into account. The first bar graph shows the minimum, average and maximum RTT on a day-to-day basis for each day the measurement was run. The minimum and maximum values are respectively the smallest and biggest RTTs for the specific day, and the average value is calculated when all the RTTs for the day are added together and divided by their number. The second bar graph shows those values but calculated for specified time intervals. Instead of on a day-to-day basis this time the whole duration of the measurement is taken into account at once. The time intervals are as follows: 00am-6am, 6am-12pm, ,12pm-6pm, and 6pm-00am.

CHAPTER 5

RESULTS AND ANALYSIS

This chapter describes the results and provides analysis of the same. The Congestion Level Index is explained and the countries are divided into defined groups based on the results from the calculations of the index. This was done in order to better differentiate them for the purpose of the analysis. The chapter continues with a description of the bar graphs generated and examples are provided along with the excel tables with the extracted information from the raw data files. The analysis section of the chapter explores the worst case scenarios based on the Congestion Level Index results and provides a discussion of the main obstacles for building an Internet profile. It also provides examples in the form of bar graphs to help better visualise the scenarios explored.

5.1 INTRODUCTION

The *ping* and *traceroute* measurement results were analysed in order to reveal Internet performance variations. The focus of this work was on the variations of the Congestion Level Index (CLI), day-to-day and specific time interval RTTs. All raw data JSON and txt files from which the data was extracted can be seen in Appendix 3, 4, 5 and 6, along with all the information extracted from them.

5.2 CLI

In order to calculate the CLI between the farthest measurement points within each country, the geographical distance between them was needed. The delay is dependent not only on the networking components (e.g. routers, switches, cabling) but also is a function of distance [3]. The distance delay is given by the following formula:

$$Distance\ delay = distance/0.6c$$

where c is the speed of light in vacuum ($3*10^8$m/sec), *0.6* is a propagation factor used for the speed of light in single-mode optical fibre over that for free space or a vacuum, the *delay* is in seconds and the *distance* in meters [3]. After establishing the delay, based mainly on the distance between the end-points, the ratio of *observed one-way delay* to *distance delay* was derived. This ratio represents the CLI.

Two different CLIs were calculated as a result from the *ping* and *traceroute* measurement delays. They were combined by adding the two final values and dividing them by two in order to build a better and more accurate picture for the performance profile. Two different tables were constructed for the measurements in the countries within the EU and another one for the measurements from Ireland to the destinations outside the EU.

By incorporating the CLI into the performance evaluation, a better insight of each country's Internet performance is received. All results for the countries within the EU can be seen in Table 5.1 and the results for the countries outside the EU can be seen in Table 5.2. By comparing the CLIs calculated and taking into account how many measurements were conducted for the specific country, the countries were separated into 6 different groups according to various factors. The comparison of the CLIs was based on their symmetry. If one of the CLIs, for a group of measurements on a specific path and opposite directions (e.g. west/east and east/west), is nearly twice or more than the CLI in the opposite directions, then they can be considered asymmetric. In case they were the same or nearly the same, then they are considered symmetric.

The groups are as follows:

1. There are measurements in the four directions (north/south, south/north, east/west, west/east) and they are all asymmetric.

2. There are measurements for the four directions and they are all symmetric.

3. There are measurements in the four directions where one group of them (e.g. east/west, west/east) is symmetric and the other one is asymmetric.

4. There are only measurements that are asymmetric for one group and no or not enough data for the other one (e.g. asymmetric results for north/south, south/north, and no data for the other two measurements). (Here are included countries with less than four measurements)

5. There are only measurements that are symmetric for one group and no or not enough data for the other one.

6. There are measurements in only one direction and no data for the other one so they can't be compared to anything.

Another two groups were made for the results from Limerick to countries outside the EU. As there are fewer measurements here and they are only one way measurements, and as such, symmetry is not a factor. Here, the only thing that separates them is the difference between CLIs. The groups are as follows:

1. There is a big difference in the CLIs between the two destination points in the same country.

2. There is no or almost no difference between the CLIs of the two destination points in the selected country.

In the tables below it can be noticed that some of the CLIs are marked with yellow colour. This indicated that they are asymmetric. Red colour was used for some of the CLIs. It represents that there was a big difference between the two CLIs combined that were as a result from the *ping* and the *traceroute* measurements.

Table 5.1: Congestion Level Index results for countries within the EU

Country	Direction	Distance [m]	Distance Delay [ms]	CLI
Austria	North-South	186684	1.0	**11.8**
	South-North	186684	1.0	**12.4**
	West-East	512336	2.8	**7.4**
	East-West	512336	2.8	**5.4**
Belgium	North-South	205702	1.1	**11.6**
	South-North	205702	1.1	**15.6**
	West-East	205651	1.1	**13.4**
	East-West	205651	1.1	**9.1**
Bulgaria	North-South	141866	0.8	**17.2**
	South-North	141866	0.8	**17.9**
	West-East	377366	2.1	**6.1**
	East-West	377366	2.1	**4.1**
Cyprus	North-South	61421	0.3	**38.1**
	South-North	61421	0.3	**71.3**
Czech Republic	North-South	207771	1.2	**6.1**
	South-North	207771	1.2	**6.6**
	West-East	371097	2.1	**5.7**
	East-West	371097	2.1	**2.8**
Denmark	North-South	186078	1.0	**16.1**
	South-North	186078	1.0	**3.9**
	West-East	123509	0.7	**74.8**
	East-West	123509	0.7	**30.6**
Estonia	North-South	163805	0.9	**3.6**
Finland	North-South	705496	3.9	**2.6**
	South-North	705496	3.9	**3.1**
	West-East	410474	2.3	**2.6**
	East-West	410474	2.3	**54.6**
France	North-South	835638	4.6	**10.9**
	South-North	835638	4.6	**6.7**
	West-East	901689	5.0	**4.7**

	East-West	901689	5.0	**2.2**
Germany	North-South	734671	4.1	**5.4**
	South-North	734671	4.1	**2.9**
	West-East	537549	3.0	**8.1**
	East-West	537549	3.0	**6.1**
Greece	North-South	303499	1.7	**4.7**
	South-North	303499	1.7	**3.9**
Hungary	North-South	170370	0.9	**3.6**
	South-North	170370	0.9	**11.2**
	West-East	372294	2.1	**10.2**
	East-West	372294	2.1	**9.8**
Ireland	West-East	175236	1.0	**18.1**
	East-West	175236	1.0	**9.7**
Italy	North-South	992401	5.5	**6.4**
	South-North	992401	5.5	**3.5**
	West-East	226607	1.3	**17.9**
	East-West	226607	1.3	**36.8**
Latvia	North-South	191378	1.1	**2.7**
	South-North	191378	1.1	**5.8**
Lithuania	North-South	130160	0.7	**11.6**
	South-North	130160	0.7	**8.4**
	West-East	175560	1.0	**7.6**
	East-West	175560	1.0	**7.3**
Luxembourg	North-South	36443	0.2	**141.9**
	South-North	36443	0.2	**65.5**
	West-East	20041	0.1	**94.0**
Netherlands	North-South	223970	1.2	**13.7**
	South-North	223970	1.2	**18.0**
	West-East	176762	1.0	**8.6**
	East-West	176762	1.0	**19.5**
Poland	North-South	505082	2.8	**2.6**
	South-North	505082	2.8	**4.4**
	West-East	558764	3.1	**8.2**
	East-West	558764	3.1	**4.8**
Portugal	North-South	275602	1.5	**5.0**
	South-North	275602	1.5	**6.6**
	East-West	168000	0.9	**16.6**
Romania	North-South	358496	2.0	**6.5**
	South-North	358496	2.0	**2.9**

	West-East	452604	2.5	**6.4**
	East-West	452604	2.5	**4.5**
Slovakia	North-South	120095	0.7	**23.5**
	West-East	361273	2.0	**4.5**
Slovenia	North-South	53804	0.3	**36.2**
	West-East	132137	0.7	**11.1**
Spain	North-South	738800	4.1	**7.9**
	South-North	818499	4.5	**9.5**
	West-East	911965	5.1	**9.6**
	East-West	911965	5.1	**6.3**
Sweden	North-South	1264158	7.0	**2.5**
	South-North	1264158	7.0	**3.7**
	West-East	450670	2.5	**4.5**
	East-West	450670	2.5	**2.4**
United Kingdom	North-South	586556	3.3	**6.4**
	South-North	586556	3.3	**5.6**
	West-East	622507	3.5	**7.7**
	East-West	622507	3.5	**6.2**

The countries from the table above were separated in the groups already described and they are as follows:

1. Denmark;

2. Austria; Belgium; Bulgaria; Lithuania; Spain; United Kingdom;

3. Czech Republic; Finland; France; Germany; Hungary; Italy; Netherlands; Poland; Romania; Sweden;

4. Cyprus; Ireland; Latvia; Luxembourg;

5. Greece; Portugal;

6. Estonia; Slovakia; Slovenia;

Table 5.2: Congestion Level Index for measurements from Ireland to countries outside the EU

Country	Direction	Distance [m]	Distance Delay [ms]	CLI
Australia	Limerick to Canberra	17428363	96.8	**1.7**
	Limerick to Melbourne	17414103	96.7	**1.9**
Brazil	Limerick to Brasilia	8503113	47.2	**2.9**
	Limerick to Sao Paulo	9249071	51.4	**2.5**
Canada	Limerick to Ottawa	4778646	26.5	**1.8**
	Limerick to Toronto	5131667	28.5	**2.1**
China	Limerick to Beijing	8450364	46.9	**3.7**
	Limerick to Shanghai	9520391	52.9	**3.5**
India	Limerick to Mumbai	7778862	43.2	**3.1**
	Limerick to New Delhi	7257415	40.3	**5.9**
Japan	Limerick to Kanagawa	9757856	54.2	**2.5**
	Limerick to Tokyo	9739778	54.1	**2.6**
Russia	Limerick to Moscow	2973437	16.5	**2.3**
	Limerick to Saint Petersburg	2495380	13.9	**2.3**
Saudi Arabia	Limerick to Jeddah	5336047	29.6	**3.6**
	Limerick to Riyadh	5543623	30.8	**2.7**
South Africa	Limerick to Cape Town	9989869	55.5	**1.9**
	Limerick to Johannesburg	9473675	52.6	**2.9**
UAE	Limerick to Abu Dhabi	6072097	33.7	**2.4**
	Limerick to Dubai	6069165	33.7	**2.3**
USA	Limerick to Los Angeles	8231838	45.7	**1.8**
	Limerick to New York City	4981974	27.7	**1.6**
	Limerick to Washington DC	5310792	29.5	**2.0**

The above countries were separated into the two described groups and they are:

1. India;

2. Australia; Brazil; Canada; China; Japan; Russia; Saudi Arabia; South Africa; UAE; USA;

As mentioned previously, the CLI represents the ratio of *observed one-way delay* to *distance delay*. The *observed one-way delay* was derived from the raw data collected

and calculated with the help of the Java code. It is the average RTT, for the whole period of the measurement, divided by two. For the ping measurements, it was put at the bottom of the excel sheet showing the day-to-day minimum, average and maximum times (can be found in Appendix 3 and 4), and for the traceroute measurements it was put in a separate file showing the average RTTs for all hops (can be found in Appendix 5 and 6).In Table 5.3 an example of how the CLI for each measurement is calculated is shown.

Table 5.3: Example of CLI calculation for United Kingdom

Country	Direction	Distance [m]	Distance Delay [ms]	PING			TRACEROUTE			CLI
				RTT [ms]	One-Way Delay [ms]	CLI	RTT [ms]	One-Way Delay [ms]	CLI	
United Kingdom	North-South	586556	3.3	42.5	21.2	6.5	41.5	20.7	6.4	**6.4**
	South-North	586556	3.3	34.5	17.3	5.3	38.7	19.3	5.9	**5.6**
	West-East	622507	3.5	51.9	26.0	7.5	54.6	27.3	7.9	**7.7**
	East-West	622507	3.5	48.3	24.1	7.0	38.2	19.1	5.5	**6.2**

The full table with all calculations of CLIs for the countries within the EU and for the countries outside the EU can be seen in Appendix 1.

5.3 DAY-TO-DAY AND TIME INTERVAL ROUND-TRIP-TIME

Bar graphs were generated for both *traceroute* (Appendix 5 and 6) and *ping* (Appendix 3 and 4) measurements, which graphs represent the RTTs from two different points of view. On one side, one can see the minimum, average and maximum RTT for each day of the duration of the measurement, and on the other the RTT is summarised in predetermined time intervals. This gives more detailed information on the performance of the network throughout the whole month on a day-to-day basis. All graphs were directly generated with purpose-built Java code. (Appendix 2). This code harnesses the data required and stores it into a separate excel sheet. Then with the values from that excel sheet a bar graph is being created. Examples of the different types of bar graphs can be seen from Figure 5.1 to Figure 5.16. They represent the graphs generated from the *ping* (Figure 5.1 to Figure 5.8) and *traceroute* (Figure 5.9 to Figure 5.16) measurements for Denmark. The initial tables with the information from which the graphs were generated are shown from Table 4.4 to Table 5.19 respectively. From Table 5.4 to Table 5.11 are shown the *ping* results, and from Table 5.12 to Table 5.19 are shown the *traceroute* results.

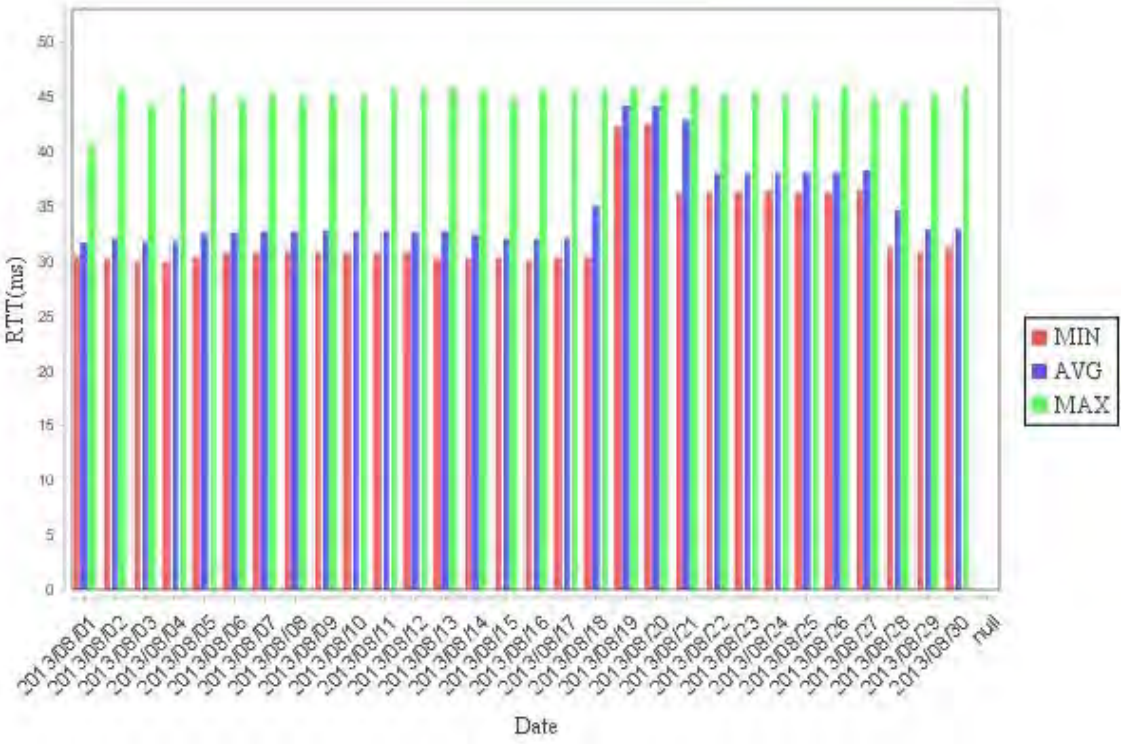


Figure 5.1: Ping Day-to-Day MIN, AVG and MAX RTTs from North to South for Denmark

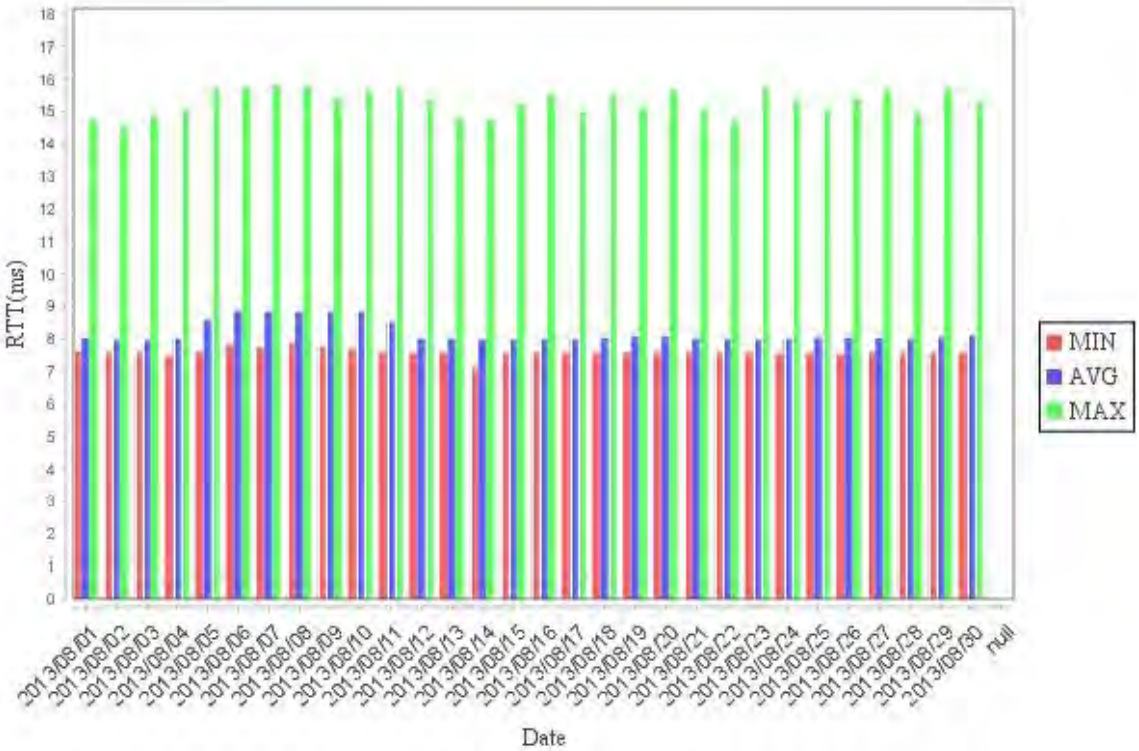


Figure 5.2: Day-to-Day MIN, AVG and MAX RTTs from South to North for Denmark

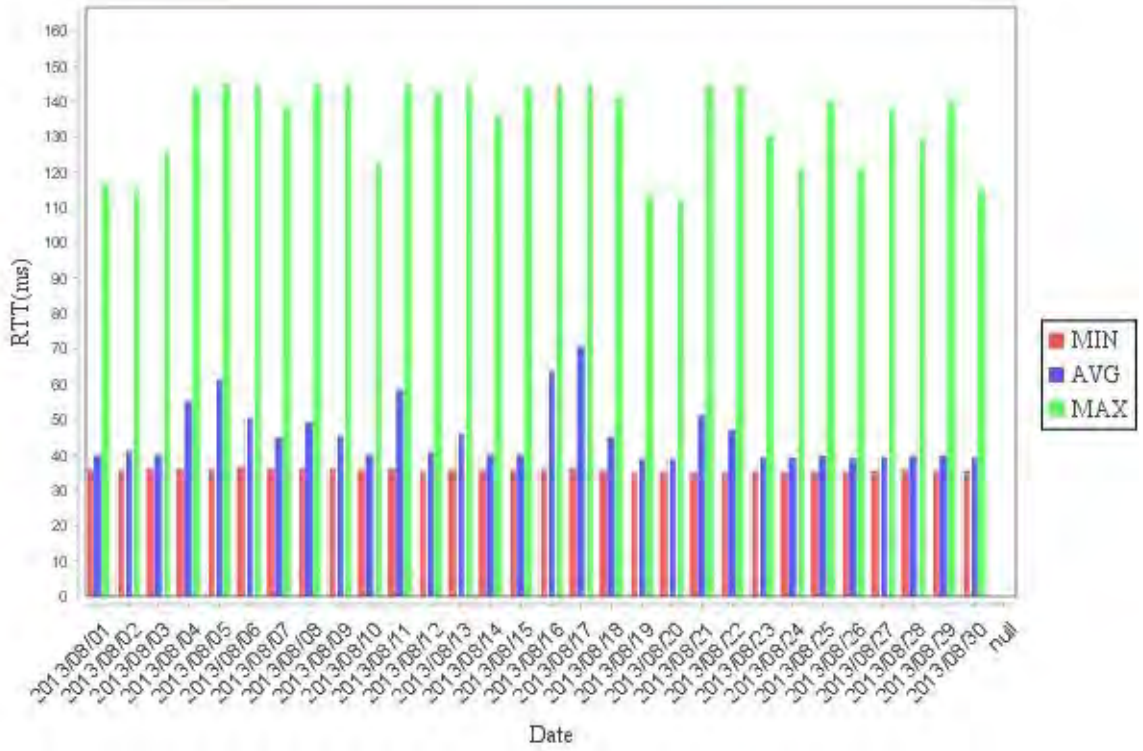


Figure 5.3: Ping Day-to-Day MIN, AVG and MAX RTTs from East to West for Denmark

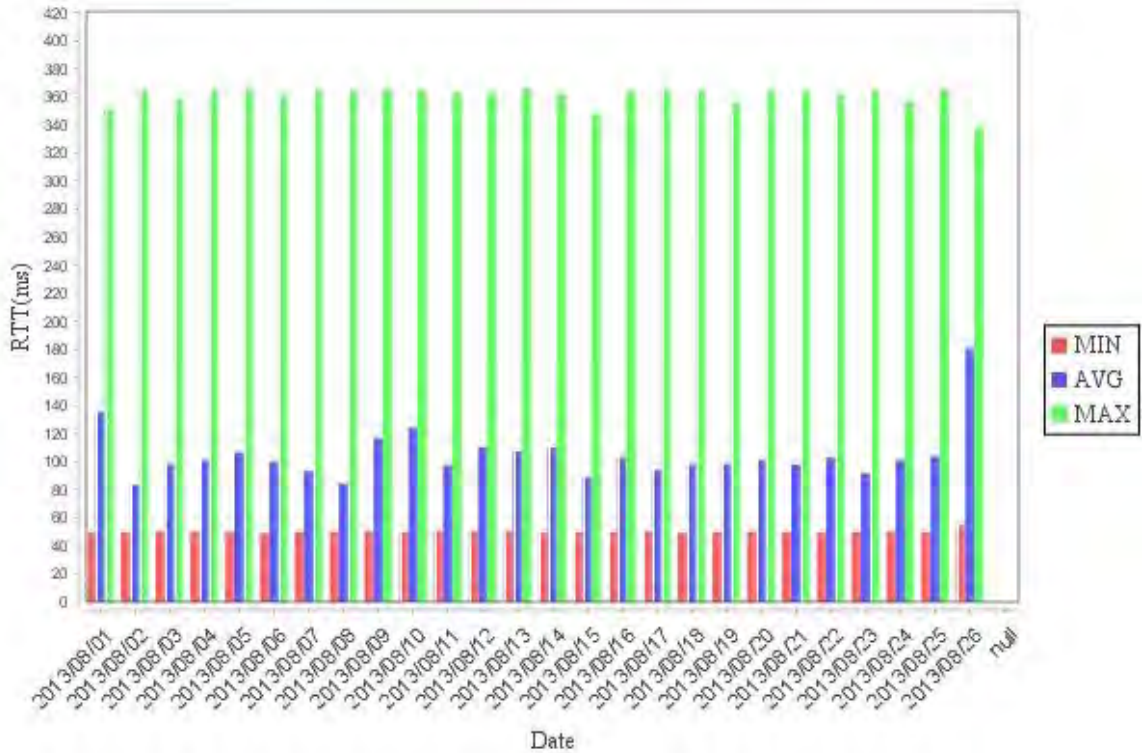


Figure 5.4: Ping Day-to-Day MIN, AVG and MAX RTTs from West to East for Denmark

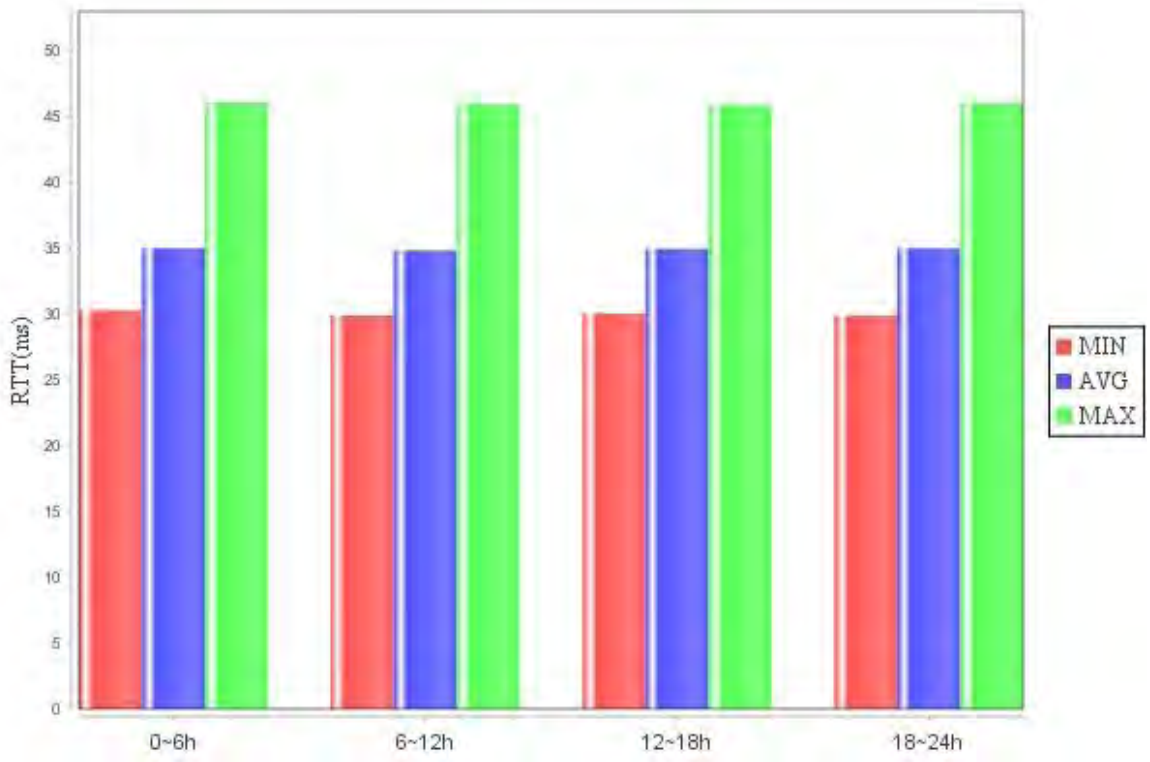


Figure 5.5: Ping MIN, AVG and MAX RTTs for time intervals from North to South for Denmark

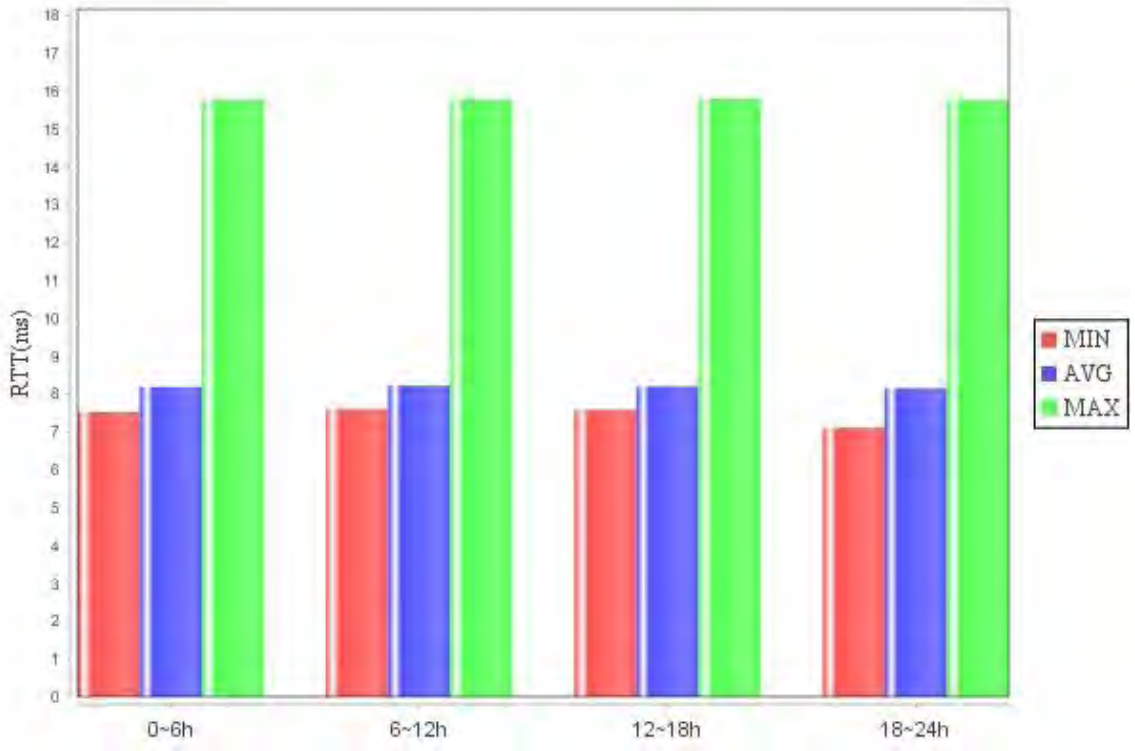


Figure 5.6: Ping MIN, AVG and MAX RTTs for time intervals from South to North for Denmark

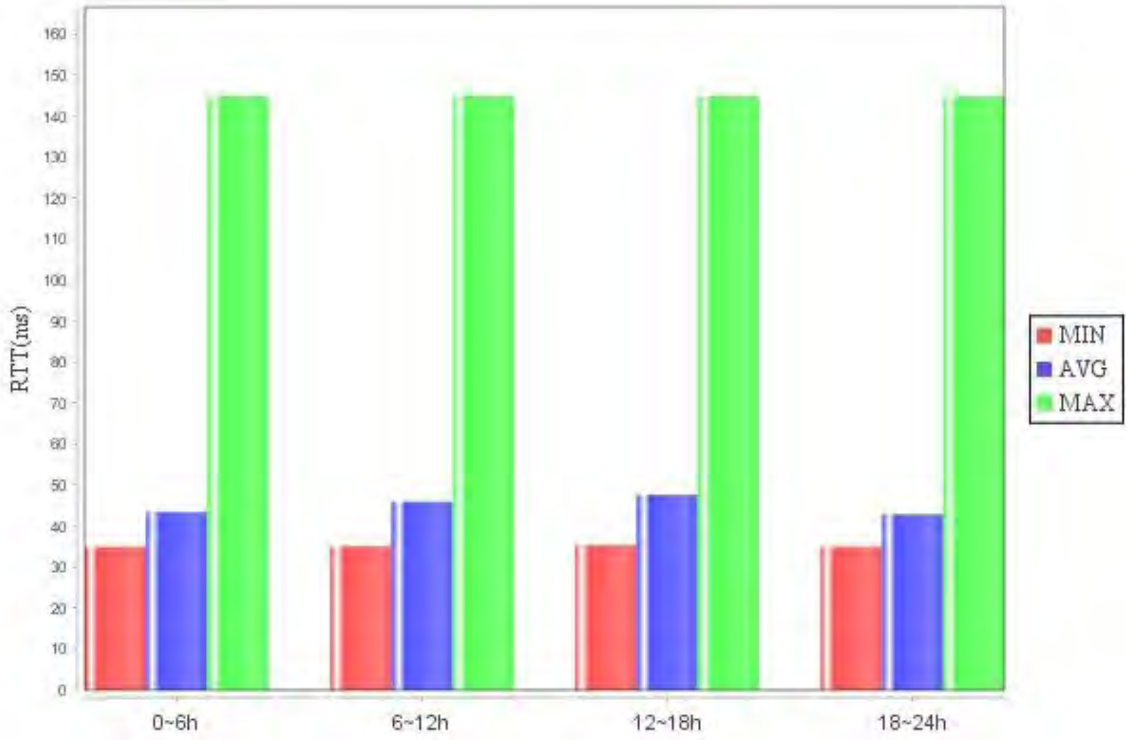


Figure 5.7: Ping MIN, AVG and MAX RTTs for time intervals from East to West for Denmark

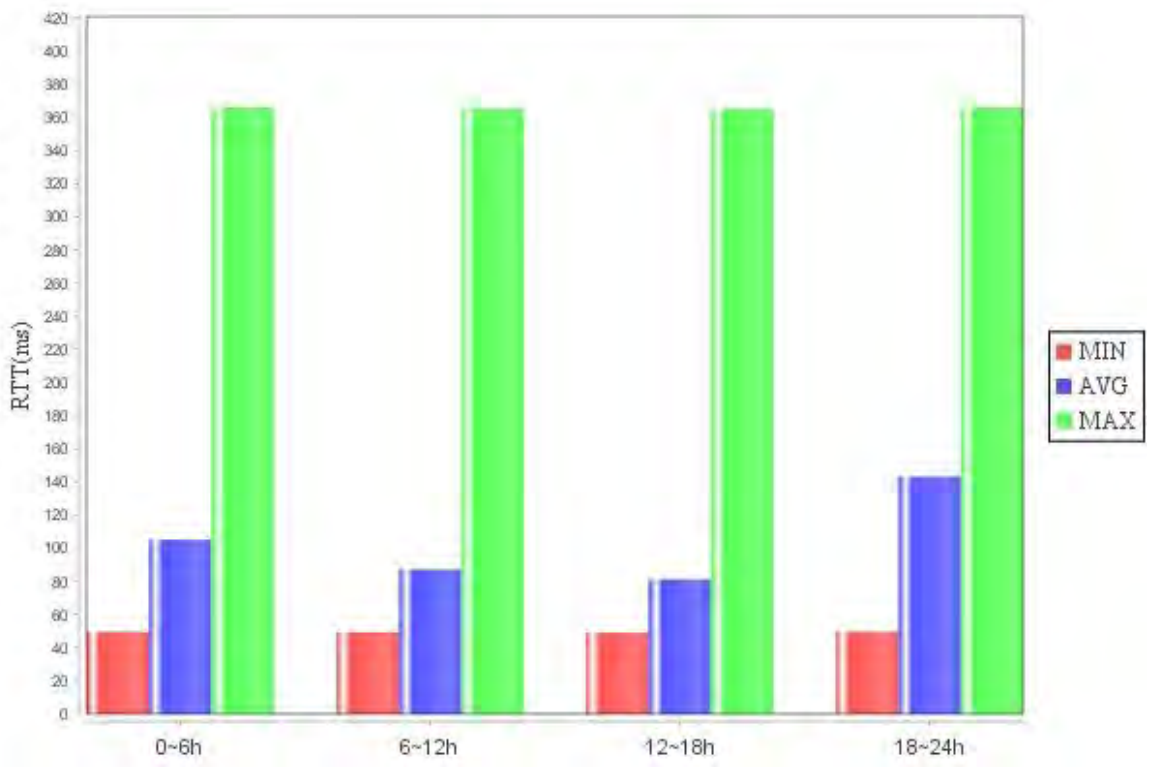


Figure 5.8: Ping MIN, AVG and MAX RTTs for time intervals from West to East for Denmark

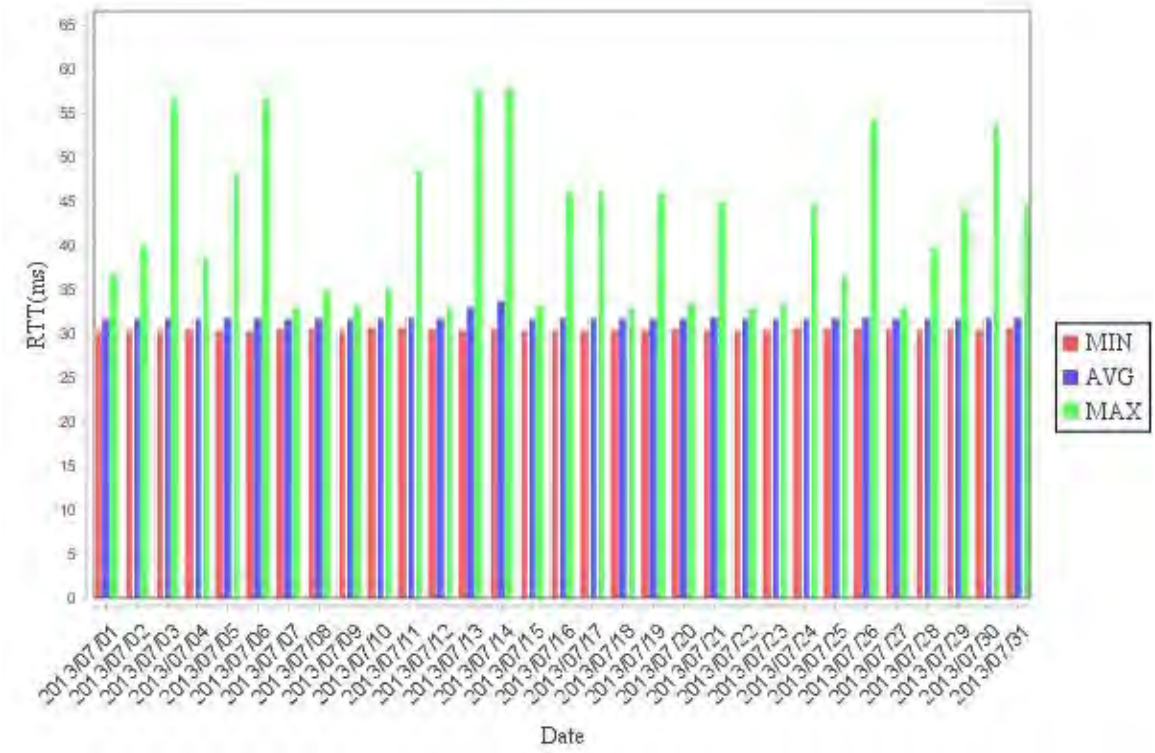


Figure 5.9: Traceroute Day-to-Day MIN, AVG and MAX RTTs from North to South for Denmark

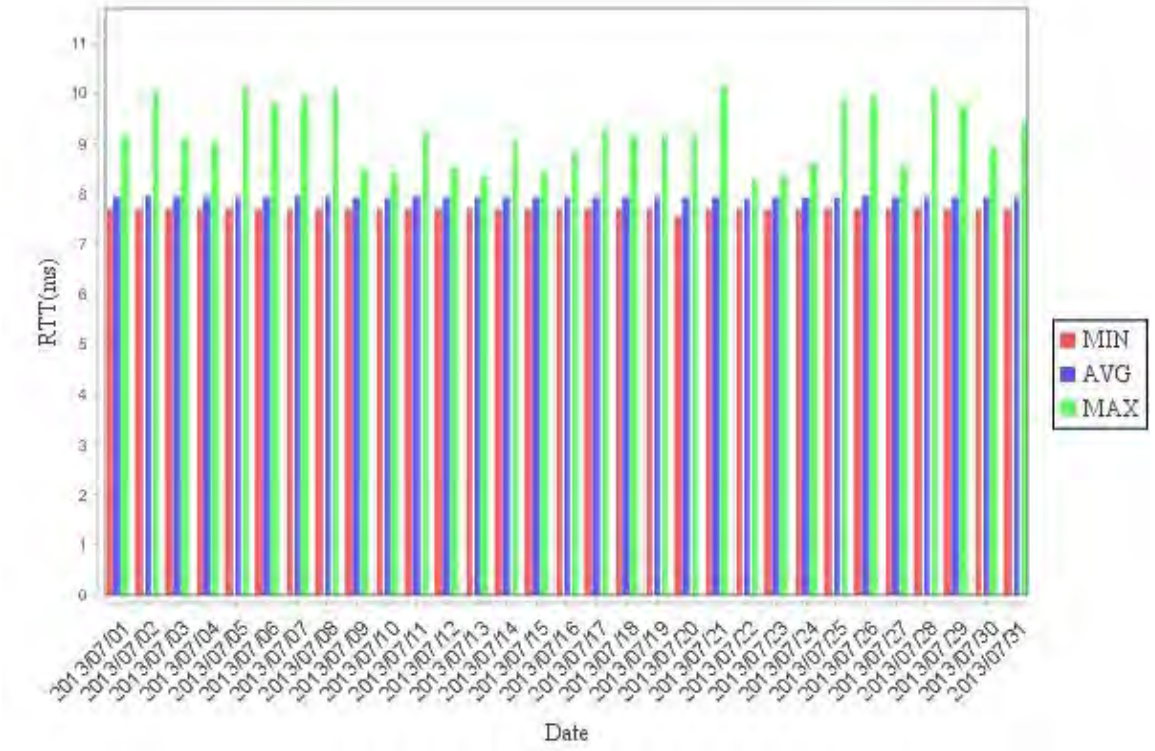


Figure 5.10 Traceroute Day-to-Day MIN, AVG and MAX RTTs from South to North for Denmark

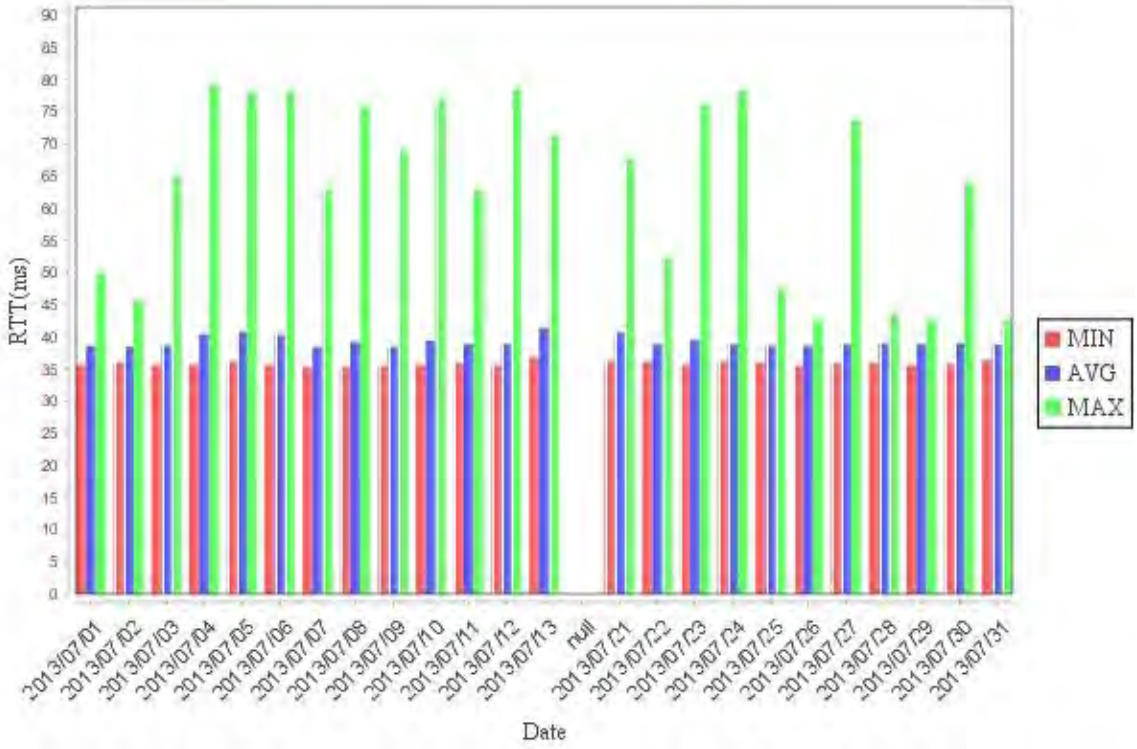


Figure 5.11 Traceroute Day-to-Day MIN, AVG and MAX RTTs from East to West for Denmark

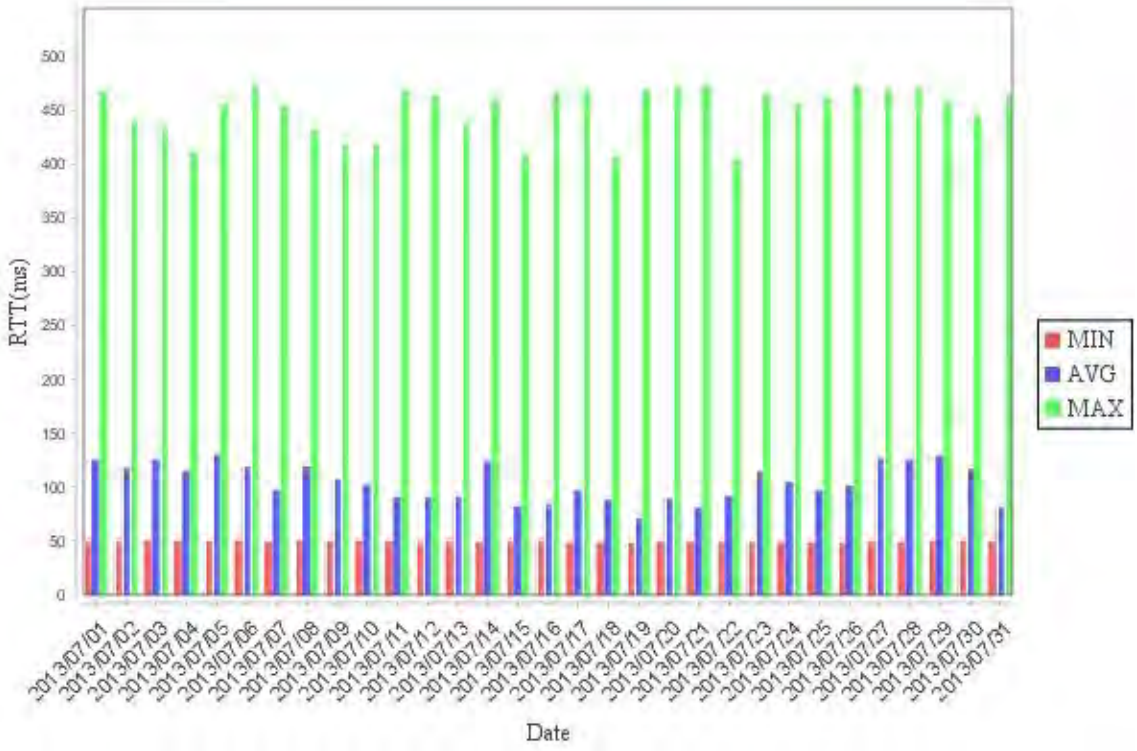


Figure 5.12 Traceroute Day-to-Day MIN, AVG and MAX RTTs from West to East for Denmark

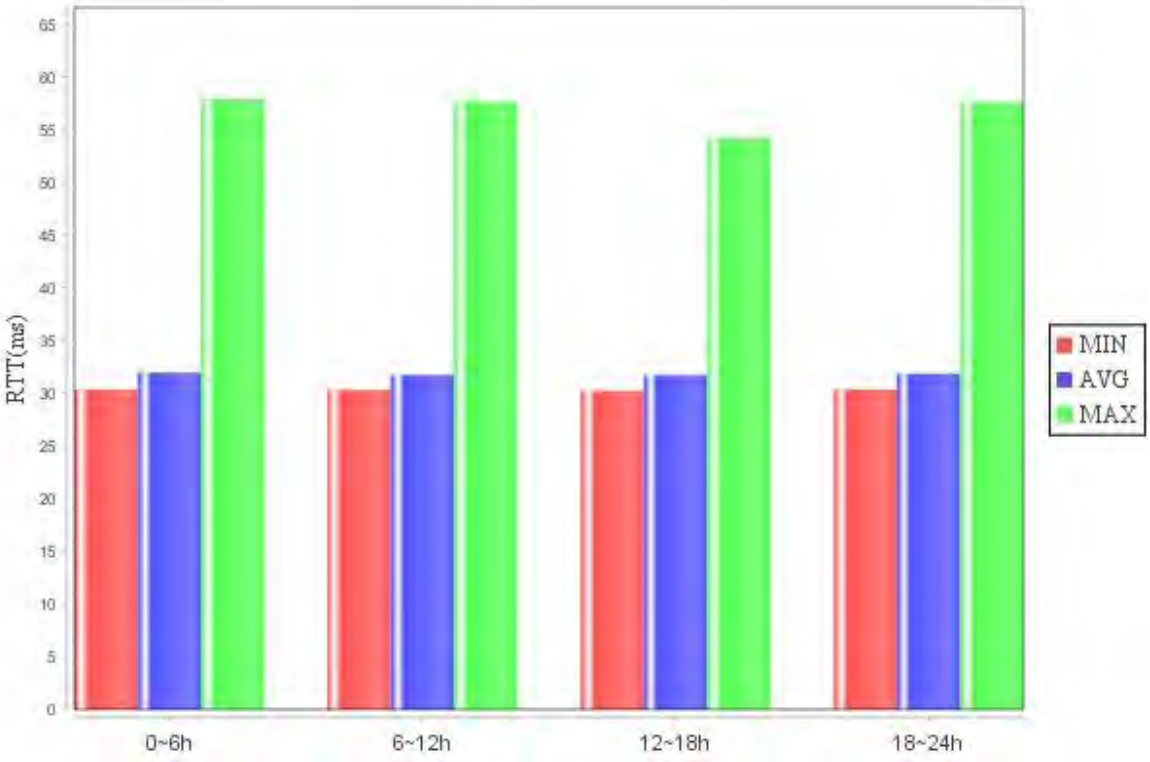


Figure 5.13 Traceroute MIN, AVG and MAX RTTs for time intervals from North to South for Denmark

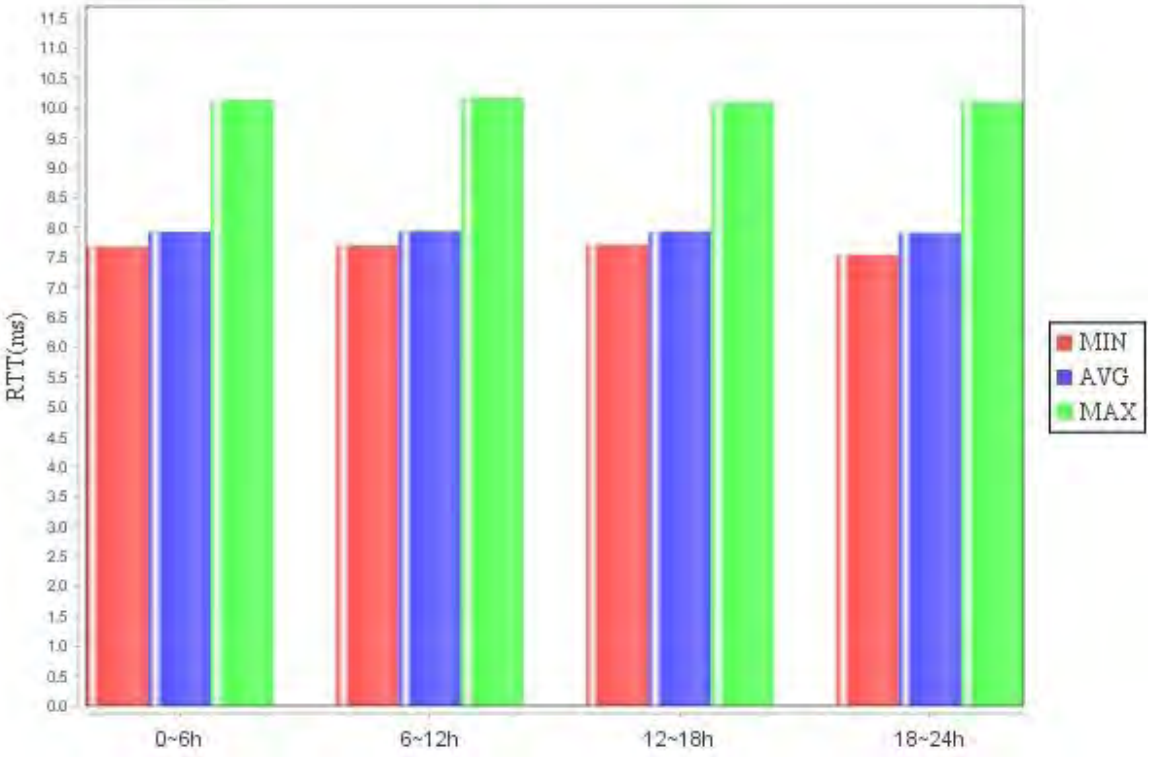


Figure 5.14 Traceroute MIN, AVG and MAX RTTs for time intervals from South to North for Denmark

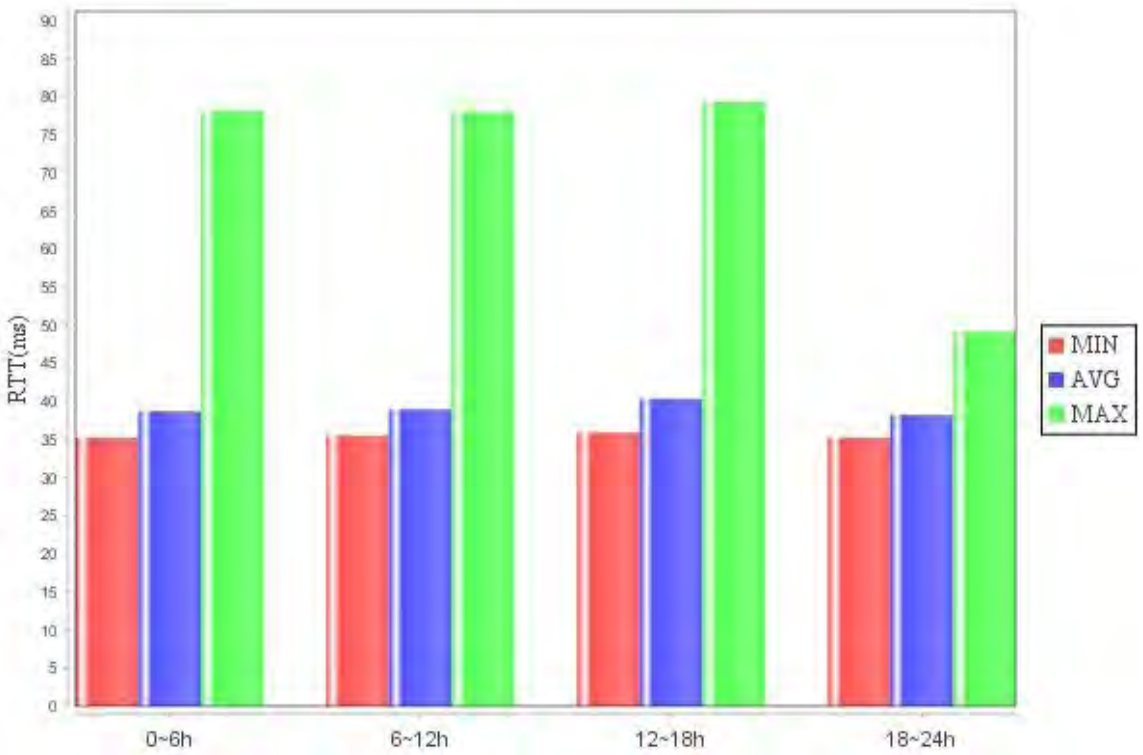


Figure 5.15 Traceroute MIN, AVG and MAX RTTs for time intervals from East to West for Denmark

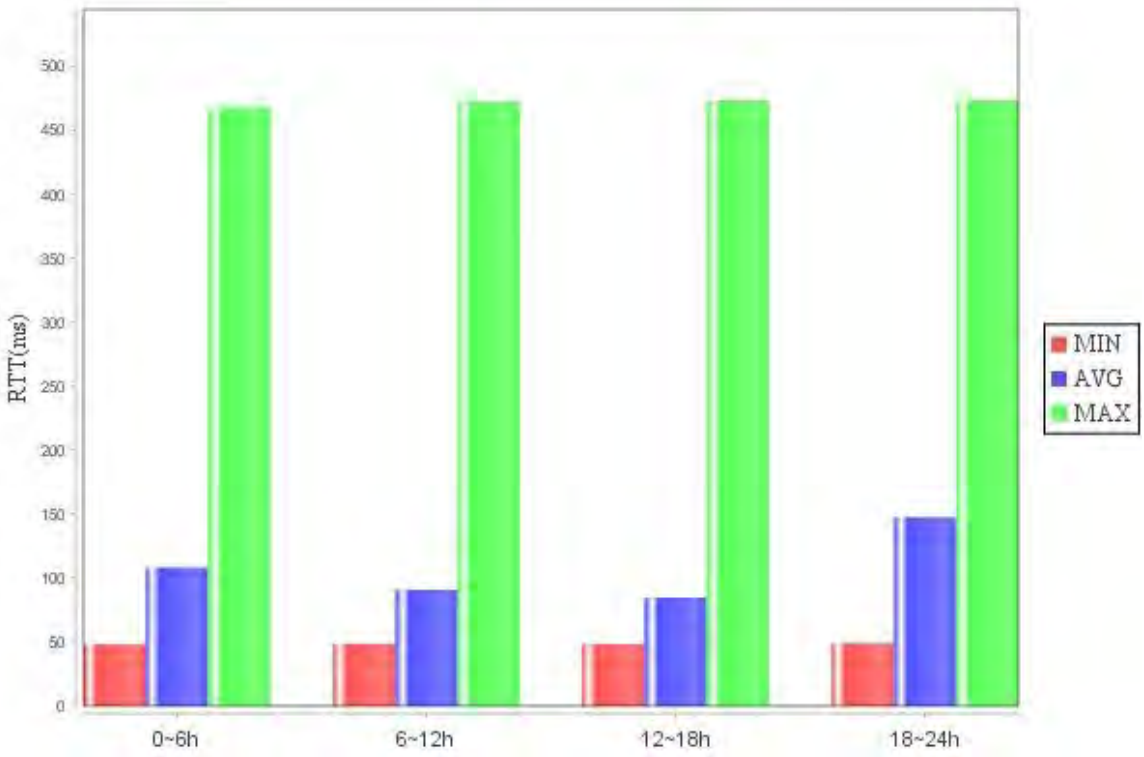


Figure 5.16 Traceroute MIN, AVG and MAX RTTs for time intervals from West to East for Denmark

Table 5.4: Day-to-Day MIN, AVG and MAX RTTs from North to South for Denmark (ping)

Date	Min	Avg	Max
2013/08/01	30.401	31.666	40.631
2013/08/02	30.271	32.047	45.66
2013/08/03	29.875	31.778	44.207
2013/08/04	29.868	31.834	46.083
2013/08/05	30.352	32.507	45.296
2013/08/06	30.755	32.558	44.818
2013/08/07	30.811	32.692	45.253
2013/08/08	30.928	32.659	45.06
2013/08/09	30.826	32.755	45.224
2013/08/10	30.793	32.712	45.162
2013/08/11	30.755	32.738	45.824
2013/08/12	30.884	32.603	45.723
2013/08/13	30.225	32.743	45.749
2013/08/14	30.211	32.427	45.536
2013/08/15	30.31	32.026	44.851
2013/08/16	30.031	31.972	45.606
2013/08/17	30.336	32.067	45.471
2013/08/18	30.333	35.053	45.705
2013/08/19	42.281	44.154	45.786
2013/08/20	42.495	44.136	45.587
2013/08/21	36.227	42.901	46.072
2013/08/22	36.326	37.976	45.183
2013/08/23	36.335	38.014	45.455
2013/08/24	36.435	38.023	45.259
2013/08/25	36.247	38.063	44.918
2013/08/26	36.27	38.027	45.979
2013/08/27	36.527	38.271	44.95
2013/08/28	31.203	34.64	44.383
2013/08/29	30.828	32.894	45.25
2013/08/30	31.144	32.955	45.886

Table 5.5: Day-to-Day MIN, AVG and MAX RTTs from South to North for Denmark (ping)

Date	Min	Avg	Max
2013/08/01	7.603	8.004	14.759
2013/08/02	7.593	7.977	14.515
2013/08/03	7.58	7.964	14.86
2013/08/04	7.461	8.014	15.012
2013/08/05	7.595	8.574	15.722
2013/08/06	7.813	8.861	15.767
2013/08/07	7.728	8.828	15.797
2013/08/08	7.848	8.814	15.778
2013/08/09	7.768	8.817	15.411
2013/08/10	7.696	8.842	15.638
2013/08/11	7.585	8.518	15.754
2013/08/12	7.571	8.012	15.381
2013/08/13	7.576	8.001	14.803
2013/08/14	7.105	7.981	14.72
2013/08/15	7.583	7.971	15.222
2013/08/16	7.589	8.002	15.49
2013/08/17	7.577	7.988	14.963
2013/08/18	7.57	8.004	15.483
2013/08/19	7.589	8.069	15.072
2013/08/20	7.58	8.067	15.667
2013/08/21	7.594	7.996	15.024
2013/08/22	7.595	7.992	14.696
2013/08/23	7.583	7.958	15.762
2013/08/24	7.54	7.993	15.361
2013/08/25	7.573	8.033	15.039
2013/08/26	7.518	8.011	15.374
2013/08/27	7.585	8.016	15.664
2013/08/28	7.577	7.989	14.895
2013/08/29	7.566	8.052	15.687
2013/08/30	7.582	8.099	15.293

Table 5.6: Day-to-Day MIN, AVG and MAX RTTs from East to West for Denmark (ping)

Date	Min	Avg	Max
2013/08/01	36.105	39.899	116.687
2013/08/02	35.686	41.094	114.788
2013/08/03	36.217	40.023	125.422
2013/08/04	36.015	55.216	143.595
2013/08/05	36.029	61.216	144.864
2013/08/06	36.645	50.459	144.82
2013/08/07	36.174	44.978	138.256
2013/08/08	36.293	49.174	144.878
2013/08/09	36.29	45.467	144.852
2013/08/10	35.772	40.123	122.465
2013/08/11	36.142	58.434	144.875
2013/08/12	35.274	40.922	142.176
2013/08/13	35.873	45.95	144.645
2013/08/14	35.702	40.303	135.907
2013/08/15	35.857	40.128	144.156
2013/08/16	36.015	63.534	144.846
2013/08/17	36.357	70.733	144.857
2013/08/18	35.798	45.113	141.121
2013/08/19	34.967	38.884	113.019
2013/08/20	35.287	38.849	111.963
2013/08/21	34.964	51.225	144.787
2013/08/22	35.096	47.199	144.442
2013/08/23	35.312	39.338	130.41
2013/08/24	35.006	39.225	120.505
2013/08/25	35.293	39.734	140.084
2013/08/26	35.208	39.121	120.768
2013/08/27	35.383	39.361	137.636
2013/08/28	36.048	39.55	129.143
2013/08/29	35.517	39.674	140.109
2013/08/30	35.358	39.298	115.031

Table 5.7: Day-to-Day MIN, AVG and MAX RTTs from West to East for Denmark (ping)

Date	Min	Avg	Max
2013/08/01	49.546	135.43	350.732
2013/08/02	49.262	83.453	365.286
2013/08/03	49.785	97.956	358.365
2013/08/04	50.009	101.324	366.01
2013/08/05	49.927	106.447	365.892
2013/08/06	49.025	99.706	360.783
2013/08/07	49.84	93.135	365.14
2013/08/08	49.834	84.575	364.822
2013/08/09	50.22	116.503	365.275
2013/08/10	49.769	124.524	365.314
2013/08/11	50.341	97.493	363.152
2013/08/12	50.058	110.379	362.28
2013/08/13	50.009	107.339	365.925
2013/08/14	49.159	110.03	361.524
2013/08/15	49.764	88.721	347.204
2013/08/16	49.673	103.194	364.003
2013/08/17	49.972	93.895	365.577
2013/08/18	49.18	98.116	364.861
2013/08/19	49.449	98.054	355.856
2013/08/20	49.848	101.512	365.149
2013/08/21	49.801	97.647	365.188
2013/08/22	49.378	102.91	361.476
2013/08/23	49.689	91.845	364.894
2013/08/24	49.834	100.969	356.625
2013/08/25	49.767	103.681	365.421
2013/08/26	55.12	181.075	337.697
	0	0	0
	0	0	0
	0	0	0
	0	0	0

Table 5.8: MIN, AVG and MAX RTTs for time intervals from North to South for Denmark (ping)

Time	Min	Avg	Max
0~6h	30.264	35.008	46.083
6~12h	29.875	34.842	45.949
12~18h	30.031	34.944	45.886
18~24h	29.868	35.007	46.032

Table 5.9: MIN, AVG and MAX RTTs for time intervals from East to West for Denmark (ping)

Time	Min	Avg	Max
0~6h	34.964	43.389	144.864
6~12h	35.067	45.945	144.878
12~18h	35.358	47.637	144.857
18~24h	35.006	42.87	144.787

Table 5.10: MIN, AVG and MAX RTTs for time intervals from South to North for Denmark (ping)

Time	Min	Avg	Max
0~6h	7.518	8.179	15.778
6~12h	7.594	8.218	15.773
12~18h	7.573	8.195	15.797
18~24h	7.105	8.148	15.767

Table 5.11: MIN, AVG and MAX RTTs for time intervals from West to East for Denmark (ping)

Time	Min	Avg	Max
0~6h	49.37	105.012	365.892
6~12h	49.262	87.134	365.421
12~18h	49.025	81.059	365.188
18~24h	49.546	143.19	366.01

Table 5.12: Day-to-Day MIN, AVG and MAX RTTs from North to South for Denmark (traceroute)

Date	Min	Avg	Max
2013/07/01	30.347	31.63	36.736
2013/07/02	30.341	31.652	39.948
2013/07/03	30.304	31.742	56.594
2013/07/04	30.471	31.673	38.585
2013/07/05	30.336	31.766	48.04
2013/07/06	30.231	31.727	56.593
2013/07/07	30.486	31.583	32.976
2013/07/08	30.55	31.729	34.943
2013/07/09	30.324	31.647	33.091
2013/07/10	30.672	31.748	35.321
2013/07/11	30.625	31.861	48.443
2013/07/12	30.492	31.692	32.875
2013/07/13	30.342	33.01	57.588
2013/07/14	30.486	33.704	57.905
2013/07/15	30.337	31.692	33.225
2013/07/16	30.337	31.782	46.121
2013/07/17	30.341	31.754	46.124
2013/07/18	30.416	31.671	32.833
2013/07/19	30.358	31.696	45.949
2013/07/20	30.489	31.691	33.445
2013/07/21	30.347	31.826	44.909
2013/07/22	30.366	31.659	32.835
2013/07/23	30.408	31.687	33.563
2013/07/24	30.502	31.704	44.772
2013/07/25	30.551	31.727	36.54
2013/07/26	30.576	31.776	54.42
2013/07/27	30.556	31.651	32.821
2013/07/28	30.345	31.648	39.77
2013/07/29	30.459	31.701	44.12
2013/07/30	30.341	31.837	53.65
2013/07/31	30.628	31.787	44.558

Table 5.13: Day-to-Day MIN, AVG and MAX RTTs from South to North for Denmark (traceroute)

Date	Min	Avg	Max
2013/07/01	7.704	7.939	9.148
2013/07/02	7.696	7.958	10.04
2013/07/03	7.696	7.941	9.095
2013/07/04	7.702	7.931	9.01
2013/07/05	7.706	7.934	10.171
2013/07/06	7.688	7.931	9.807
2013/07/07	7.705	7.945	9.931
2013/07/08	7.7	7.947	10.094
2013/07/09	7.695	7.918	8.483
2013/07/10	7.702	7.915	8.415
2013/07/11	7.704	7.942	9.224
2013/07/12	7.701	7.924	8.533
2013/07/13	7.715	7.929	8.35
2013/07/14	7.678	7.921	9.052
2013/07/15	7.703	7.926	8.431
2013/07/16	7.689	7.926	8.815
2013/07/17	7.708	7.922	9.281
2013/07/18	7.695	7.926	9.147
2013/07/19	7.708	7.93	9.167
2013/07/20	7.54	7.916	9.157
2013/07/21	7.703	7.921	10.137
2013/07/22	7.702	7.906	8.287
2013/07/23	7.683	7.921	8.368
2013/07/24	7.692	7.915	8.602
2013/07/25	7.708	7.916	9.888
2013/07/26	7.709	7.957	9.943
2013/07/27	7.713	7.928	8.551
2013/07/28	7.713	7.942	10.092
2013/07/29	7.701	7.925	9.761
2013/07/30	7.699	7.93	8.944
2013/07/31	7.691	7.929	9.462

Table 5.14: Day-to-Day MIN, AVG and MAX RTTs from East to West for Denmark (traceroute)

Date	Min	Avg	Max
2013/07/01	35.602	38.552	49.92
2013/07/02	36.056	38.417	45.644
2013/07/03	35.599	38.666	65.087
2013/07/04	35.703	40.342	79.337
2013/07/05	36.135	40.845	77.998
2013/07/06	35.562	40.288	78.081
2013/07/07	35.238	38.386	62.829
2013/07/08	35.212	39.183	75.641
2013/07/09	35.402	38.38	69.097
2013/07/10	35.6	39.35	77.005
2013/07/11	35.942	38.848	62.761
2013/07/12	35.506	38.864	78.408
2013/07/13	36.772	41.357	71.166
	0	0	0
	0	0	0
	0	0	0
	0	0	0
	0	0	0
	0	0	0
	0	0	0
2013/07/21	36.242	40.651	67.654
2013/07/22	36.074	38.828	52.129
2013/07/23	35.669	39.471	75.983
2013/07/24	36.144	38.875	78.158
2013/07/25	35.955	38.605	47.367
2013/07/26	35.486	38.575	42.534
2013/07/27	35.853	38.841	73.543
2013/07/28	35.942	38.969	43.576
2013/07/29	35.522	38.83	42.496
2013/07/30	35.738	39.022	63.862
2013/07/31	36.367	38.774	42.591

Table 5.15: Day-to-Day MIN, AVG and MAX RTTs from West to East for Denmark (traceroute)

Date	Min	Avg	Max
2013/07/01	49.413	125.651	467.294
2013/07/02	49.347	117.82	440.122
2013/07/03	50.56	125.933	432.947
2013/07/04	50.031	114.949	409.915
2013/07/05	49.67	129.919	455.177
2013/07/06	50.218	119.003	473.393
2013/07/07	49.238	97.409	454.649
2013/07/08	50.564	119.484	431.345
2013/07/09	49.317	108.116	417.894
2013/07/10	49.994	102.546	418.052
2013/07/11	49.573	90.993	468.009
2013/07/12	48.8	90.402	463.013
2013/07/13	49.378	91.278	437.368
2013/07/14	48.934	124.017	458.901
2013/07/15	49.355	82.128	408.682
2013/07/16	50.099	84.125	466.328
2013/07/17	48.76	97.089	468.857
2013/07/18	48.792	88.419	406.588
2013/07/19	48.132	70.928	469.164
2013/07/20	49.347	89.768	472.335
2013/07/21	49.156	81.231	473.004
2013/07/22	48.978	91.529	404.854
2013/07/23	48.838	114.931	464.196
2013/07/24	48.627	104.508	456.176
2013/07/25	48.525	97.038	461.13
2013/07/26	48.351	101.229	473.399
2013/07/27	49.075	126.814	469.256
2013/07/28	48.702	124.765	470.957
2013/07/29	50.012	129.403	457.968
2013/07/30	49.711	117.218	445.222
2013/07/31	49.332	81.295	462.827

Table 5.16: MIN, AVG and MAX RTTs for time intervals from North to South Denmark (traceroute)

Time	Min	Avg	Max
0~6h	30.337	31.946	57.905
6~12h	30.304	31.738	57.645
12~18h	30.231	31.741	54.164
18~24h	30.336	31.854	57.588

Table 5.17: MIN, AVG and MAX RTTs for time intervals from South to North for Denmark (traceroute)

Time	Min	Avg	Max
0~6h	7.678	7.928	10.137
6~12h	7.695	7.941	10.171
12~18h	7.711	7.937	10.094
18~24h	7.54	7.906	10.092

Table 5.18: MIN, AVG and MAX RTTs for time intervals from East to West for Denmark (traceroute)

Time	Min	Avg	Max
0~6h	35.238	38.718	78.158
6~12h	35.522	38.962	77.998
12~18h	35.934	40.365	79.337
18~24h	35.212	38.223	49.113

Table 5.19: MIN, AVG and MAX RTTs for time intervals from West to East for Denmark (traceroute)

Time	Min	Avg	Max
0~6h	48.132	108.036	468.009
6~12h	48.627	90.683	472.335
12~18h	48.351	84.734	473.393
18~24h	49.241	147.428	473.399

If a measurement for a day or a couple of days is missing, it is represented in the graphs as "null" and the dates are skipped. The same is represented in the tables as an empty space where the date is supposed to be, and as 0 where the RTT is.

5.4 RESULTS ANALYSIS

The main QoS metrics used are *bandwidth*, *link stability*, *delay* and *packet loss*. In this research, the main focus relates to the *delay component*, or the so called 'Round-Trip-Time'. The exact QoS requirements are hard to pinpoint, especially having in mind the fact that there is a wide variation based on the country involved. Depending on the underlying network infrastructure, service providers, number of users and the traffic, comparing the QoS for two separate countries is a difficult task. The results can serve as a benchmark for future reference when conducting more measurements, and comparison could be made to see if there are any improvements or new issues across the network. With the help of the CLI index, one can investigate this and compare the different results for a specific country, based on the symmetry of the results, and come to a conclusion if there are any faults in any of the regions. Simply, an asymmetric result illustrates a problem along the path the measurement was conducted. Exploring the bottleneck further facilitates the search for the issue causing the problem. The issues may vary depending on the region and users. End-users use different applications, each of which generates different types of traffic. In order for the ISPs to provide the best QoS, they need to know what applications require higher prioritisation, as some applications would prove to be more sensitive to latency, jitter etc.

The information acquired can be useful to network administrators, ISPs and any other interested party. By exploring the CLI the fault can be located, examined and alleviated. In case of an asymmetric result, for opposite direction of the country under consideration, it's easy to see which CLI is bigger and that his path would be the one experiencing the faults. Starting from cabling, switches and routers, going through the telephone network, up to the final connecting components located closer to the end-user, any possibility should be explored as any of these might be in poor condition and

causing problems and unsatisfied customers. Looking closely into the CLI index groups, the worst cases are explored. Worst cases are considered countries where the difference between CLIs for a set of opposite directions is the biggest. In group one there is only one country which is Denmark for which all CLIs are considered asymmetric. The ratio between the North and South CLIs is 4.12:1 and for the East and West one it is 2.4:1. The CLI in the south/north direction is considerably smaller than the one for north/south, and the east/west is with noticeable difference from the west/east one. They both show much better performance, which shows that there's a problem existing in the opposite directions. Looking into the next group, which is group 3, there are two different countries consisting of a worst case scenario for each set of opposite directions. The CLI asymmetric difference for Hungary's north/south to south/north results is in a ratio of 3.1:1. The other country of interest would be Finland with 21:1 ratio difference between the CLIs in the other set of opposite directions: east/west and west/east. Again two countries include the worst case scenarios for group 4. There Luxembourg has a 2.16:1 ratio difference for the north/south and south/north CLIs while Ireland is with 1.86:1 margin between the east/west and west/east CLIs. Those very obvious differences can be caused due to various factors as already mentioned, from poor network condition, through number of users and traffic, to faults in networking equipment.

Further exploration of the problems across the network can be done by looking into the bar graphs. They visualise the delay from a source to a destination. The day-to-day bar graphs show the different RTTs for every day of the month and by exploring patterns and dependencies one can conclude what days of the week or the month, the network was the busiest, unavailable or if any problems occurred. With the help of the summarised time interval bar graphs for the duration of the whole measurement it is possible to see if there's a difference between the RTTs for different times of the day.

For the worst case scenarios of CLI asymmetry, the bar graphs can be used to better visualise it. Only the time interval bar graphs from the ping results are shown here as they summarise the RTTs for the whole period of time in a concise way. There is no need of showing the traceroute ones as they'd be very similar as there is no big

difference in the CLI results from the ping and traceroute measurements for those specific countries . For a reference all bar graphs can be seen in Appendix 3, 4, 5 and 6.

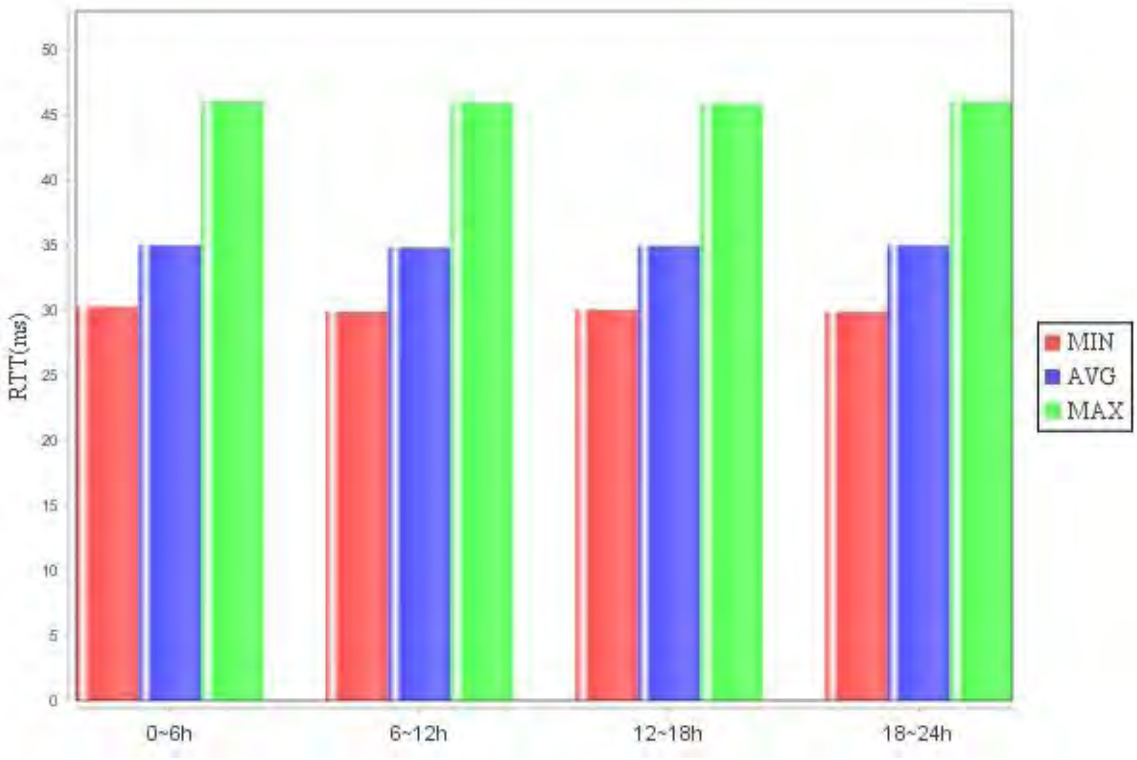


Figure 5.17: Denmark's North to South time interval MIN, AVG and MAX RTTs

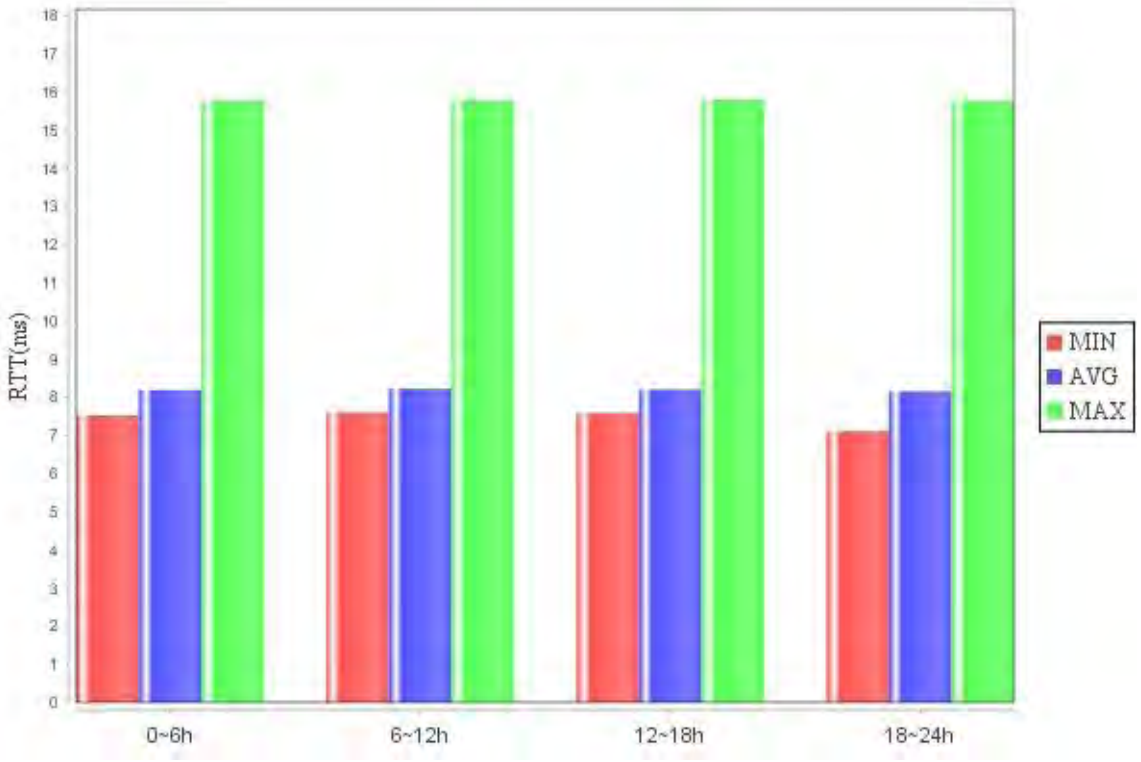


Figure 5.18: Denmark's South to North time interval MIN, AVG and MAX RTTs

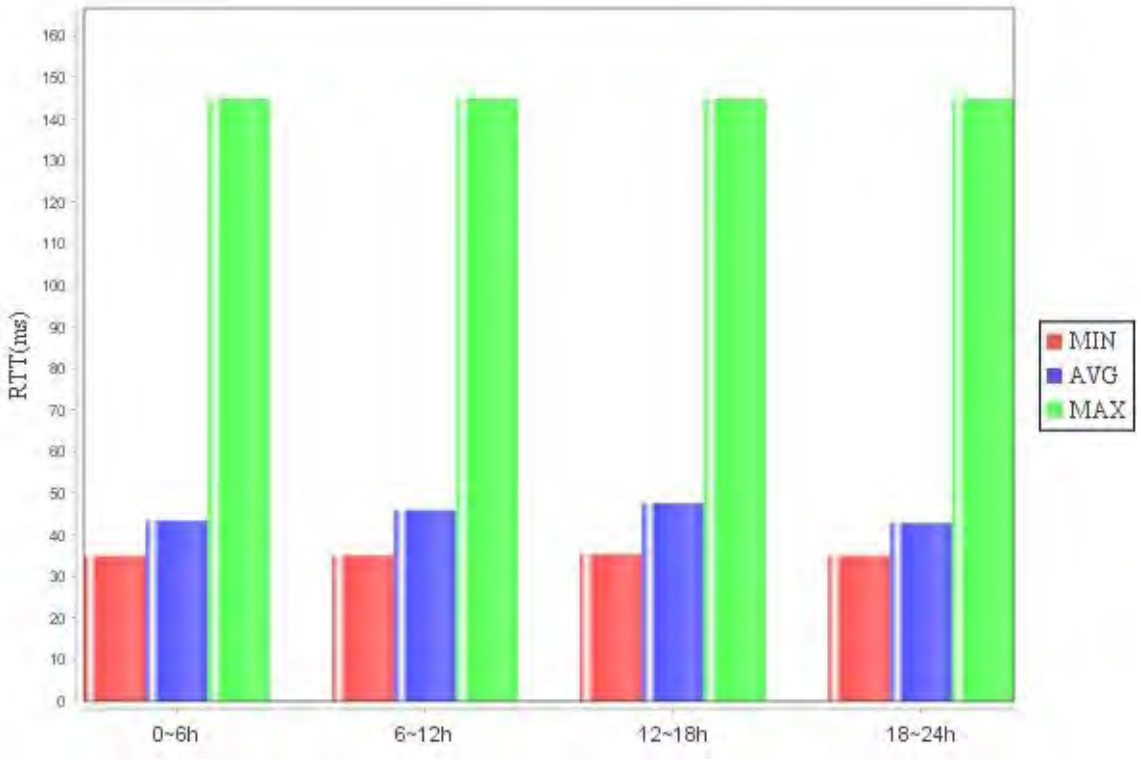


Figure 5.19: Denmark's East to West 30 day MIN, AVG and MAX RTTs

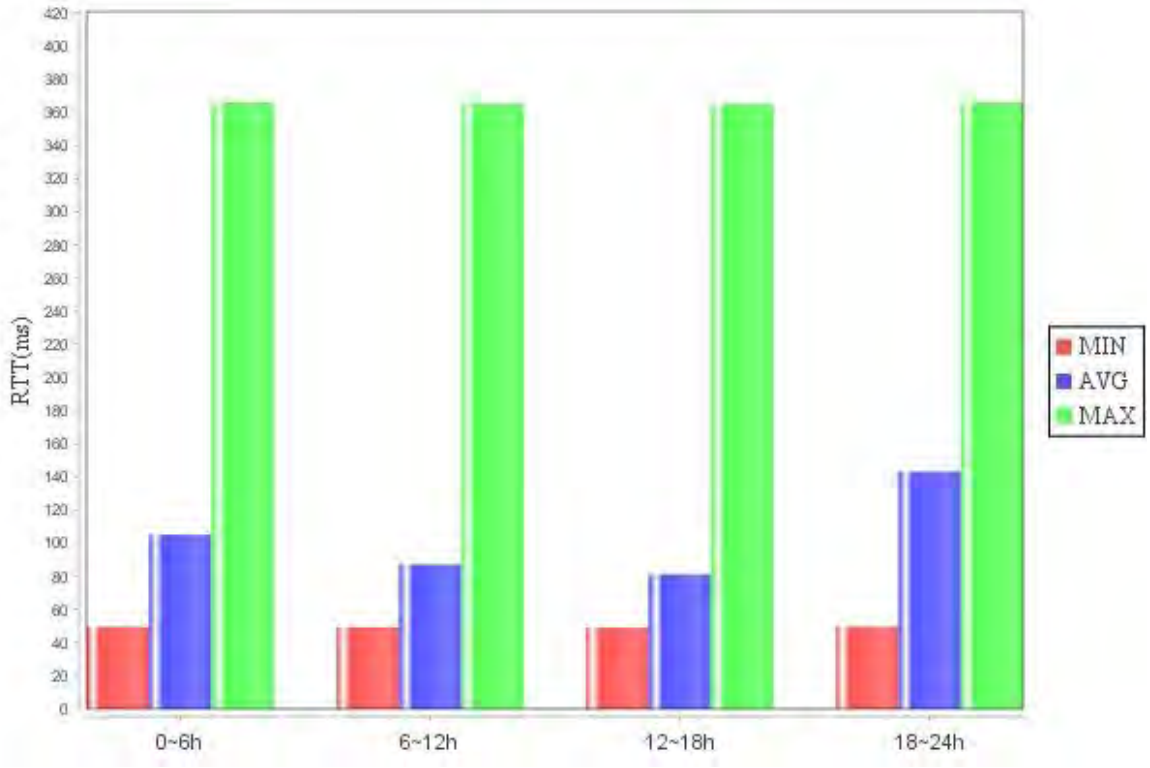


Figure 5.20: Denmark's West to East time interval MIN, AVG and MAX RTTs

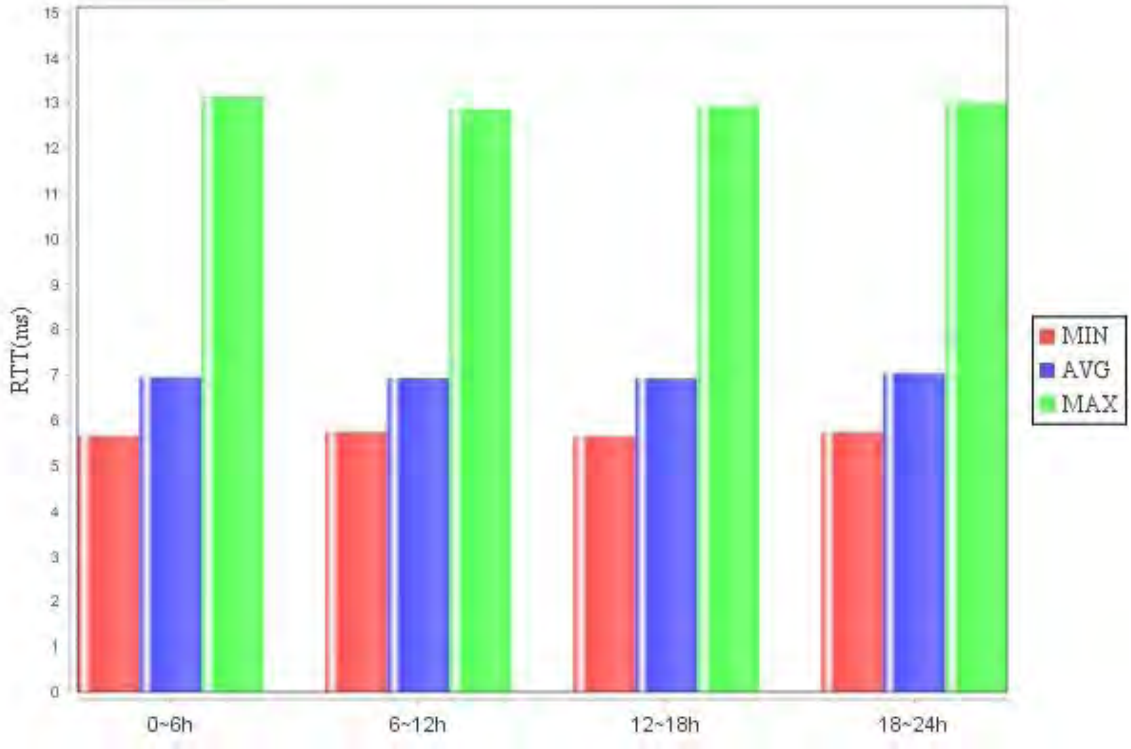


Figure 5.21: Hungary's North to South time interval MIN, AVG and MAX RTTs

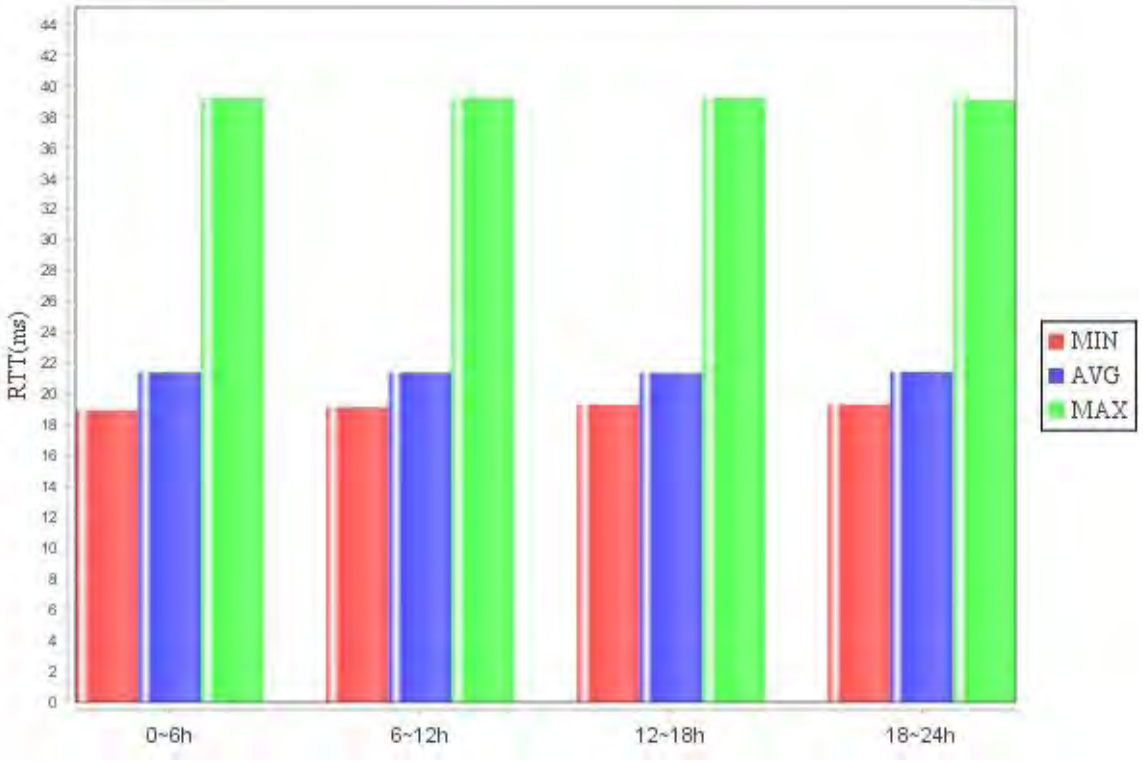


Figure 5.22: Hungary's South to North time interval MIN, AVG and MAX RTTs

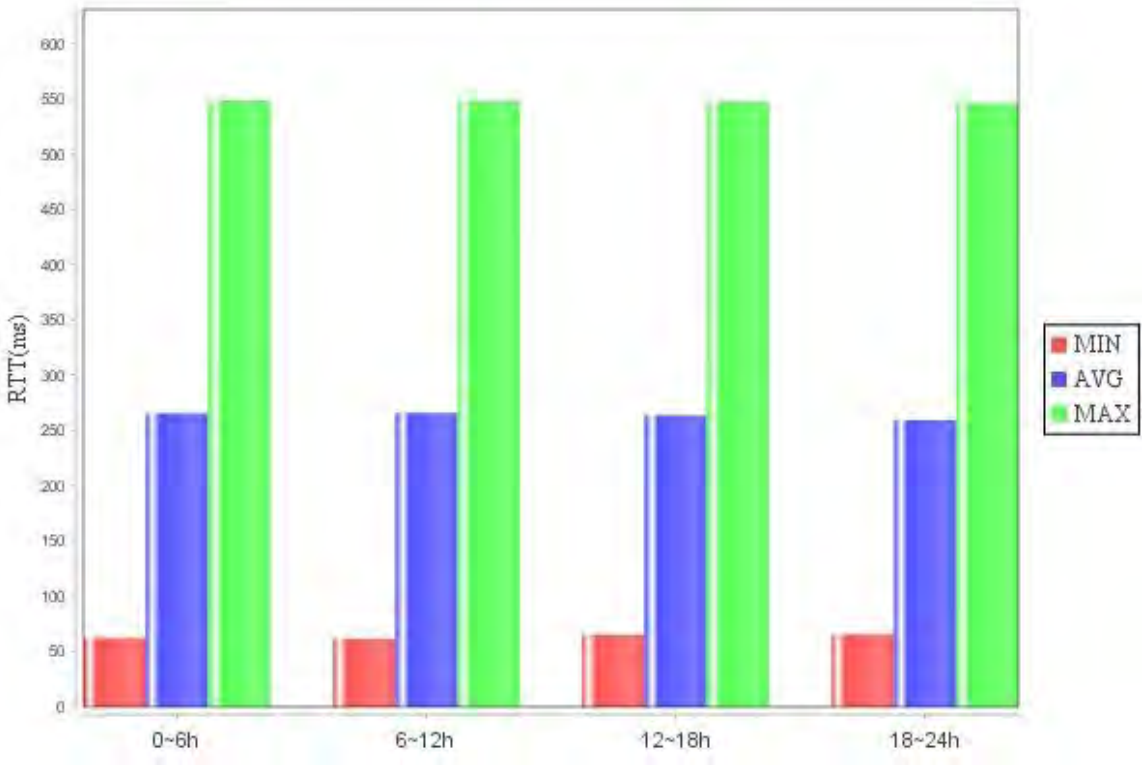


Figure 5.23: Finland's East to West time interval MIN, AVG and MAX RTTs

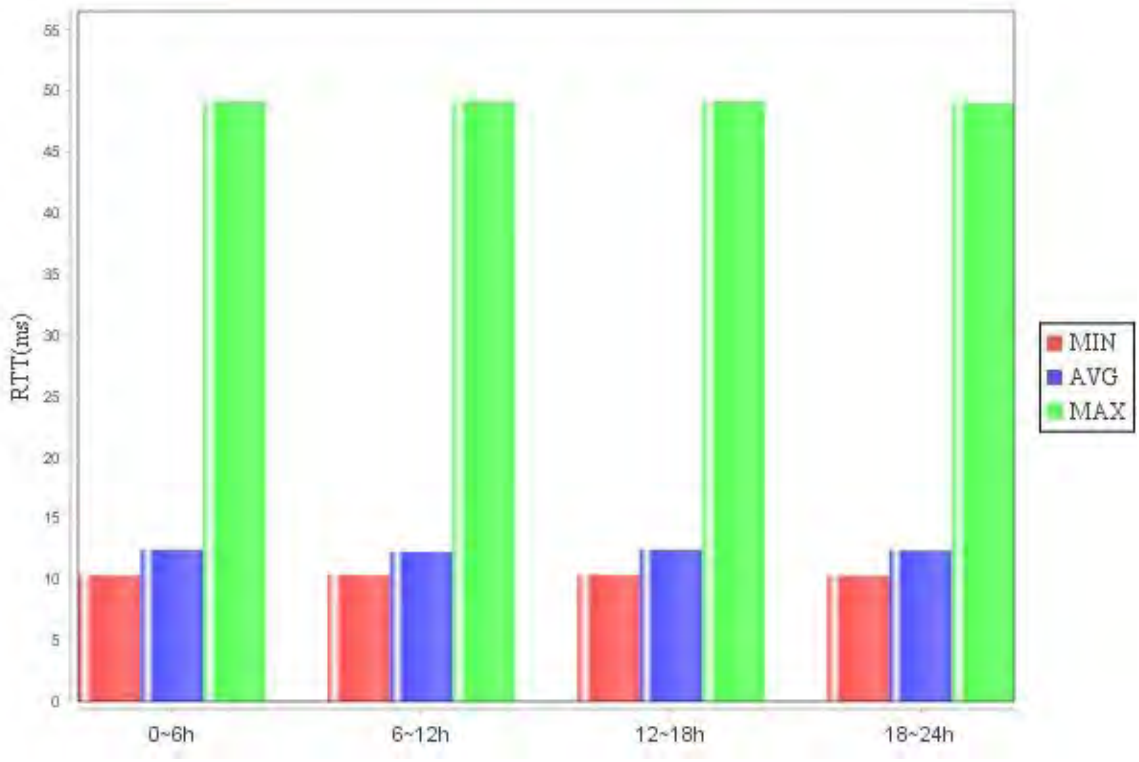


Figure 5.24: Finland's West to East time interval MIN, AVG and MAX RTTs

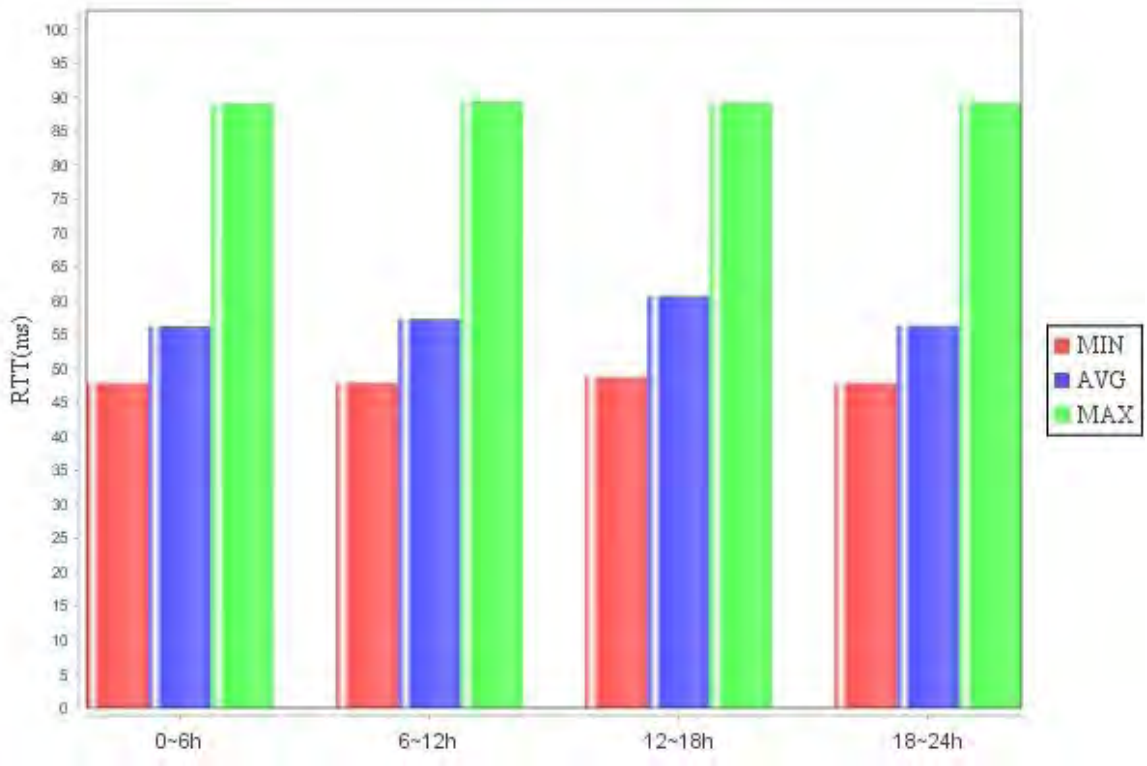


Figure 5.25: Luxembourg's North to South time interval MIN, AVG and MAX RTTs

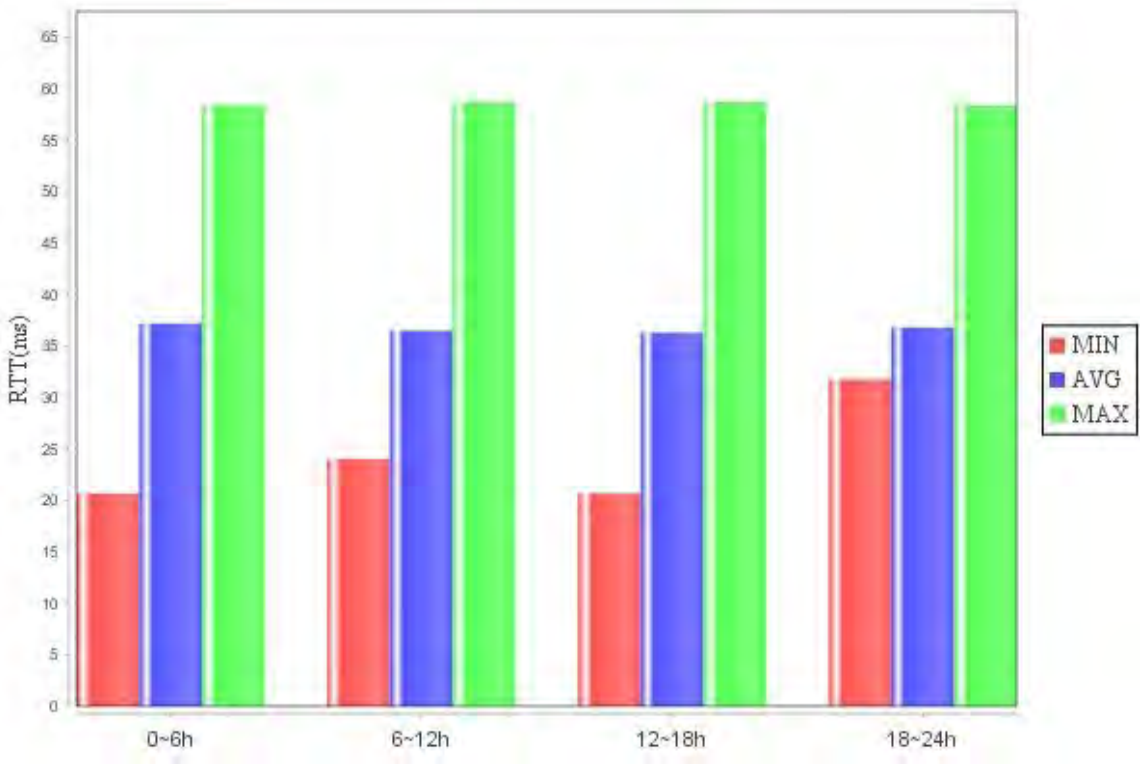


Figure 5.26: Luxembourg's South to North time interval MIN, AVG and MAX RTTs

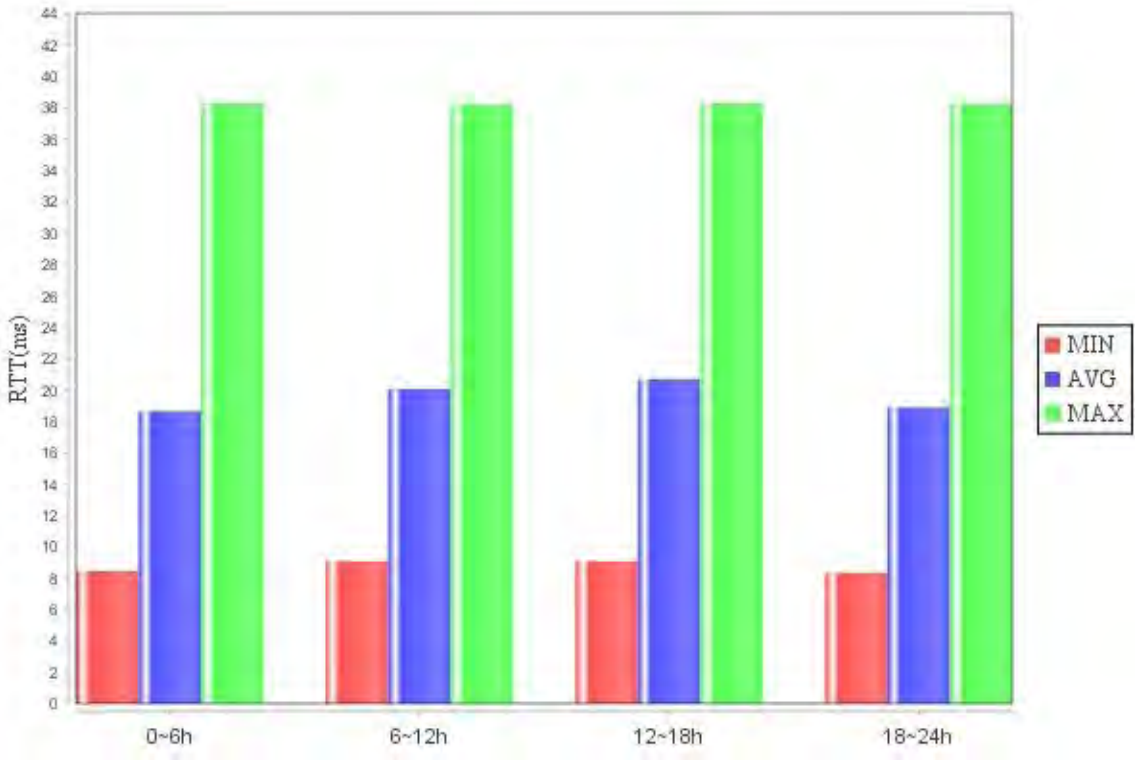


Figure 5.27: Ireland's East to West time interval MIN, AVG and MAX RTTs

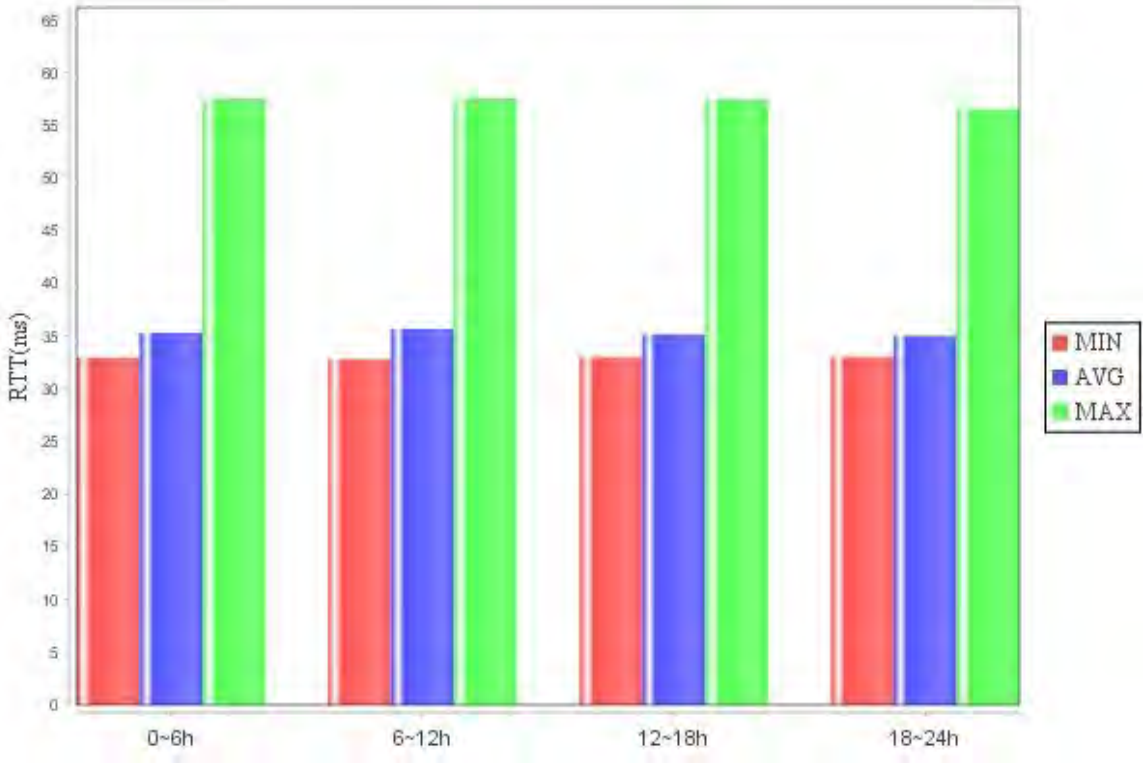


Figure 5.28: Ireland's West to East time interval MIN, AVG and MAX RTTs

In problematic countries, or specific measurements, the problem along the path can be located through further examination into the *traceroute* result. By exploring the RTTs of the different hops, it becomes a trivial matter to locate the source of a problem and discern the requisite actions in order to improve the performance.

As mentioned previously, the QoS profile for each country is very different as there's no defined set of rules in order to measure this profile. For an ISP, QoS would be the provision of different priority to different applications in order to achieve the best service quality. From an end-user's point of view, for example, it would be their speed of downloading and ability to stream videos and music. There are many countries with different regions covered by various ISPs, which causes confusion for customers as well as different QoS provisioning and therefore performance on the site of the end-users. Customers can only explore and in most cases understand the financial aspect of the market and compare ISPs depending on the price. By conducting these measurements, the results can be made available to any interested party in order to explore and make improvements where needed. All the information can be uploaded on a dedicated

website in order to help in achieving better QoS provisioning and performance. The results provide a benchmark, for almost all countries within the EU, which can be used for future reference and improvements. Ireland's performance profile is acquired and its Internet connectivity to the capitals and financial centres of some of the biggest countries in the world.

In the future, further studies can be conducted for longer periods of time or over different paths of interest. With the help of a dedicated website for the purpose, all measurements can be uploaded regularly, presenting the most recent results. They are made easy to understand even for customers. In case of more than one ISP in a country, with further exploration, measurements can be performed only in the region it is covering, depending on probes availability. Afterwards, the performance and results of the different ISPs can be compared. This will make it much easier for customers to compare various carriers, instead of looking into different sets of data on different websites, which can be very confusing for them. The QoS information will be made widely available by publishing regular reports for periods of time. Even further development of such a website could be the provisioning of dedicated measurements ordered by an interested party (private or public organizations, ISPs, end-users etc.) along paths of their interest. Results could be made public or private depending on the measurement itself. Such a development would provide more results and help building a much better understanding and overview of the QoS performance over specific paths nationally and internationally.

CHAPTER 6

CONCLUSION AND FUTURE WORK

The research presented in this thesis focuses on using active network probing strategy for the analysis of performance of national networks and international links. Being part of the RIPE Atlas project, the biggest active measurement network existing, gives the opportunity for those measurements to be conducted [95]. The innovative approach is the focus on building specific countries' Internet performance profiles by conducing *ping* and *traceroute* measurements within the borders of those predetermined countries. Those measurements were done on a cross-country basis for the four furthest north, south, east and west points available for the purpose. Another novel aspect is the establishing of Ireland's current state of the Internet by conducting measurements from the probe located in the UL TRC to important destinations around the globe [94].

The active measurements were conducted for month long time intervals. The goal was to obtain valuable information for the performance and the state of the network for the countries of interest. The measurements were broken into three groups, each one for the duration of one month. This had to be done due to the limitation of how many measurements can be performed simultaneously. The first group of measurements conducted in July 2013 consisted of *traceroute* measurements in the countries within the EU. The second group consisted of *ping* measurements for the same countries for the month of August 2013. The final third group was with both *traceroute* and *ping*

measurements from the probe in UL to various destination points consisting of capitals and financial centres of countries around the world [94].

The data from all the measurements was in JSON file format. In order to transform it and extract the needed information for the research, a java code was designed specifically for the purpose of extracting information and summarizing it. Bar graphs with the minimum, average and maximum RTT were generated and a CLI index was calculated to help discover patterns and dependencies, and get better understanding about the Internet performance. The results from the experiment were used to serve as a benchmark and also used to investigate faults throughout the networks within the countries explored. In many aspects this information is very useful especially to the ISPs in those countries. By exploring the CLI index and the bar graphs generated from the raw data extracted, one can locate faults across the networks tested and get much better understanding of the QoS for the particular path or the whole country.

Different limitations were met while undertaking this research, such as the credit system RIPE Atlas is using for conducting measurements, and the limits on number of measurements that can be conducted simultaneously. As a result of those the research needed a set time limit and separation of the measurements conducted. Despite those limitations, the results from the experiment offer a different perspective on how the Internet performance can be explored using an innovative and creative approach.

The experiment could be conducted for longer periods of time in the future and the findings from this research can be used as a benchmark to compare and look for improvements. Countries need to constantly seek for improvements in the quality of the Internet offered. The measurements can be conducted over differing and more links exploring new paths within the countries and building a more detailed profile. As the RIPE Atlas network is constantly growing there will be new probes one will be able to use in order to achieve those goals. A website with findings from the experiment can be created and give the wider public information about the performance and QoS of the investigated national networks. By updating it with regular reports and using the results from this and similar research efforts as a reference point, interested individuals or

organizations will be able to look into findings that are not available anywhere else. Even further, if ISPs want their networks externally analysed, an option for private measurements could be made available and the finding to be sent in private or with the option to be made publicly available for comparison.

As a result of the experiment a large set of data was collected. It was visualized to better show patterns and dependencies. The CLI metric calculated helped explore the symmetry for measurements in opposite directions for the countries. With the novel aspect of using active measurements for building an internet performance profile by using a cross-country model where north, south, east and west are used as vantage points, the research benefits from the large set of measurements collected and a great set of visualized results. As is the goal of the research, these results contribute towards the establishment of a benchmark for future reference and are showing the current state of the Internet performance in the selected locations.

REFERENCES

1. Coates, M.J., et al., *Internet tomography.* Signal Processing Magazine, 2002. **19**: p. 47-65.
2. Lawrence, E., G. Michailidis, and V. Nair, *Fast, moment-based estimation methods for delay tomography.* Transactions on signal processing, 2008.
3. Moloisane, A., I. Ganchev, and M. O'Droma, *Internet Tomography: An Introduction to Concepts, Techniques, Tools and Applications* 2013: Cambridge Scholars Publishing.
4. Lawrence, E., et al., *Network tomography: a review and recent developments.* 2005.
5. Tsang, Y., M.J. Coates, and R. Nowak, *Network Delay Tomography.* IEEE Trans. Signal Processing, special issue on Signal Processing in Networking, 2003. **51**(8): p. 2125-2136.
6. Xi, B., G. Michailidis, and V.N. Nair, *Estimating network loss rates using active tomography.* 2005.
7. Castro, R., et al., *Network tomography: recent developments.* Statistical Science, 2003. **19**: p. 499-517.
8. Chen, Y., D. Bindel, and R.H. Katz. *Tomography-based Overlay Network Monitoring*. in *Proceedings of the 3rd ACM SIGCOMM Conference on Internet Measurement 2003*. 2003.
9. Pasztor, A., *Accurate active measurement in the Internet and its applications*, in *Department of Electrical and Electronic Engineering*. 2003, The University of Melbourne.
10. CAIDA, *CAIDA Metrics Working Group, "Network Measurement FAQ, draft," [Online]. Available - http://www.caida.org/outreach/metricswg/faq-03.xml [Last Accessed: June 2013].*
11. Martin, H.S., A. McGregor, and J.G. Cleary, *Analysis of Internet Delay Times*, in *The First Passive and Active Measurement Workshop*. 2000.
12. Claffy, K., G. Miller, and K. Thompson, *The nature of the beast: recent traffic measurements from an Internet backbone.* INET '98, 1998.
13. Agrawal, S., K. Naidu, and R. Rastogi, *Diagnosing link-level anomalies using passive probes*, in *IEEE INFOCOM 2007*. 2007.
14. Nguyen, H.X. and P. Thiran, *Binary versus analogue path monitoring in IP networks.* Lecture Notes in Computer Science, 2005. **3431**: p. 97.
15. Caceres, R., et al., *Multicast-based inference of network internal loss characteristics.* IEEE Transactions on Information Theory, 1999. **45**.
16. Bu, T., et al., *Network tomography on general topologies*, in *SIGMETRICS*. 2002.
17. Shavitt, Y., et al., *Computing the unmeasured: An algebraic approach to internet mapping*, in *IEEE INFOCOM*. 2001.
18. Padmanabhan, V.N., L. Qiu, and H.J. Wang, *Server-based inference of internet performance*, in *IEEE INFOCOM*. 2003.
19. Chen, Y., et al., *An algebraic approach to practical and scaleable overlay network monitoring*, in *SIGCOMM*. 2004.
20. Duffield, N.G., *Simple network performance tomography*, in *Internet Measurement Conference* 2003.
21. Nguyen, H. and P. Thiran, *Using end-to-end data to infer lossy links in sensor networks*, in *IEEE INFOCOM*. 2006: Barcelona.
22. Nguyen, H.X. and P. Thiran, *Active measurement for multiple link failures diagnosis in IP networks*, in *Passive & Active Measurement Workshop*. 2004.

23. Chaudet, C., et al., *Optimal positioning of active and passive monitoring devices*, in *CoNEXT*. 2005.
24. Suh, K., et al., *Locating network monitors: Complexity, heuristics and coverage*, in *IEEE INFOCOM*. 2005.
25. Cantieni, G.R., et al., *Reformulating the monitor placement problem: Optimal network-wide sampling*, in *Intel Research Technical Report*. 2005.
26. Endace, *Endace: Accelerated network security, [Online]. Available - http://www.endace.com/ [Last Accessed: June 2013].*
27. Feige, U., *A threshold of ln n for approximating set cover (preliminary version)*, in *STOC*. 1996.
28. Paxson, V., *End-to-end routing behavior in the Internet.* IEEE/ACM Transactions on Networking, 1997. **5**: p. 601-615.
29. Vardi, Y., *Estimating source-destination traffic intensities from link data.* Journal of the American Statistical Association, 1996. **91**: p. 365-377.
30. Tebaldi, C. and M. West, *Bayesian inference of network traffic using link count data (with discussion).* Journal of the American Statistical Association, 1998. **93**: p. 557-576.
31. Cao, J., et al., *Time varying network tomography: router link data.* Journal of the American Statistical Association, 2000. **95**: p. 1063-1072.
32. Zhang, Y., et al. *Fast accurate computation of large scale IP traffic matrices from link loads*. in *ACM Sigmetrics* 2003. San Diego, CA.
33. Zhang, Y., et al. *An information theoretic approach to traffic matrix estimation*. in *ACM SIGCOMM* 2003. Karlsruhe, Germany.
34. Soule, A., et al. *How to identify and estimate the largest traffic matrix elements in a dynamic environment*. in *ACM Sigmetrics* 2004. New York.
35. Liang, G., N. Taft, and B. Yu, *A fast lightweight approach to origin destination IP traffic estimation using partial measurements.* 2005.
36. Roughan, M. *First order characterization of Internet traffic matrices*. in *ISI*. 2005. Sydney, Australia.
37. Eriksoon, B., P. Barford, and R. Nowak, *Network discovery from passive measurements*, in *SIGCOMM*. 2008: Seattle, Washington, USA.
38. Alderson, D., et al., *Understanding Internet Topology: Principles, Models and Validation*, in *IEEE/ACM Transactions on Networking*. 2005.
39. Faloutsos, C., M. Faloutsos, and P. Faloutsos. *On Power-Law Relationships of Internet Topology*. in *ACM SIGCOMM* 1999. Cambridge, MA.
40. Govindan, R. and H. Tangmunarunkit. *Heuristics for Internet Map Discovery*. in *IEEE INFOCOM* 2000. Tel Aviv, Israel.
41. Gueye, B., et al., *Constraint-Based Geolocation of Internet Hosts*, in *IEEE/ACM Transactions on Networking*. 2006.
42. Spring, N., R. Mahajan, and D. Wetherall. *Measuring ISP Topologies with Rocketfuel*. in *ACM SIGCOMM* 2002. Pittsburgh, PA.
43. Jin, C., H. Wang, and K. Shin. *Hop-Count Filtering: An Erective Defense Against Spoofed Trffic.* in *IEEE INFOCOM* 2003. San Francisco, CA.
44. Krishnamurthy, B. and J. Wang. *On Network-Aware Clustering of Web Clients*. in *ACM SIGCOMM*. 2000. Stockholm, Sweeden.
45. Madadevan, P., et al. *Orbis: Rescaling Degree Correlations to Generate Annotated Internet Topologies*. in *ACM SIGCOMM* 2007. Kyoto, Japan.
46. CAIDA. *The Skitter Project, [Online]. Available - http://www.caida.org/tools/measurement/skitter/. [Last Accessed: July 2013]*.
47. Jain, A.K. and M. Figueiredo, *Unsupervised learning of finite mixture models.* IEEE Transaction on Pattern Analysis and Machine Intelligence, 2002. **24**(3): p. 381-396.

48. Kalidindi, S. and M.J. Zekauskas, *Surveyor: An Infrastructure for Internet Performance Measurements*, in *INET*. 1999.
49. McGregor, T. and H.-W. Braun, *Balancing Cost and Utility in Active Monitoring: The AMP example*, in *INET*. 2000: Tokyo, Japan.
50. McGregor, T., H.-W. Braun, and J. Brown, *The NLANR Network Analysis Infrastructure.* IEEE Communications Magazine special issue on "Network Traffic Measurements and Experiments, 2000.
51. Uijterwaal, H. and O. Kolkman, *RIPE-158: Internet Delay Measurements using Test Traffic: Design Note*. 1997, RIPE NCC.
52. Bolot, J.C., *Characterizing end-to-end packet delay and loss in the internet.* High-Speed Networks, 1993. **2**(3).
53. Carter, R.L. and M.E. Crovella, *Measuring bottleneck link speed in packet-switched networks.* Performance Evaluation 1996. **27&28**: p. 297-318.
54. Ribeiro, V., et al. *Multifractal crosstraffic estimation*. in *ITC Specialist Seminar*. 2000. Monterey, USA.
55. Cottrell, L., *Comparison of some Internet Active End-to-end Performance Measurement projects, [Online]. Available - http://www.slac.stanford.edu/comp/net/wan-mon/iepm-cf.html. [Last accessed: July 2013].* 1999.
56. Cardwell, N., S. Savage, and T. Anderson. *Modeling TCP Latency*. in *INFOCOM*. 2000.
57. Mathis, M., et al., *The Macroscopic Behavior of the TCP Congestion Avoidance Algorithm.* Computer Communications Review, 1997. **27**(3).
58. Padhye, J., et al. *Modeling TCP Throughput: A Simple Model and its Empirical Validation*. in *ACM SIGCOMM* 1998.
59. Padhye, J. and S. Floyd. *On Inferring TCP Behavior*. in *ACM SIGCOMM* 2001.
60. Veres, A. and M. Boda. *Tha Chaotic Nature of TCP Congestion Control*. in *Infocom*. 2000.
61. Siegell, B.S., *Felix Project: Topology Discovery from One-way Delay Measurements*, in *The First Passive and Active Measurement Workshop*. 2000: Hamilton, New Zealand.
62. Claffy, K., *Internet measurement and data analysis: topology, workload, performance and routing statistics*, in *NAE workshop*. 1999.
63. Horneffer, M., *Assessing Internet Performance Metrics Using Large-Scale TCP SYNbased Measurements*, in *The First Passive and Active Measurement Workshop*. 2000: Hamilton, New Zealand.
64. Cheng, H.S., et al., *Measuring IP Network Performance - The SingAREN Approach*, in *The First Passive and Active Measurement Workshop*. 2000: Hamilton, New Zealand.
65. Cottrell, L. and W. Matthews, *Comparison of Surveyor and PingER, [Online]. Available - http://www.slac.stanford.edu/comp/net/wan-mon/surveyor-vs-pinger.html. [Last accessed: August 2013].* 1999.
66. Matthewss, W. and L. Cottrell, *The PingER Project: Active Internet Performance Monitoring for the HENP Community.* IEEE Communications Magazine special issue on "Network Traffic Measurements and Experiments", 2000.
67. Luckie, M.J., A.J. McGregor, and H.-W. Braun, *Towards Improving Packet Probing Techniques*, in *ACM SIGCOMM*. 2001: San Francisco, CA.
68. Jacobson, V., *Patchar - a tool to infer characteristics of internet paths, [Online]. Available - ftp://ftp.ee.lbl.gov/pathchar/msri-talk.pdf. [Last accessed: August 2013].* 1997.
69. Lawrence, E., G. Michailidis, and V. Nair, *Statistical inverse problems in active network tomography.* 2007.
70. Xi, B., *Estimating internal link loss rates using active network tomography*. 2004, The University of Michigan.

71. Presti, L., et al., *Multicast-based inference of network-internal delay distributions.* IEEE Transactions on Networking 2000. **10**: p. 761–775.
72. Liang, G. and B. Yu, *Maximum pseudo likelihood estimation in network tomography.* IEEE Transactions on Signal Processing 2003. **51**: p. 2043–2053.
73. Shih, M.-F. and A.O. Hero, *Unicast-based inference of network link delay distributions with finite mixture models.* IEEE Transactions on Signal Processing 2003. **51**: p. 2219–2228.
74. Lawrence, E., G. Michailidis, and V.N. Nair, *Network delay tomography using flexicast experiments.* Journal of the Royal Statistical Society: Series B (Statistical Methodology), 2006. **68**(5): p. 785-813.
75. Coates, M.J. and R.D. Nowak. *Network loss inference using unicast end-to-end measurement*. in *ITC Conference on IP Traffic, Modelling and Management*. 2000.
76. Xi, B., G. Michailidis, and V.N. Nair. *Least squares estimates of network link loss probabilities using end-to-end multicast measurements*. in *37th Conference on Information Sciences and Systems*. 2003.
77. Michailidis, G., V.N. Nair, and B. Xi, *Fast least squares algorithms for estimating and monitoring network link Losses based on active probing schemes.* 2005.
78. Lawrence, E., G. Michailidis, and V.N. Nair, *Local Area Network analysis using end-to-end delay tomography.* 2005.
79. Lawrence, E., G. Michailidis, and V.N. Nair, *Flexicast delay tomography.* 2005.
80. Tian, G., et al. *Routing topology identification based on end-to-end measurements*. in *IEEE International Conference on Information and Automation* 2008. Zhangjiajie, China.
81. Bestavros, A., J. Byers, and K. Harfoush, *Inference and labeling of metric-induced network topologies.* IEEE Transactions on Parallel and Distributed Systems, 2005. **16**(11): p. 1053-1065.
82. Castro, R., M. Coates, and R. Nowak, *Likelihood Based Hierarchical Clustering.* IEEE Transactions on signal processing, 2004. **52**(8): p. 2308-2321.
83. Coates, M., et al. *Maximum Likelihood Network Topology Identification from Edge-based Unicast Measurements*. in *ACM SIGMETRICS international conference on Measurement and modeling of computer systems*. 2002.
84. Coates, M., M. Rabbat, and R. Nowak. *Merging logical topologies using end-to-end measurements*. in *SIGCOMM Conference on Internet Measurement*. 2003. Miami, FL.
85. Duffield, N., J. Horowitz, and Presti, *Multicast Topology Inference from End-to-end Loss.* IEEE Transactions on Information Theory, 2002. **48**(1): p. 26-45.
86. Hui, T. and S. Hong, *An improved algorithm for multicast topology discovery from end-to-end measurements.* International Journal of Communication Systems, 2006. **19**(8): p. 935-953.
87. Rabbat, M., M. Coates, and R. Nowak, *Multiple Source Internet Tomography.* IEEE Journal on Selected Areas in Communications, 2006. **24**(12): p. 2221-2234.
88. Tsang, Y., et al. *Network radar: Tomography from round trip time measurements*. in *ACM SIGCOMM Conference on Internet Measurement*. 2004. Taormina, Sicily, Italy.
89. Brodie, M., I. Rish, and S. Ma. *Optimizing probe selection for fault localization*. in *Distributed Systems: Operation and Management*. 2001. Nancy, France.
90. Rish, I., M. Brodie, and S. Ma. *Accuracy vs. Efficiency Trade-offs in Probabilistic Diagnosis*. in *The Eighteenth National Conference on Artificial Intelligence* 2002. Edmonton, Alberta, Canada.
91. Brodie, M., et al., *Active probing strategies for problem diagnosis in distributed systems.* 2003.
92. CAIDA. *http://www.caida.org/projects/ark/, [Online]. [Last Accessed: July 2013]*.

93. RIPE, *https://atlas.ripe.net/about/, [Online]. [Last Accessed: August 2013]*.
94. N. Todorov, I. Ganchev, M. O'Droma. 2014. "*Internet Performance Profiling of Countries*". 4th UL-NUIG Alliance Postgraduate Research Day 2014, p. x-1. 20 May, Limerick, Ireland.
95. N. Todorov, I. Ganchev, M. O'Droma. 2013. "*Internet Tomography aiding Traffic Engineering*". Book of Abstracts, NUIG-UL Alliance Engineering, Informatics and Science Research Day 2013, p. 10. 11 April, Galway, Ireland.

APPENDIX LISTING

The following appendices are contained within the attached DVD. The contents of each of these appendices is here briefly set out.

APPENDIX 1: CLI RESULTS

This appendix consists of the full table with CLI results for all countries within the EU and for the destinations outside the EU.

APPENDIX 2: JAVA CODE

In Appendix 2, the java code for processing the raw data, extracting the needed information and generating the bar graphs, can be found.

APPENDIX 3: PING RESULTS FOR THE COUNTRIES WITHIN THE EU

Appendix 3 consists of the tables and bar graphs extracted from the *ping* raw data files for the countries within the EU. It is divided in folders for each country, and subfolders for all the directions where measurements were conducted. Inside those folders can be found the raw data files, bar graphs and the excel tables used for generating them.

APPENDIX 4: PING RESULTS FOR THE DESTINATIONS OUTSIDE THE EU

Appendix 4 consists of the tables and bar graphs extracted from the *ping* raw data files for measurements to destinations outside the EU. It is divided in folders for each country, and subfolders for the two destinations (e.g. capital and financial centre). Inside those folders can be found the raw data files, bar graphs and the excel tables used for generating them.

APPENDIX 5: TRACEROUTE RESULTS FOR THE COUNTRIES WITHIN THE EU

Appendix 5 consists of the tables and bar graphs extracted from the *traceroute* raw data files for the countries within the EU. It is divided in folders for each country, and subfolders for all the directions where measurements were conducted. Inside those folders can be found the raw data files, bar graphs and the excel tables used for generating them.

APPENDIX 6: TRACEROUTE RESULTS FOR THE DESTINATIONS OUTSIDE THE EU

Appendix 6 consists of the tables and bar graphs extracted from the *traceroute* raw data files for measurements to destinations outside the EU. It is divided in folders for each country, and subfolders for the two destinations (e.g. capital and financial centre). Inside those folders can be found the raw data files, bar graphs and the excel tables used for generating them.